

ASTRONOMICAL SPECTROSCOPY

(based on lectures from Prof. Piero Rafanelli and Dr. Giovanni La Mura)

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Fundamentals of atomic spectroscopy

1.1 Quantum numbers definition

Let us consider an atom or an ion with 1 valence electron. Valence electrons are defined as those electrons which lie outside the atomic core, that consists of the stable group of electrons filling a complete shell. The elements that have only 1 valence electron are H and alkaline metals (Li, Na, K, Rb, Cs, Fr). Ions with only 1 electron are, instead, He^+ , Li^{++} etc.

The solution of the time-independent Schrödinger equation:

$$-\frac{\hbar^2}{2\mu}\nabla^2\psi - \frac{1}{4\pi\epsilon_0}\frac{Ze^2}{r}\psi = E\psi$$

gives a set of equations, called wave functions (ψ). Each wave function is associated with an energy level and describes the spatial probability distribution of the electrons, that is the probability of finding an electron of a certain energy at some position in space. A wave function for an electron in an atom is called an atomic orbital.

The state of an atom or an ion with only 1 valence electron is defined by the following quantum numbers:

- n principal quantum number ($n = 1, 2, 3, \dots, \infty$)
It defines the energy level in which the electron can be found and the size of the orbital.
- ℓ azimuthal quantum number ($\ell = 0, 1, \dots, n - 1$)
It is related to the angular momentum associated to the orbital motion and the states with $\ell = 0, 1, 2, 3, 4, \dots, n - 1$, are also named s (sharp), p (principal), d (diffuse), f (fundamental), etc.
- m_ℓ magnetic quantum number ($-\ell \leq m_\ell \leq \ell$)
It specifies the orientation in space of an orbital of a given $n\ell$.
- s spin quantum number ($s = \pm 1/2$)
It specifies the orientation of the spin axis of an electron.

The Pauli exclusion principle states that no two electrons in the same atom can have identical values for all four of their quantum numbers. This means that no more than two electrons can occupy the same orbital, and that two electrons in the same orbital must have opposite spins.

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In addition we can consider the total angular momentum $j = \ell + s$ (also called inner quantum number), that assumes m_j values, where $-j \leq m_j \leq j$ is called secondary total angular momentum quantum number.

The angular momentum quantum numbers ℓ , s and j are associated with the actual angular momentum vectors $\vec{L}$, $\vec{S}$ and $\vec{J}$ by the relations:

$$\begin{aligned} |\vec{L}| &= \sqrt{\ell(\ell+1)}\hbar \\ |\vec{S}| &= \sqrt{s(s+1)}\hbar \\ |\vec{J}| &= \sqrt{j(j+1)}\hbar \end{aligned}$$

Electron states are defined by means of their quantum numbers n , ℓ , s , j (and m_j if a magnetic field is present) and each state corresponds to a different binding energy.

If we consider, for example, an atom of Na I ($Z = 11$), the electron configuration is

$$1s^2 2s^2 2p^6 3s^1$$

where $1s^2 2s^2 2p^6$ is the electron configuration of the atomic core, while $3s^1$ represents the valence electron in its ground state.

1.2 Atomic term symbol

If an atom or an ion has more than 1 valence electron, the interaction between the vectors of the system leads to coupling of the electron properties: spin-spin, orbit-orbit, spin-orbit. In light atoms (generally $Z \leq 30$) the **spin-orbit coupling** is weaker than the other two and it follows the **Russell–Saunders coupling (LS)** scheme, while in heavier elements with larger nuclear charge, it becomes as strong as the other two and the interaction is described by the **$j - j$ coupling**.

In case of LS coupling we can perform a vectorial sum of the orbital and spin angular momenta and, thus, define the total azimuthal quantum number:

$$\vec{L} = \sum_i \vec{\ell}_i$$

assuming $(2L + 1)$ values $M_L = -L, -L + 1, \dots, L - 1, L$, and the total spin quantum number:

$$\vec{S} = \sum_i \vec{s}_i$$

assuming $(2S + 1)$ values $M_S = -S, -S + 1, \dots, S - 1, S$.

The sums are calculated on the valence electrons only. In particular, when:

$$\begin{aligned} i = 1 &\Rightarrow L_1 = \ell_1 & S_1 = s_1 \\ i = 2 &\Rightarrow L_2 = \ell_1 + \ell_2, \dots, |\ell_1 - \ell_2| & S_2 = s_1 + s_2, \dots, |s_1 - s_2| \\ i = 3 &\Rightarrow L_3 = L_2 + \ell_3, \dots, |L_2 - \ell_3| & S_3 = S_2 + s_3, \dots, |S_2 - s_3| \end{aligned}$$

1.2. ATOMIC TERM SYMBOL

We can therefore define the term state of an ion as it is described by the quantum number set nLS . Then S and L couple together and form the total angular momentum J :

$$\vec{J} = \vec{L} + \vec{S}$$

Each nLS term naturally splits into $r = 2J + 1$ sub-levels, characterized by different values of J :

$$\text{if } L \geq S \quad J = L + S, L + (S - 1), \dots, L - (S - 1), L - S \quad r = 2S + 1$$

$$\text{if } L < S \quad J = S + L, S + (L - 1), \dots, S - (L - 1), S - L \quad r = 2L + 1$$

so that, in absence of any external field, the term nLS is split in a multiplet $nLSJ$. Each multiplet term is represented by the so called **term symbol**:

$$n^{(2S+1)}L_J$$

We also define the **parity of a state** $(-1)^{\sum_i \ell_i}$ to be odd (o) when it is equal to -1 or even (e) when it is equal to $+1$, so that the term is represented by:

$$n^{(2S+1)}L_J^{o,e}$$

It is easy to see that in case of an odd number of electrons in odd orbitals (that is orbitals with $\ell = 1, 3, 5, \dots$) the term symbol is odd, while an even number of electrons in odd orbitals will make an even term symbol.

In the presence of an external magnetic field, each multiplet term splits in $g = 2J + 1$ simple terms, due to the various possible orientations of the angular momentum $\vec{J}$ with respect to the field lines.

If there is no external magnetic field, we still have to weigh each energy levels g times to take into account the effect of degeneracy, that is the number of quantum states having different ℓm_ℓ s with the same energy. g is called **statistical weight** or **multiplicity of level**, and it corresponds to how many electrons you can put in the level before Pauli exclusion principle is violated.

The statistical weight of a total term LS is calculated as the sum of the weights of its multiplet terms LSJ :

$$\begin{aligned} g &= \sum_{J=L-S}^{L+S} (2J + 1) = 2 \sum_{J=L-S}^{L+S} J + (2S + 1) \\ &= 2[(L - S) + (L - S + 1) + \dots + (L + S - 1) + (L + S)] + (2S + 1) \\ &= 2L(2S + 1) + (2S + 1) \\ &= (2L + 1)(2S + 1) \end{aligned}$$

Finally, in order to decide what is the **fundamental level**, that is the level having the lowest energy value, we need to apply the **Hund's rules** (valid for the Russell-Saunders coupling):

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1. select the term with the highest S ;
2. given S , select the term with the highest L ;
3. given S and L ,
 - select $J = |L - S|$, if less than half of the shell is occupied;
 - select $J = S$, if half of the shell is occupied;
 - select $J = L + S$, if more than half of the shell is occupied;

1.3 Hydrogen and Helium

In the case of the **hydrogen** atom, there is only 1 electron with spin $s = 1/2$. The ground state is $n = 1$, therefore:

$$L = \ell_1 = 0 \quad S = s_1 = \frac{1}{2} \quad J = \frac{1}{2} \Rightarrow 1^2S_{1/2}^e$$

and the statistical weight is $g_1 = 2$.

At level $n = 2$, we can have $\ell_1 = 0$ that gives $2^2S_{1/2}^e$, and $\ell_1 = 1$ that gives:

$$L = \ell_1 = 1 \quad S = s_1 = \frac{1}{2} \quad J = \frac{3}{2}, \frac{1}{2} \Rightarrow 2^2P_{3/2}^o, 2^2P_{1/2}^o$$

whose statistical weights are 4 and 2 respectively. The total multiplicity of level 2 is $g_2 = 2 + 4 + 2 = 8$. In the hydrogen atom, each total term LS corresponds to a statistical weight of:

$$g_n = \sum_{L=0}^{n-1} 2(2L+1) = 2n^2$$

Therefore all the LS terms with the same principal quantum number n have the same energy. This occurs because the hydrogen atom has only one electron and there is no mantle affecting the binding energy according to the orbit's eccentricity. Of course, this situation changes for more complex ions.

The **helium** atom case is more complex.

The ground state configuration is $1s^2$. The two electrons are not independent because, due to the Pauli exclusion principle, their spins must be anti-parallel. Therefore, we have:

$$\left. \begin{array}{l} \ell_1 = \ell_2 = 0 \Rightarrow L = 0 \\ s_1 = \frac{1}{2} \quad s_2 = -\frac{1}{2} \Rightarrow S = 0 \end{array} \right\} J = 0 \Rightarrow 1^1S_0^e$$

with statistical weight $g_1 = 1$.

If we consider the excited configuration $1s^1 2s^1$, the two electrons are now independent because they do not occupy the same orbital. Therefore, we have:

$$\left. \begin{array}{l} \ell_1 = \ell_2 = 0 \Rightarrow L = 0 \\ s_1 = \frac{1}{2} \quad s_2 = \frac{1}{2} \Rightarrow S = 0, 1 \end{array} \right\} J = 0, 1 \Rightarrow 2^3S_1^e, 1^1S_0^e$$

with multiplicity $g = 3$ and $g = 1$.

1.4 Two *p* electrons

In case the two electrons are **independent**, that is they are not in the same orbital (np, mp):

$$\begin{aligned} \ell_1 = 1 \quad \ell_2 = 1 &\Rightarrow L = 2, 1, 0 \\ s_1 = \frac{1}{2} \quad s_2 = \frac{1}{2} &\Rightarrow S = 1, 0 \end{aligned}$$

To calculate J , it is necessary to combine the different values of L and S :

$$\begin{aligned} L = 2 \quad S = 1 &\Rightarrow J = 3, 2, 1 \Rightarrow {}^3D_3, {}^3D_2, {}^3D_1 \\ L = 2 \quad S = 0 &\Rightarrow J = 2 \Rightarrow {}^1D_2 \\ L = 1 \quad S = 1 &\Rightarrow J = 2, 1, 0 \Rightarrow {}^3P_2, {}^3P_1, {}^3P_0 \\ L = 1 \quad S = 0 &\Rightarrow J = 1 \Rightarrow {}^1P_1 \\ L = 0 \quad S = 1 &\Rightarrow J = 1 \Rightarrow {}^3S_1 \\ L = 0 \quad S = 0 &\Rightarrow J = 0 \Rightarrow {}^1S_0 \end{aligned}$$

In general, given an orbital quantum number ℓ the total number of electrons that can be hosted is $t = 2(2\ell + 1)$ because of the magnetic quantum number m_ℓ . If n_e is the number of electrons, the possible combinations taking into account the Pauli exclusion principle, which restricts the number of allowed configurations, are:

$$N = \binom{t}{n_e} = \frac{t!}{n_e!(t - n_e)!}$$

In case we have two **equivalent** *p* electrons with the same principal and orbital quantum numbers (np^2), there are 15 possible combinations:

+1	↑	↑		↓	↓		↑↓	↑	↑			↓	↓		
0	↑		↑	↓		↓		↓		↑		↑		↓	↑↓
-1		↑	↑		↓	↓			↓	↓	↑↓		↑	↑	
M_L	1	0	-1	1	0	-1	2	1	0	-1	-2	1	0	-1	0
M_S	1	1	1	-1	-1	-1	0	0	0	0	0	0	0	0	0

The configurations corresponding to $M_L = \pm 2, \pm 1, 0$ ($L = 2$) and $M_S = 0$ ($S = 0$) imply the existence of a term 1D_2 , those corresponding to $M_L = \pm 1, 0$ ($L = 1$) and $M_S = \pm 1, 0$ ($S = 1$) three terms ${}^3P_2, {}^3P_1, {}^3P_0$, and finally the $M_L = 0$ ($L = 0$) and $M_S = 0$ ($S = 0$) a term 1S_0 . Following the Hund's rules, the fundamental level is: 3P_0 .

1.5 Three *p* electrons

A similar approach can be used for three independent electrons (n_1p, n_2p, n_3p) and three equivalent electrons (np^3). In the last case there are 20 possible combinations:

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+1	↑	↓	↑	↑	↓	↑↓	↑↓	↑		↑
0	↑	↓	↑	↓	↑	↑		↑↓	↑↓	
-1	↑	↓	↓	↑	↑		↑		↑	↑↓
M_L	0	0	0	0	0	2	1	1	-1	1
M_S	3/2	-3/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2

+1		↑	↓	↓	↑↓	↑↓	↓		↓	
0	↑	↓	↑	↓	↓		↑↓	↑↓		↓
-1	↑↓	↓	↓	↑		↓		↓	↑↓	↑↓
M_L	-2	0	0	0	2	1	1	-1	1	-2
M_S	1/2	-1/2	-1/2	-1/2	-1/2	-1/2	-1/2	-1/2	-1/2	-1/2

The configurations corresponding to $M_L = \pm 2, \pm 1, 0$ ($L = 2$) and $M_S = \pm 1/2$ ($S = 1/2$) imply the existence of two terms $^2D_{3/2}$, $^2D_{5/2}$, those corresponding to $M_L = \pm 1, 0$ ($L = 1$) and $M_S = \pm 1/2$ ($S = 1/2$) two terms $^3P_{1/2}$, $^3P_{3/2}$, and finally the $M_L = 0$ ($L = 0$) and $M_S = \pm 3/2, \pm 1/2$ ($S = 3/2$) a term $^4S_{3/2}$. The fundamental level is: $^4S_{3/2}$.

1.6 Four p electrons

We consider here only the four equivalent electrons (np^4) case, for which there are again 15 possible combinations:

+1	↑↓	↑↓	↑↓	↑↓	↑	↑	↓	↓	↑	↑	↓	↓	↑↓		↑↓
0	↑	↑	↓	↓	↑↓	↑↓	↑↓	↑↓	↑	↓	↑	↓	↑↓	↑↓	
-1	↑	↓	↑	↓	↑	↓	↑	↓	↑↓	↑↓	↑↓	↑↓		↑↓	↑↓
M_L	1	1	1	1	0	0	0	0	-1	-1	-1	-1	2	-2	0
M_S	1	0	0	-1	1	0	0	-1	1	0	0	-1	0	0	0

The configurations corresponding to $M_L = \pm 2, \pm 1, 0$ ($L = 2$) and $M_S = 0$ ($S = 0$) imply the existence of a term 1D_2 , those corresponding to $M_L = \pm 1, 0$ ($L = 1$) and $M_S = \pm 1, 0$ ($S = 1$) three terms 3P_2 , 3P_1 , 3P_0 , and finally the $M_L = 0$ ($L = 0$) and $M_S = 0$ ($S = 0$) a term 1S_0 . Following the Hund's rules, the fundamental level is: 3P_2 .

1.7 Energy levels and Grötrian diagrams

The cases presented in the last section correspond to the ground and low excited states of **carbon atom** (Fig. 1.1), **nitrogen atom** (Fig. 1.2), and **oxygen atom** (Fig. 1.3).

Each term is associated to an energy level which is usually expressed in eV or in cm^{-1} units and the plots showing spectroscopic terms vs. energy are named **energy level diagrams**.

It is easy to convert from cm^{-1} to eV, and viceversa.

1.7. ENERGY LEVELS AND GRÖTRIAN DIAGRAMS

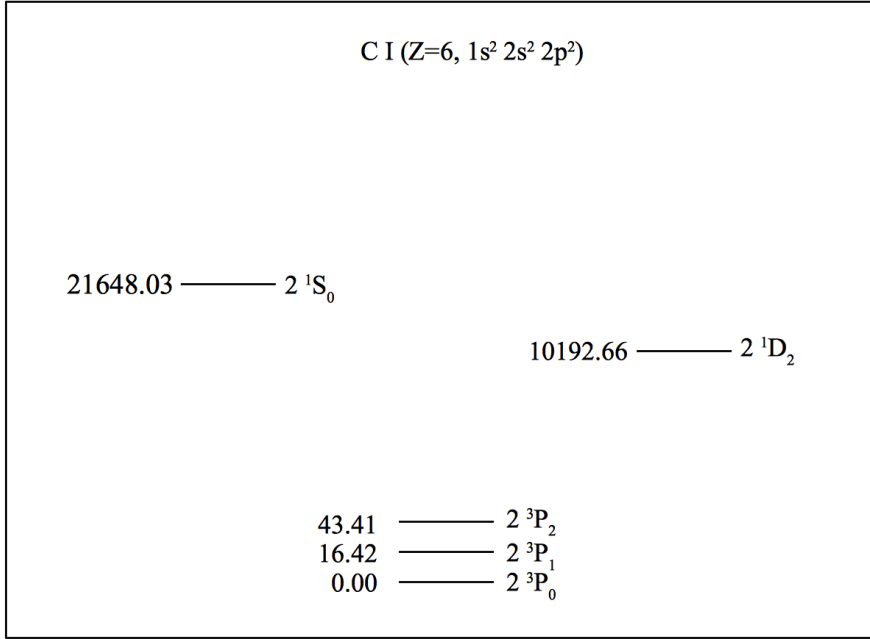


Figure 1.1: Ground and low excited levels of C I. Energy values are in cm⁻¹ units.

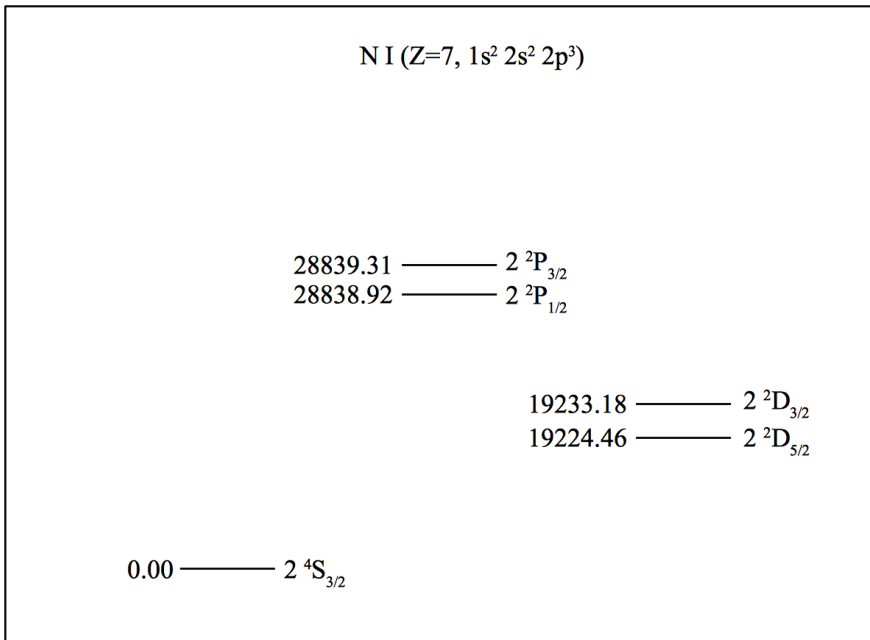


Figure 1.2: Ground and low excited levels of N I. Energy values are in cm⁻¹ units.

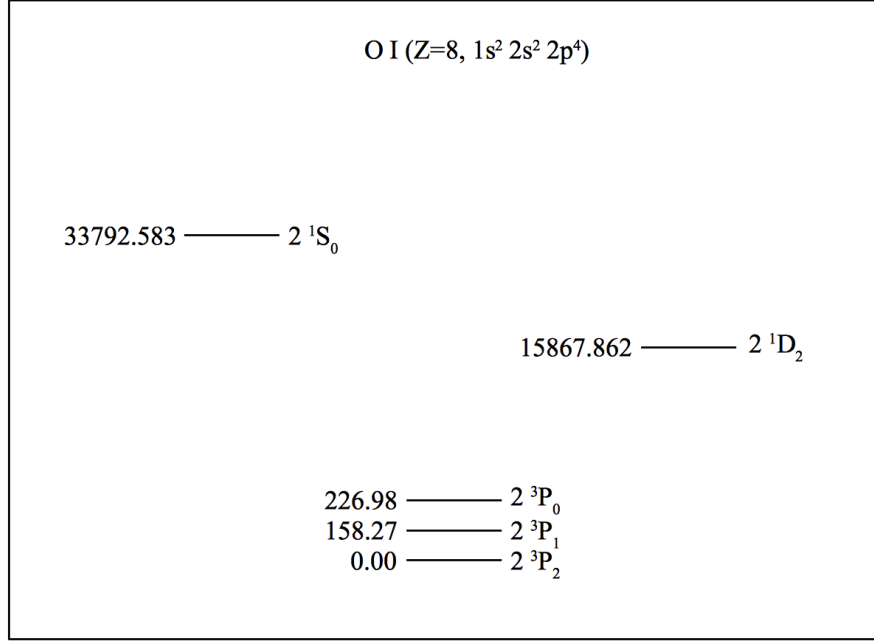


Figure 1.3: Ground and low excited levels of O I. Energy values are in cm^{-1} units.

$$E = h\nu = \frac{hc}{\lambda} = \frac{1.239842 \cdot 10^{-4}}{\lambda} \quad [\text{eV}]$$

where λ is in cm units. Therefore an energy level of 10^4 cm^{-1} ($\lambda = 1 \mu\text{m}$) corresponds to about 1.24 eV, or equally 1 eV corresponds to about 8065.54 cm^{-1} .

If we want now to calculate the wavelength corresponding to the energy necessary to move an electron from a bound state m to another bound state n :

$$\lambda_{mn} = \frac{12398.42}{\Delta E_{mn}} \quad [\text{\AA}]$$

where ΔE_{mn} is the energy difference (in eV) between level m and level n . Plots showing the possible electron transitions between energy levels of an atom and their corresponding wavelengths are named **Grötrian diagrams**.

1.8 Selection rules

Not all possible electron transitions are allowed. Indeed for transitions involving energy exchanges with radiation, through the emission or absorption of photons, the following general **selection rules** apply:

1. no restriction on Δn
2. only transitions connecting states with the different parity are permitted ($e \leftrightarrow o$ or $\Delta(\sum_i \ell_i) = \pm 1, \pm 3, \dots$)

3. $\Delta J = \pm 1, 0$ but $J = 0 \rightarrow J = 0$ is forbidden
4. $\Delta M_J = \pm 1, 0$ but $M_J = 0 \rightarrow M_J = 0$ is forbidden, if it is also $\Delta J = 0$
5. $\Delta L = \pm 1, 0$ but it must always be $\Delta \ell = \pm 1$ for the jumping electron
6. $\Delta S = 0$ (if $L \geq S$ the transitions between states with different multiplicity are forbidden)

These conditions hold in electric dipole approximation, 5 and 6 are specifically valid in Russell-Saunders coupling (and not in case of $j - j$ coupling). Magnetic dipole radiation is 5 orders of magnitude weaker and electric quadrupole radiation is 8 orders of magnitude weaker.

1.9 Examples

1.9.1 Hydrogen

We can calculate the spectral terms corresponding to the different location of the unique electron in the orbitals of the atom.

$$\begin{array}{llll}
 1s^1 & \Rightarrow & L = 0 & S = 1/2 \Rightarrow 1 \ ^2S_{1/2}^e \\
 2s^1 & \Rightarrow & L = 0 & S = 1/2 \Rightarrow 2 \ ^2S_{1/2}^e \\
 2p^1 & \Rightarrow & L = 1 & S = 1/2 \Rightarrow 2 \ ^2P_{1/2}^o, 2 \ ^2P_{3/2}^o \\
 3s^1 & \Rightarrow & L = 0 & S = 1/2 \Rightarrow 3 \ ^2S_{1/2}^e \\
 3p^1 & \Rightarrow & L = 1 & S = 1/2 \Rightarrow 3 \ ^2P_{1/2}^o, 3 \ ^2P_{3/2}^o \\
 3d^1 & \Rightarrow & L = 2 & S = 1/2 \Rightarrow 3 \ ^2D_{3/2}^e, 3 \ ^2D_{5/2}^e \\
 \dots & & &
 \end{array}$$

On the basis of the selection rules: $\Delta n = \text{any integer}$, $\Delta \ell = \pm 1$, $\Delta s = 0$, therefore when we consider the fundamental level $n = 1$, only transitions $1s \leftrightarrow 2p, 3p, 4p$, etc. are **permitted**. These transitions form the well known **Lyman series**, whose spectral lines belong to the UV wavelength range. In particular,

$$\text{Ly}\alpha \Rightarrow 1 \ ^2S_{1/2} \leftrightarrow \begin{cases} 2 \ ^2P_{1/2} \\ 2 \ ^2P_{3/2} \end{cases}$$

while the transition $1 \ ^2S_{1/2} \leftrightarrow 2 \ ^2S_{1/2}$ is **forbidden**. The energy interval between $2 \ ^2P_{1/2}$ and $2 \ ^2P_{3/2}$ is very small, about $4.5 \cdot 10^{-5}$ eV, or about 0.0054 Å. Similarly,

$$\text{Ly}\beta \Rightarrow 1 \ ^2S_{1/2} \leftrightarrow \begin{cases} 3 \ ^2P_{1/2} \\ 3 \ ^2P_{3/2} \end{cases}$$

while the transitions $1 \ ^2S_{1/2} \leftrightarrow 3 \ ^2S_{1/2}$, $1 \ ^2S_{1/2} \leftrightarrow 3 \ ^2D_{3/2}$, $1 \ ^2S_{1/2} \leftrightarrow 3 \ ^2D_{5/2}$ are **forbidden**. The energy interval between $3 \ ^2P_{1/2}$ and $3 \ ^2P_{3/2}$ is about $1.3 \cdot 10^{-5}$ eV, or about 0.0011 Å.

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To calculate the energy variation corresponding to the transition between two levels n and m , with $m > n$, we apply the **Rydberg-Ritz formula**:

$$\Delta E = R_H \left(\frac{1}{n^2} - \frac{1}{m^2} \right)$$

where R_H is known as **Rydberg constant** for hydrogen. 1 R corresponds to $3.289841960355 \cdot 10^{15}$ Hz, or 911.267051 Å, or 13.6056923 eV.

Therefore, if $n = 1$ and $m = 2, 3, 4, \dots, \infty$, we have:

n	m	ΔE_{mn} (eV)	λ_{mn} (Å)	Line
1	2	10.2	1215.7	Ly α
	3	12.09	1025.7	Ly β
	4	12.75	972.5	Ly γ
	5	13.05	949.7	Ly δ
	...	...	...	
	∞	13.6	911.3	Ly $_{head}$

In case we consider level $n = 2$ as reference, we obtain the **Balmer series**, whose spectral lines belong to the optical wavelength range. Pay attention that the wavelengths are calculated in air (and not in vacuum as in the previous case), therefore considering the refractive index of air ($n \simeq 1.000277$). Only transitions between $2s \leftrightarrow 3p, 4p, 5p, \dots$, or $2p \leftrightarrow 3s, 4s, 5s, \dots$, or $2p \leftrightarrow 3d, 4d, 5d, \dots$ are permitted. In particular,

$$H\alpha \Rightarrow 2^2S_{1/2} \leftrightarrow \begin{cases} 3^2P_{1/2} \\ 3^2P_{3/2} \end{cases}$$

$$H\alpha \Rightarrow 2^2P_{1/2} \leftrightarrow \begin{cases} 3^2S_{1/2} \\ 3^2D_{3/2} \end{cases}$$

$$H\alpha \Rightarrow 2^2P_{3/2} \leftrightarrow \begin{cases} 3^2S_{1/2} \\ 3^2D_{3/2} \\ 3^2D_{5/2} \end{cases}$$

When we consider level $n = 3$ as reference, we obtain the **Paschen series**, and when we consider level $n = 4$ as reference, we obtain the **Brackett series**: Their spectral lines belong both to the near infrared wavelength range.

1.9.2 Helium

In this case we have two electrons to allocate in different orbitals. We calculate the spectral terms keeping one electron at the fundamental level and moving the other.

$$1s^2 \Rightarrow \begin{matrix} \ell_1 = 0, \ell_2 = 0 \\ s_1 = \frac{1}{2}, s_2 = -\frac{1}{2} \end{matrix} \Rightarrow \begin{matrix} L = 0 \\ S = 0 \end{matrix} \Rightarrow J = 0 \Rightarrow 1^1S_0^e$$

$$1s^1 2s^1 \Rightarrow \begin{matrix} \ell_1 = 0, \ell_2 = 0 \\ s_1 = \frac{1}{2}, s_2 = \frac{1}{2} \end{matrix} \Rightarrow \begin{matrix} L = 0 \\ S = 0, 1 \end{matrix} \Rightarrow \begin{matrix} J = 0 \\ J = 1 \end{matrix} \Rightarrow \begin{matrix} 2^1S_0^e \\ 2^3S_1^e \end{matrix}$$

1.9. EXAMPLES

n	m	ΔE_{mn} (eV)	λ_{mn} (Å)	
2	3	1.89	6562.8	H α
	4	2.55	4861.3	H β
	5	2.86	4340.4	H γ
	6	3.02	4101.7	H δ
	7	3.12	3970.0	H ϵ
	8	3.19	3889.0	H8
	9	3.23	3835.3	H9
	10	3.26	3797.9	H10
	...	...	...	
	∞	3.4	3646.0	H $_{head}$
3	4	0.66	18751	Pa α
	5	0.97	12818	Pa β
	6	1.13	10938	Pa γ
	7	1.23	10049	Pa δ
	...	...	...	
	∞	1.51	8203	Pa $_{head}$
4	5	0.31	40522	Br α
	6	0.47	26258	Br β
	7	0.57	21661	Br γ
	8	0.64	19445	Br δ
	...	...	...	
	∞	0.85	14580	Br $_{head}$

$$1s^1 2p^1 \Rightarrow \begin{matrix} \ell_1 = 0, \ell_2 = 1 \\ s_1 = \frac{1}{2}, s_2 = \frac{1}{2} \end{matrix} \Rightarrow \begin{matrix} L = 1 \\ S = 0, 1 \end{matrix} \Rightarrow \begin{matrix} J = 1 \\ J = 2, 1, 0 \end{matrix} \Rightarrow \begin{matrix} 2^1 P_1^o \\ 2^3 P_{2,1,0}^o \end{matrix}$$

$$1s^1 3s^1 \Rightarrow \begin{matrix} \ell_1 = 0, \ell_2 = 0 \\ s_1 = \frac{1}{2}, s_2 = \frac{1}{2} \end{matrix} \Rightarrow \begin{matrix} L = 0 \\ S = 0, 1 \end{matrix} \Rightarrow \begin{matrix} J = 0 \\ J = 1 \end{matrix} \Rightarrow \begin{matrix} 3^1 S_0^e \\ 3^3 S_1^e \end{matrix}$$

$$1s^1 3p^1 \Rightarrow \begin{matrix} \ell_1 = 0, \ell_2 = 1 \\ s_1 = \frac{1}{2}, s_2 = \frac{1}{2} \end{matrix} \Rightarrow \begin{matrix} L = 1 \\ S = 0, 1 \end{matrix} \Rightarrow \begin{matrix} J = 1 \\ J = 2, 1, 0 \end{matrix} \Rightarrow \begin{matrix} 3^1 P_1^o \\ 3^3 P_{2,1,0}^o \end{matrix}$$

$$1s^1 3d^1 \Rightarrow \begin{matrix} \ell_1 = 0, \ell_2 = 2 \\ s_1 = \frac{1}{2}, s_2 = \frac{1}{2} \end{matrix} \Rightarrow \begin{matrix} L = 2 \\ S = 0, 1 \end{matrix} \Rightarrow \begin{matrix} J = 2 \\ J = 3, 2, 1 \end{matrix} \Rightarrow \begin{matrix} 3^1 D_2^e \\ 3^3 D_{3,2,1}^e \end{matrix}$$

and so on. The spectral terms can be subdivided in those having $2S + 1 = 1$, called **singlets**, and those having $2S + 1 = 3$, called **triplets**. The first level over the fundamental one is $2^3 S_1$ that has energy 19.82 eV and the second level is $2^1 S_0$ with energy 20.62 eV. The following table lists important transitions of He I in the visible range. Another important line is He II $\lambda 4686$, the Pa α transition of helium in hydrogen-like configuration.

CHAPTER 1. FUNDAMENTALS OF ATOMIC SPECTROSCOPY

n	m	E_n (eV)	E_m (eV)	λ_{mn} (Å)	Line
2^1S_0	4^1P_1	20.62	23.74	3964.73	He I λ 3965
2^1P_1	4^1D_2	21.22	23.74	4921.93	He I λ 4922
2^1S_0	3^1P_1	20.62	23.09	5015.68	He I λ 5016
2^1P_1	3^1D_2	21.22	23.07	6678.15	He I λ 6678
2^3S_1	$4^3P_{2,1,0}$	19.82	23.70	3187.75	He I λ 3187
2^3S_1	$3^3P_{2,1,0}$	19.82	23.00	3888.65	He I λ 3889
$2^3P_{2,1,0}$	$5^3D_{3,2,1}$	20.96	24.04	4026.19	He I λ 4026
$2^3P_{2,1,0}$	$4^3D_{3,2,1}$	20.96	23.74	4471.48	He I λ 4471
$2^3P_{2,1,0}$	4^3S_1	20.96	23.59	4713.14	He I λ 4713
$2^3P_{2,1,0}$	$3^3D_{3,2,1}$	20.96	23.07	5875.62	He I λ 5876
$2^3P_{2,1,0}$	3^3S_1	20.96	22.72	7065.19	He I λ 7065
2^3S_1	$2^3P_{2,1,0}$	19.82	20.96	10830.3	He I λ 10830

1.9.3 Oxygen

We have already seen that the electron configuration of O I, $1s^2 2s^2 2p^4$, gives rise to the following spectral terms: $2^3P_{2,1,0}$, 2^1D_2 , and 2^1S_0 , where the first is the fundamental. Since we have four electrons in odd orbitals, all these terms are even. Therefore the transitions listed in the following table are **forbidden**.

n	m	E_n (eV)	E_m (eV)	λ_{mn} (Å)	Line
2^1D_2	2^1S_0	1.97	4.19	5577.34	O I λ 5577
2^3P_1	2^1D_2	0.00	1.97	6300.3	O I λ 6300
2^3P_0	2^1D_2	0.02	1.97	6363.78	O I λ 6364

1.9.4 Sodium

The Na I atom has 1 valence electron of configuration $3s^1$, therefore:

$$L = 0 \quad S = \frac{1}{2} \quad J = \frac{1}{2} \quad \Rightarrow \quad 3^2S_{1/2}^e$$

which is the fundamental level. But when the electron is in a higher angular momentum state, the situation may be different, as, for example $3p^1$:

$$L = 1 \quad S = \frac{1}{2} \quad J = \begin{cases} L + S = 3/2 \\ L - S = 1/2 \end{cases} \quad \Rightarrow \quad 3^2P_{1/2}^o, 3^2P_{3/2}^o$$

The transitions of the electron between the S level and the P levels correspond to the well known Na I doublet:

1.9. EXAMPLES

n	m	E_n (eV)	E_m (eV)	λ_{mn} (Å)	Line
$3^2S_{1/2}$	$3^2P_{3/2}$	0.00	2.104	5889.95	Na I λ 5890
$3^2S_{1/2}$	$3^2P_{1/2}$	0.00	2.102	5895.92	Na I λ 5896

2

Population of energy levels

We consider a plasma in **thermodynamic equilibrium (TE)** when it is located in a closed system and each energy exchange process is balanced by its reverse process. The plasma is characterized by a temperature (T) and a pressure (P), and it emits black-body radiation according to the Planck's law. The population of the excited and ionized states follow the Boltzmann and Saha equations (see next sections) and the velocity distribution is Maxwellian:

$$\phi(v, T)dv = \frac{4}{\sqrt{\pi}} \left(\frac{m}{2k_B T} \right)^{3/2} v^2 \exp \left(-\frac{mv^2}{2k_B T} \right) dv$$

Actually a real astrophysical plasma deviates from TE. However, in many cases it is possible to adopt the **local thermodynamic equilibrium (LTE)** approximation, which assumes that a plasma can be locally described by the previously mentioned relations. This is true when elastic collisions in which particles exchange kinetic energy are much more frequent than collisions producing excitation or ionization, that is when the radiation processes can be neglected with respect to the collision processes. Therefore LTE requires high electron density N_e .

If the electron density is not sufficient to have LTE, but still so high that the excited states can be in equilibrium among each other (while the ground states are not), the heavy particles gain less energy than electrons and the plasma cannot be longer described by a single temperature. In this case the plasma is in **partial local thermodynamic equilibrium (PLTE)**.

In addition it is extremely useful to define the **equivalent thermodynamic equilibrium (ETE)**, that is the case in which there is no transformation of kinetic energy in other energy forms. It corresponds to TE, where the temperature is the kinetic temperature of the gas (therefore following the Maxwellian velocity distribution), the electron density is equal to the electron density of the gas.

2.1 Boltzmann equation

If we consider a sample of atoms and ions, we can introduce the following definitions:

- N_{in} number of i -times ionized atoms in the energy level n per unit volume (cm^{-3})

CHAPTER 2. POPULATION OF ENERGY LEVELS

- $N_i = \sum_n N_{in}$ number of i -times ionized atoms per unit volume (cm^{-3})
- $N = \sum_i N_i$ number of atoms and ions of an element per unit volume (cm^{-3})
- χ_{in} excitation energy of the level n from the ground state of the i -times ionized atom (eV)
- $\chi_i = \chi_{i\infty}$ ionization energy from the ground state of the i -times ionized atom (eV)
- g_{in} statistical weight of the n level in the i -times ionized atom

As already mentioned, in TE conditions the distribution of the ions of a given element in the available energy levels is given by the **Boltzmann equation**:

$$\frac{N_{in}}{N_{i1}} = \frac{g_{in}}{g_{i1}} \exp\left(-\frac{\chi_{in}}{k_B T}\right) \quad (2.1)$$

where $k_B = 1.38065 \cdot 10^{-16} \text{ erg K}^{-1} = 8.617733 \cdot 10^{-5} \text{ eV K}^{-1}$ is the Boltzmann constant. For computational purposes, it is useful to express this equation in logarithmic form:

$$\log \frac{N_{in}}{N_{i1}} = \log \frac{g_{in}}{g_{i1}} - \chi_{in} \frac{5039.544}{T} \quad (2.2)$$

It can be easily seen from Eq. 2.1 that the population of excited levels n decreases exponentially with their excitation energy, though an increase of temperature brings to an higher population of the excited levels (Fig. 2.1), because of the higher average energy of particles, that can therefore fill more easily the excitation energy gap through collisional processes.

Summing Eq. 2.1 over all the excitation states n , we have that:

$$\frac{N_i}{N_{i1}} = \frac{1}{g_{i1}} \sum_n g_{in} \exp\left(-\frac{\chi_{in}}{k_B T}\right)$$

and dividing Eq. 2.1 by this result, we obtain:

$$\frac{N_{in}}{N_i} = \frac{g_{in}}{u_i(T)} \exp\left(-\frac{\chi_{in}}{k_B T}\right) \quad (2.3)$$

where

$$u_i(T) = \sum_n g_{in} \exp\left(-\frac{\chi_{in}}{k_B T}\right)$$

is called **partition function**.

In principle $u_i(T)$ is a diverging series, since we sum from $n = 1$ to $n = \infty$, but actually we must take into account the effects of spectral line broadening (such as natural broadening, collisional broadening or Doppler broadening) which limit the number of discrete energy levels to a finite n_* , above which any further level melts in a continuous band.

Different methods and approximations have been used to calculate $u_i(T)$ (see for example Irwin, A.W. 1981, ApJS, 45, 621; Cardona, O. et al. 2010, ApJ, 711, 239; D'Ammando, G. et al. 2013, Physics of Plasmas, 20). Fig. 2.2 shows

2.1. BOLTZMANN EQUATION

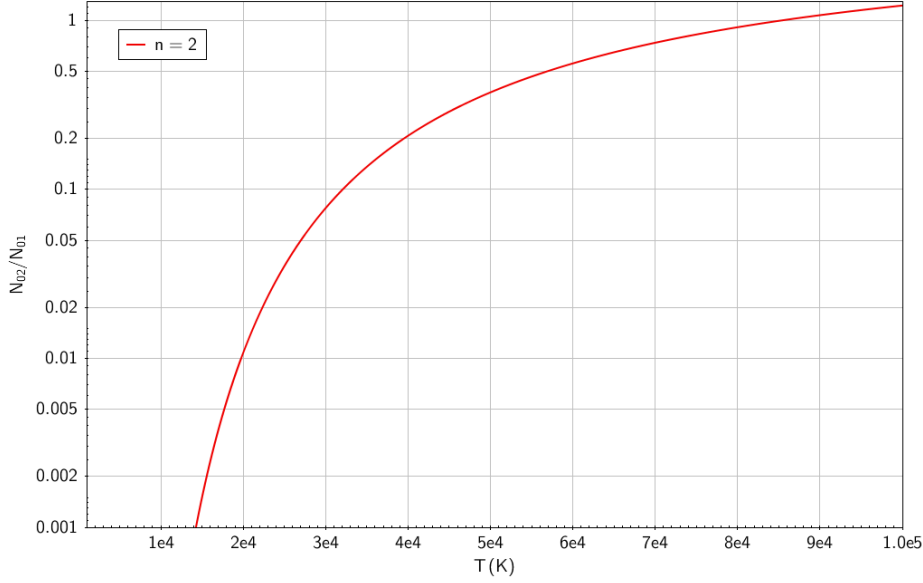


Figure 2.1: Boltzmann equation for hydrogen. The curve shows the fraction of H atoms populated at level $n = 2$ with the respect to the level $n = 1$.

two examples of the partition functions of H I and Na I obtained by applying the approximation given by Irwin (1981) and valid for temperature values between 1000 K and 16000 K:

$$\ln u_i(T) = \sum_{i=0}^5 a_i (\ln T)^i$$

The partition function of H I is about 2 within this temperature range. This is obvious because:

$$u_0(T) = g_{01} + g_{02} \exp\left(-\frac{\chi_{02}}{k_B T}\right) + g_{03} \exp\left(-\frac{\chi_{03}}{k_B T}\right) + g_{04} \exp\left(-\frac{\chi_{04}}{k_B T}\right) + \dots$$

Assuming $T = 10^4$ K, we have:

$$\begin{aligned} u_0(T) &= 2 + 8 \exp\left(-\frac{10.2}{0.8617733}\right) + 18 \exp\left(-\frac{12.09}{0.8617733}\right) + 32 \exp\left(-\frac{12.75}{0.8617733}\right) + \dots \\ &= 2 + 5.79 \cdot 10^{-5} + 1.45 \cdot 10^{-5} + 1.2 \cdot 10^{-5} + \dots \\ &\simeq 2 \end{aligned}$$

Therefore:

$$\frac{N_{02}}{N_0} = \frac{g_{02}}{u_0(T)} \exp\left(-\frac{\chi_{02}}{k_B T}\right) = \frac{8}{2} e^{-\frac{10.2}{0.8617733}} \simeq 2.9 \cdot 10^{-5}$$

For $T = 2 \cdot 10^4$ K, the ratio $\frac{N_{02}}{N_0}$ is about 0.01 and since from Fig. 2.1 we have $N_{02} = N_{01}$ at $T \simeq 86000$ K, at such temperature we would expect to have:

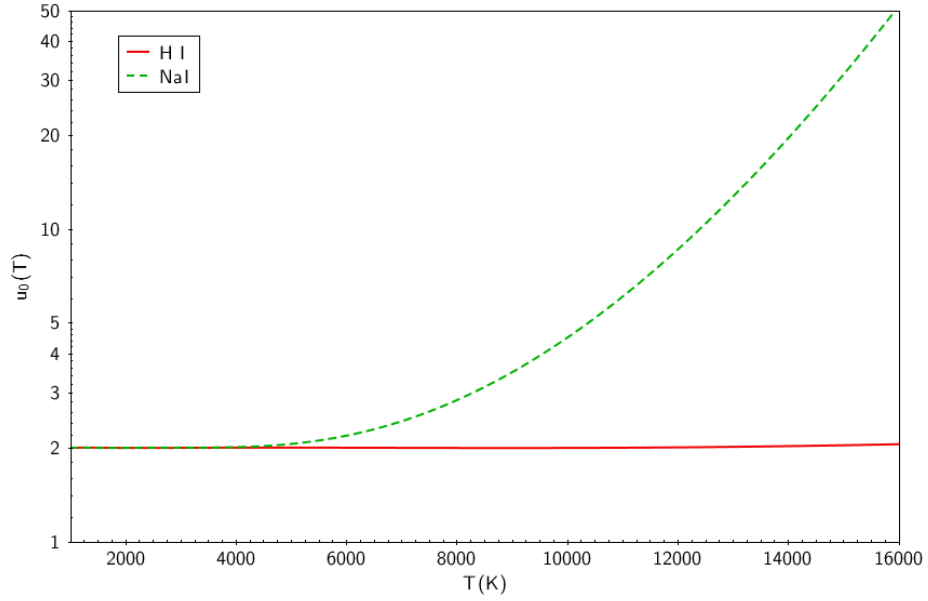


Figure 2.2: Partition function of H I (solid red) e Na I (dashed green) according to Irwin (1981).

$$\frac{N_{02}}{N_0} = \frac{N_{02}}{N_{01} + N_{02}} = 0.5$$

that is 50% of the neutral H atoms would be populated at level $n = 2$. But this is wrong because we are not taking into account the ionization effect.

2.2 Saha equation

The Saha equation is the relationship which describes the equilibrium established for the reactions of ionization and recombination and it can be regarded as the extension of the Boltzmann's formula to the region of space where the levels are continuously distributed with positive energy. Assuming that χ_0 is the ionization potential of a particular species, we can represent the state of a free electron with kinetic energy K as:

$$E = \chi_0 + K = \chi_0 + \frac{p^2}{2m_e}$$

where p is the momentum of the electron.

Because of the Heisenberg's uncertainty principle, we cannot define the properties of an electron with an accuracy better than: $dp_x dx \geq h$ we have to subdivide the six-dimensional phase space in cells with a volume of h^3 , each one being able to contain up to 2 electrons. The number of atoms in the ground level which lost an electron, whose position in the phase space is between $\vec{x}$ and $\vec{x} + d\vec{x}$, and between $\vec{p}$ and $\vec{p} + d\vec{p}$ is given by:

2.2. SAHA EQUATION

$$\frac{dN_{11}}{N_{01}} = 2 \frac{g_{11}}{g_{01}} \exp \left[-\frac{1}{k_B T} \left(\chi_0 + \frac{p_x^2 + p_y^2 + p_z^2}{2m_e} \right) \right] \frac{d^3 x d^3 p}{h^3}$$

where the factor 2 is due to the number of electrons that can share the same phase space cell. If we integrate this formula over the whole phase space, we have:

$$\frac{N_{11}}{N_{01}} = 2 \frac{g_{11}}{g_{01}} \frac{1}{h^3} \exp \left(-\frac{\chi_0}{k_B T} \right) \int_V d^3 x \int_{-\infty}^{\infty} \exp \left(-\frac{p^2}{2m_e k_B T} \right) d^3 p$$

If we consider the approximation in which we have an ion for each electron, we can use the expression:

$$V = N_e^{-1} = \frac{k_B T}{P_e}$$

where N_e is the electron density and P_e is the electron pressure.

And finally we obtain the **Saha equation**:

$$\frac{N_{11}}{N_{01}} P_e = 2 \frac{g_{11}}{g_{01}} \frac{(2\pi m_e)^{3/2} (k_B T)^{5/2}}{h^3} \exp \left(-\frac{\chi_0}{k_B T} \right) \quad (2.4)$$

or

$$\frac{N_{11}}{N_{01}} N_e = 2 \frac{g_{11}}{g_{01}} \frac{(2\pi m_e k_B T)^{3/2}}{h^3} \exp \left(-\frac{\chi_0}{k_B T} \right) \quad (2.5)$$

This relation gives the ratio between the occupation number of the ground level of the singly ionized atom, with respect to the occupation number of the ground level of the neutral atom. Summing over all the possible excitation levels and considering the more general case of subsequent ionization states, we get:

$$\frac{N_{i+1}}{N_i} P_e = 2 \frac{u_{i+1}(T)}{u_i(T)} \frac{(2\pi m_e)^{3/2} (k_B T)^{5/2}}{h^3} \exp \left(-\frac{\chi_i}{k_B T} \right) \quad (2.6)$$

which has a logarithmic expression:

$$\log \left(\frac{N_{i+1}}{N_i} P_e \right) = \log \left[2 \frac{u_{i+1}(T)}{u_i(T)} \right] - \chi_i \frac{5039.544}{T} + \frac{5}{2} \log T - 0.477 \quad (2.7)$$

where χ_i is measured in eV and P_e in dyn cm⁻².

Notice from Eq. 2.6 and 2.7 that the effect of increasing the electron pressure is to reduce the degree of ionization (recombinations are favored and there are fewer cells for free electrons), while increasing temperature raises the ionization.

We come back now to the previous H I example and we assume an electron pressure $P_e = 20 \text{ N m}^{-2}$ ($= 20 \text{ Pa}$, or $\sim 2 \cdot 10^{-4} \text{ atm}$). Typical values in stellar atmospheres are $P_e \sim 0.1 \text{ N m}^{-2}$ for cool stars and $P_e \sim 100 \text{ N m}^{-2}$ for hot stars. In case of $T = 10^4 \text{ K}$, this pressure corresponds to an electron density $N_e \sim 10^{14} \text{ cm}^{-3}$.

$$\frac{N_1}{N_0} = 2 \frac{u_1}{u_0} \frac{(2\pi m_e)^{3/2} (k_B T)^{5/2}}{h^3} \frac{e^{-\chi_0/K_B T}}{P_e}$$

CHAPTER 2. POPULATION OF ENERGY LEVELS

$u_1 = 1$ because we have only one state, $u_0 \simeq 2$ as seen before, and $\chi_0 = 13.6$ eV, therefore:

$$\frac{N_1}{N_0} \simeq 1.664 \cdot 10^{-3} T^{5/2} e^{-\frac{157814}{T}}$$

and

$$\frac{N_1}{N_{tot}} = \frac{N_1}{N_0 + N_1} = \frac{N_1/N_0}{1 + N_1/N_0}$$

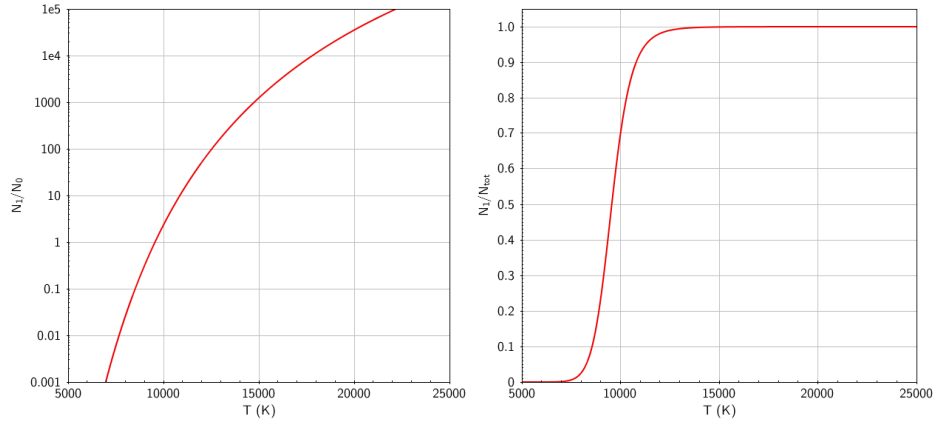


Figure 2.3: On the left the ratio between the number density of ionized H atoms and neutral H atoms. On the right the ratio between the number density of ionized H atoms and the total H atoms (ionized plus neutral).

Fig. 2.3 shows that the fraction of ionized H atoms increases rapidly with temperature. In fact already at $T \sim 9600$ K there is an equal number of ionized and neutral atoms, or in other words, 50% of H is ionized at that temperature. This fraction becomes 70% at $T = 10000$ K and at $T \sim 14000$ K, H is fully ionized.

2.3 Population of excited levels

2.3.1 Hydrogen

Now we focus on hydrogen Balmer lines ($n = 2$) and we combine Boltzmann and Saha equations to calculate the N_{02}/N_{tot} ratio.

$$\frac{N_{02}}{N_{tot}} = \frac{N_{02}}{N_0 + N_1} = \frac{N_{02}/N_0}{1 + \frac{N_1}{N_0}}$$

According to Eq. 2.3:

$$\frac{N_{02}}{N_0} = \frac{g_{02}}{u_0(T)} e^{-\chi_{02}/k_B T} \simeq \frac{8}{2} e^{-10.2/k_B T} = 4 e^{-\frac{118360.6}{T}}$$

while, according to Eq. 2.6:

2.3. POPULATION OF EXCITED LEVELS

$$\frac{N_1}{N_0} P_e = 2 \frac{u_1(T)}{u_0(T)} \frac{(2\pi m_e)^{3/2} k_B^{5/2}}{h^3} T^{5/2} e^{-\chi_0/k_B T}$$

where

$$\frac{(2\pi m_e)^{3/2} k_B^{5/2}}{h^3} \simeq 0.0333 \text{ N m}^{-2} \text{ K}^{-5/2}$$

Therefore, assuming $P_e = 20$ Pa, we have:

$$\frac{N_1}{N_0} = 2 \cdot \frac{1}{2} \cdot \frac{0.0333}{20} \cdot T^{5/2} e^{-13.6/k_B T} \simeq 1.665 \cdot 10^{-3} T^{5/2} e^{-\frac{157814.126}{T}}$$

The N_{02}/N_{tot} ratio as a function of temperature is plotted in the following Fig. 2.4. We can observe that level $n = 2$ starts to be significantly populated at temperatures higher than about 7000 K, than the number density of H atoms able to absorb photons of the Balmer series increases rapidly and reaches a maximum at $T \simeq 9900$ K, then it decreases rapidly again up to $T \simeq 15000$ K and slowly at higher temperatures. The effect of this result is that even if hydrogen Balmer lines are observed in almost all stellar spectral lines, their strength has a maximum in A0 stars (Fig. 2.5).

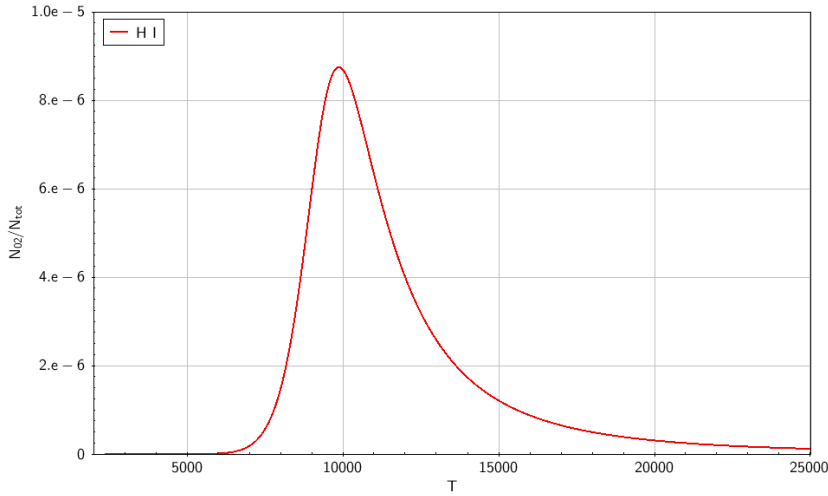


Figure 2.4: The number density of hydrogen atoms populated at level $n = 2$ over the total as a function of temperature (in K).

2.3.2 Helium

We now consider some of the He I lines listed in the previous chapter: $\lambda 4026$, $\lambda 4471$, $\lambda 4713$, and $\lambda 5876$. All of them have $2^3P_{2,1,0}$ as lower level, whose energy is about 20.964 eV.

$$\frac{N_{02}}{N_{tot}} = \frac{N_{02}}{N_0 + N_1 + N_2} = \frac{N_{02}/N_0}{1 + \frac{N_1}{N_0} \left(1 + \frac{N_2}{N_1}\right)}$$

CHAPTER 2. POPULATION OF ENERGY LEVELS

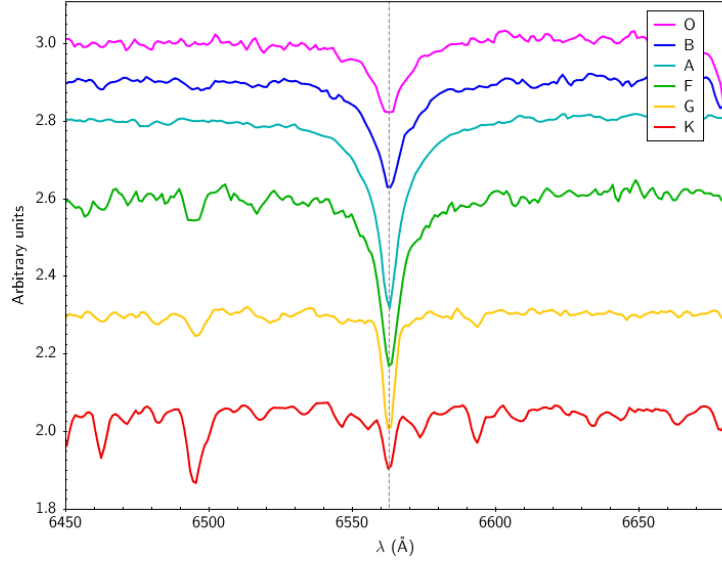


Figure 2.5: H α line in different spectral types. The Y axis is in arbitrary units to make the comparison easier.

$$\frac{N_{02}}{N_0} = \frac{g_{02}}{u_0(T)} e^{-\chi_{02}/k_B T} = \frac{9}{1} \cdot e^{-20.964/k_B T} = 9 e^{-\frac{243266.848}{T}}$$

where $g_{02} = (2L + 1)(2S + 1) = 9$, and $u_0(T) = 1$ according to Irwin (1981).

We assume again $P_e = 20$ Pa and we apply the Saha equation:

$$\frac{N_1}{N_0} \simeq 2 \cdot \frac{2}{1} \cdot \frac{0.0333}{20} \cdot T^{5/2} e^{-24.587/k_B T} \simeq 6.66 \cdot 10^{-3} T^{5/2} e^{-\frac{285307.052}{T}}$$

$$\frac{N_2}{N_1} \simeq 2 \cdot \frac{1}{2} \cdot \frac{0.0333}{20} \cdot T^{5/2} e^{-54.416/k_B T} \simeq 1.665 \cdot 10^{-3} T^{5/2} e^{-\frac{631442.167}{T}}$$

where $u_1(T) = 2$ and $u_2(T) = 1$ (Irwin 1981), $\chi_0 = 24.587$ eV and $\chi_1 = 54.416$ eV. We observe that the last population ratio is ≈ 0 up to temperatures equal to about 25000 K.

Let us consider now the He II $\lambda 4686$ line. This is a transition between the lower level 3 having energy equal to about 48.372 eV and the upper level 4 with energy of about 51.017 eV.

$$\frac{N_{13}}{N_{tot}} = \frac{N_{13}}{N_0 + N_1 + N_2} = \frac{N_{13}/N_1}{\frac{N_0}{N_1} + 1 + \frac{N_2}{N_1}}$$

$$\frac{N_{13}}{N_1} = \frac{g_{13}}{u_1(T)} e^{-\chi_{13}/k_B T} = \frac{18}{2} \cdot e^{-48.372/k_B T} = 9 e^{-\frac{561307.713}{T}}$$

2.3. POPULATION OF EXCITED LEVELS

We can now plot the populations of level 2 for He I and level 3 for He II. Fig. 2.6 shows that He I lines have their maximum intensity at temperature $T \simeq 15800$ K, that corresponds to B5 stars, while He II lines have their maximum at $T \simeq 34000$ K, that is for O stars. This is confirmed in Fig. 2.7.

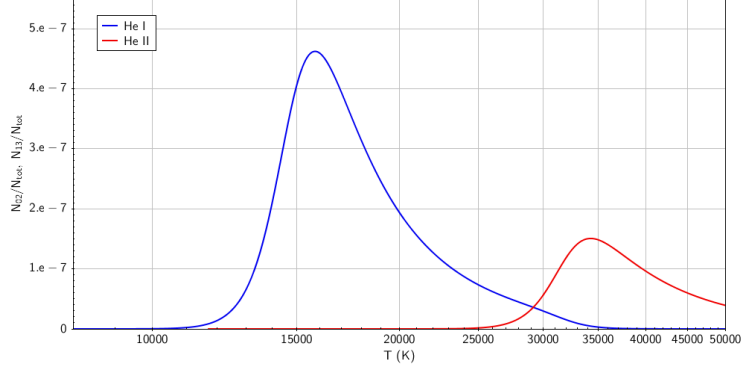


Figure 2.6: The population of level 2 in He I and level 3 in He II over the total as a function of temperature.

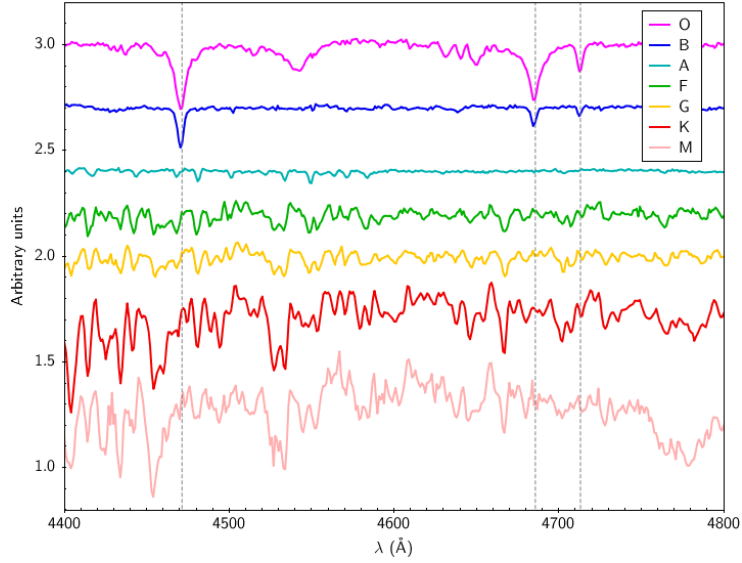


Figure 2.7: Helium lines in different spectral types. The dashed vertical lines indicate the positions of He I $\lambda 4471$, He II $\lambda 4686$, and He I $\lambda 4713$. The Y axis is in arbitrary units to make the comparison easier.

CHAPTER 2. POPULATION OF ENERGY LEVELS

2.3.3 Sodium

We consider now the already seen Na doublet $\lambda\lambda 5890, 5896$. The lower level, that is also the fundamental one, is $3^2S_{1/2}$.

$$\frac{N_{01}}{N_{tot}} \simeq \frac{N_{01}}{N_0 + N_1 + N_2} = \frac{N_{01}/N_0}{1 + \frac{N_1}{N_0} \left(1 + \frac{N_2}{N_1}\right)}$$

$$\frac{N_{01}}{N_0} = \frac{g_{01}}{u_0(T)} e^{-\chi_{01}/k_B T} = \frac{2}{u_0(T)}$$

We assume now an electron pressure $P_e = 1$ Pa:

$$\frac{N_1}{N_0} \simeq 2 \cdot \frac{u_1(T)}{u_0(T)} \cdot \frac{0.0333}{1} \cdot T^{5/2} e^{-5.139/k_B T} = 0.0666 \cdot \frac{u_1(T)}{u_0(T)} \cdot T^{5/2} e^{-\frac{59632.852}{T}}$$

$$\frac{N_2}{N_1} \simeq 2 \cdot \frac{u_2(T)}{u_1(T)} \cdot \frac{0.0333}{1} \cdot T^{5/2} e^{-47.286/k_B T} = 0.0666 \cdot \frac{u_2(T)}{u_1(T)} \cdot T^{5/2} e^{-\frac{548705.787}{T}}$$

Fig. 2.8 shows u_0 , u_1 and u_2 partition functions of Na as given by Irwin (1981). It can be seen that $u_0 = 2$ up to $T \sim 4000$ K, then it increases rapidly, while $u_1 = 1$ and u_2 ranges between 5 and 6.

Fig. 2.9 shows that the fundamental level is highly populated at low temperatures, therefore we expect to detect a strong Na doublet in G, K, and M spectral types, while these lines are expected to become gradually fainter up to be almost undetected in hot stars (Fig. 2.10).

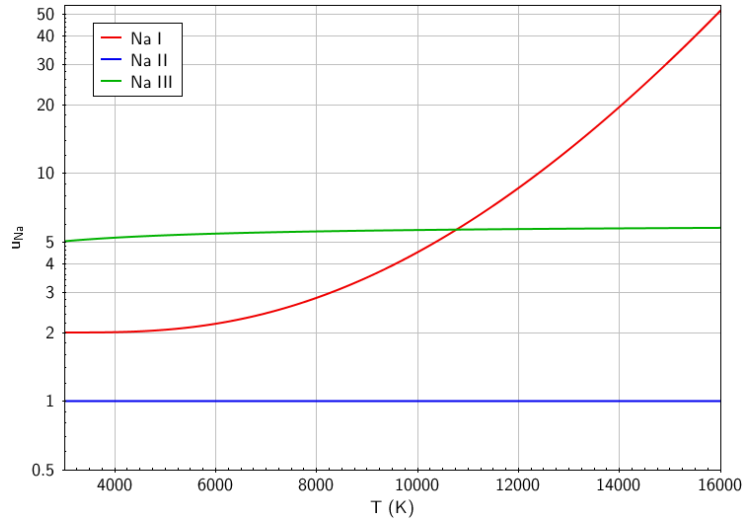


Figure 2.8: The sodium partition functions calculated by Irwin (1981).

2.3. POPULATION OF EXCITED LEVELS

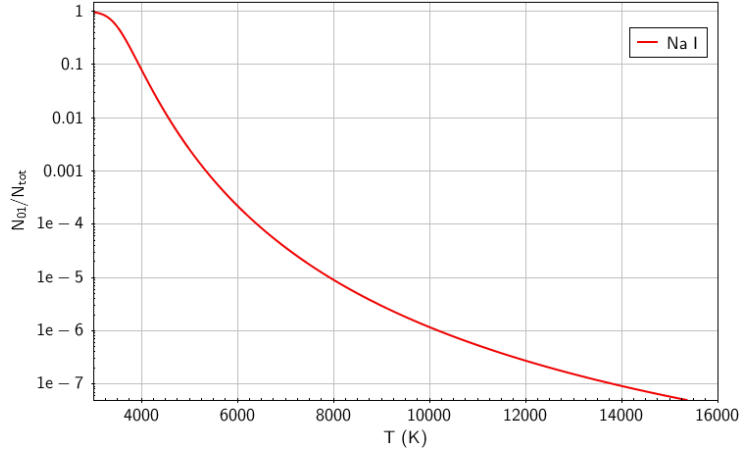


Figure 2.9: The population of the fundamental level in Na I.

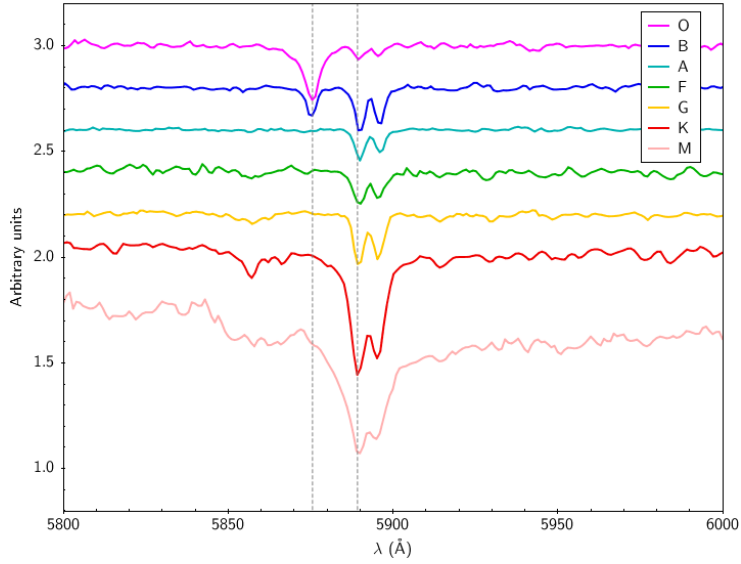


Figure 2.10: Sodium doublet $\lambda\lambda 5890, 5896$ in different spectral types. The dashed vertical lines indicate the positions of He I $\lambda 5876$ and Na I $\lambda 5890$. The Y axis is in arbitrary units to make the comparison easier.

2.3.4 Magnesium

Magnesium has atomic number $Z = 12$ and electronic configuration: $1s^2 2s^2 2p^6 3s^2$. The fundamental level of Mg I is 3^1S_0 and the first excited level is $3^3P_{0,1,2}$ having energies 2.709, 2.711 and 2.716 eV respectively. This sub-levels are sufficiently separated that the electron transitions between them and the excited level 4^3S_1 (5.108 eV) form a famous triplet of the visible spectral range: Mg I $\lambda 5175$ (5167.32, 5172.68, 5183.60 Å).

$$\frac{N_{02}}{N_{tot}} = \frac{N_{02}}{N_0 + N_1 + N_2} = \frac{N_{02}/N_0}{1 + \frac{N_1}{N_0} \left(1 + \frac{N_2}{N_1}\right)}$$

$$\frac{N_{02}}{N_0} = \frac{g_{02}}{u_0(T)} e^{-\chi_{02}/k_B T} = \frac{9}{u_0(T)} e^{-2.71/k_B T} = \frac{9}{u_0(T)} e^{-\frac{31446.785}{T}}$$

We assume again an electron pressure $P_e = 1$ Pa:

$$\frac{N_1}{N_0} \simeq 2 \cdot \frac{u_1(T)}{u_0(T)} \cdot \frac{0.0333}{1} \cdot T^{5/2} e^{-7.646/k_B T} = 0.0666 \cdot \frac{u_1(T)}{u_0(T)} \cdot T^{5/2} e^{-\frac{88724.03}{T}}$$

$$\frac{N_2}{N_1} \simeq 2 \cdot \frac{u_2(T)}{u_1(T)} \cdot \frac{0.0333}{1} \cdot T^{5/2} e^{-15.035/k_B T} = 0.0666 \cdot \frac{u_2(T)}{u_1(T)} \cdot T^{5/2} e^{-\frac{174465.837}{T}}$$

The partition functions of Mg according to Irwin (1981) are plotted in Fig. 2.11. u_0 is equal to 1 at low temperatures, then it increases, while $u_1 \simeq 2$ and $u_2 = 1$. The resulting population of the first excited level over the total population of Mg atoms is plotted in Fig. 2.12. The function shows a peak at $T \simeq 4500$ K and the level is highly populated for low temperatures, therefore the Mg I triplet is expected to be strong in K stars, but also in G and M stars. Then the triplet becomes fainter at higher temperatures (Fig. 2.13).

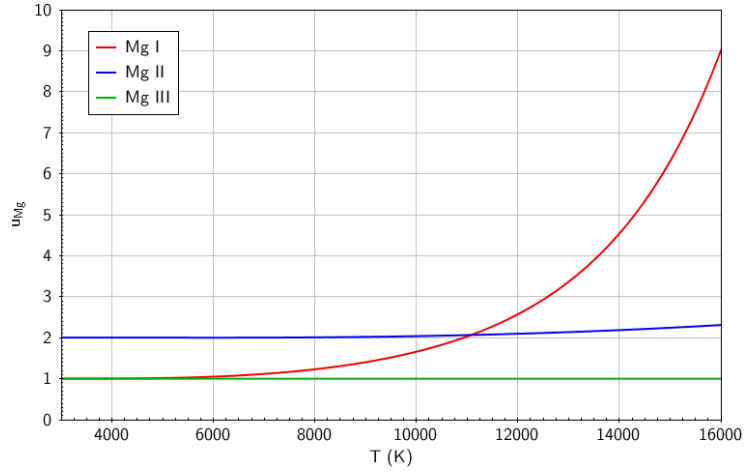


Figure 2.11: The magnesium partition functions calculated by Irwin (1981).

2.3.5 Calcium

Calcium has atomic number $Z=20$. Here we focus on Ca II that has electron configuration: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$. The fundamental level is $4^2S_{1/2}$. The first excited level is $3^2D_{3/2,5/2}$ with energies 1.69 and 1.7 eV respectively, and the

2.3. POPULATION OF EXCITED LEVELS

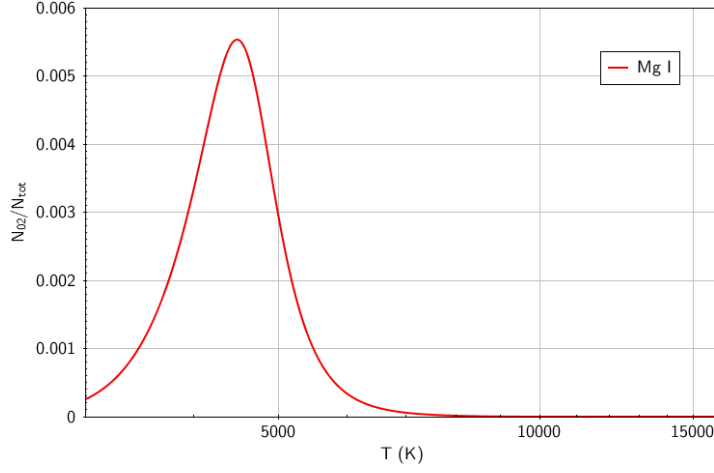


Figure 2.12: The population of the excited level $3^3P_{0,1,2}$ in Mg I.

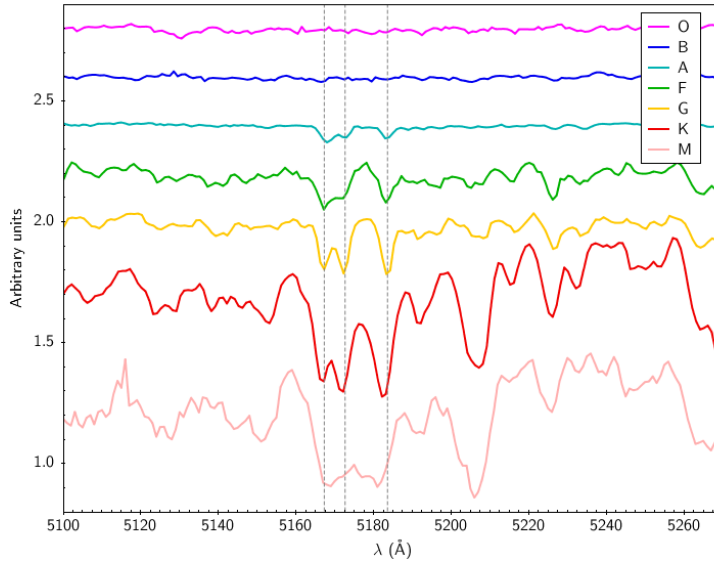


Figure 2.13: Magnesium triplet $\lambda 5175$ in different spectral types. The dashed vertical lines indicate the positions of the three lines: 5167.32, 5172.68, 5183.60 Å. The Y axis is in arbitrary units to make the comparison easier.

second is $4^2P_{1/2,3/2}$ with energies 3.12 and 3.15 eV. The transitions between levels $n = 1$ and $n = 3$ correspond to a famous doublet in the blue part of the visible range (K and H lines according to Fraunhofer), while the transitions between levels $n = 2$ and $n = 3$ correspond to a triplet in the red part of the visible range (Ca T).

The following table describes the spectral lines in details:

CHAPTER 2. POPULATION OF ENERGY LEVELS

n	m	E_n (eV)	E_m (eV)	λ_{mn} (Å)	Line
$4^2S_{1/2}$	$4^2P_{3/2}$	0.00	3.15	3933.66	Ca II λ 3934
$4^2S_{1/2}$	$4^2P_{1/2}$	0.00	3.12	3968.47	Ca II λ 3969
$3^2D_{3/2}$	$4^2P_{3/2}$	1.69	3.15	8498.02	Ca II λ 8498
$3^2D_{5/2}$	$4^2P_{3/2}$	1.7	3.15	8542.09	Ca II λ 8542
$3^2D_{3/2}$	$4^2P_{1/2}$	1.69	3.12	8662.14	Ca II λ 8662

For the H and K doublet we need to calculate the population of the fundamental level with respect to the total number density of calcium atoms.

$$\frac{N_{11}}{N_{tot}} = \frac{N_{11}}{N_0 + N_1 + N_2} = \frac{N_{11}/N_1}{1 + \frac{N_0}{N_1} + \frac{N_2}{N_1}}$$

$$\frac{N_{11}}{N_1} = \frac{g_{11}}{u_1(T)} e^{-\chi_{11}/k_B T} = \frac{2}{u_1(T)}$$

As before, we assume electron pressure $P_e = 1$ Pa.

$$\frac{N_1}{N_0} \simeq 2 \cdot \frac{u_1(T)}{u_0(T)} \cdot \frac{0.0333}{1} \cdot T^{5/2} e^{-6.113/k_B T} = 0.0666 \cdot \frac{u_1(T)}{u_0(T)} \cdot T^{5/2} e^{-\frac{70935.128}{T}}$$

$$\frac{N_2}{N_1} \simeq 2 \cdot \frac{u_2(T)}{u_1(T)} \cdot \frac{0.0333}{1} \cdot T^{5/2} e^{-11.871/k_B T} = 0.0666 \cdot \frac{u_2(T)}{u_1(T)} \cdot T^{5/2} e^{-\frac{137750.844}{T}}$$

For the Ca T triplet we need to calculate the population of the first excited level.

$$\frac{N_{12}}{N_{tot}} = \frac{N_{12}}{N_0 + N_1 + N_2} = \frac{N_{12}/N_1}{1 + \frac{N_0}{N_1} + \frac{N_2}{N_1}}$$

$$\frac{N_{12}}{N_1} = \frac{g_{12}}{u_1(T)} e^{-\chi_{12}/k_B T} = \frac{10}{u_1(T)} e^{-1.7/k_B T} = \frac{10}{u_1(T)} e^{-\frac{19726.765}{T}}$$

The partition functions u_0 , u_1 , and u_2 are plotted in Fig. 2.14 (Irwin, 1981). The plot of the populations for level 1 and 2 (Fig. 2.15) shows that Ca II lines are well detected up to A stars and become fainter in B and O stars. Indeed looking at Fig. 2.16 we observe strong Ca KH doublet in F, G and K stars. Then Ca K becomes fainter, while Ca H seems to become stronger. This effect is caused by the overlay between Ca H and H ϵ : in cool stars hydrogen lines are fainter and calcium lines are stronger, the contrary happens when the temperature increases. Fig. 2.17 shows a comparison of the red part of the visible spectrum in a G star and in a A star. The presence of the hydrogen Paschen lines in A stars makes the calcium triplet not visible.

2.3. POPULATION OF EXCITED LEVELS

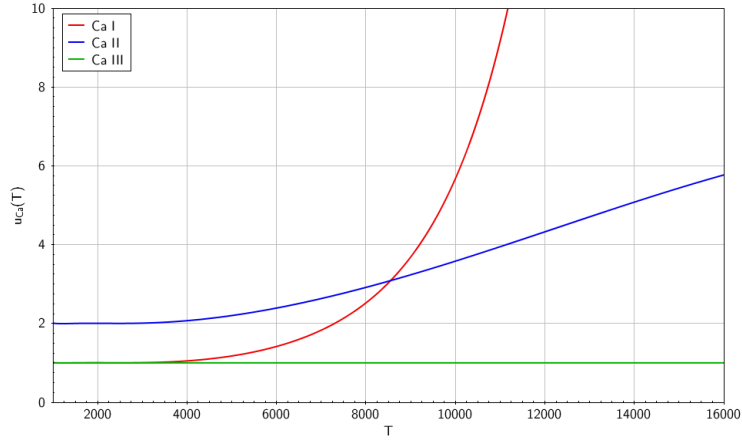


Figure 2.14: The calcium partition functions calculated by Irwin (1981).

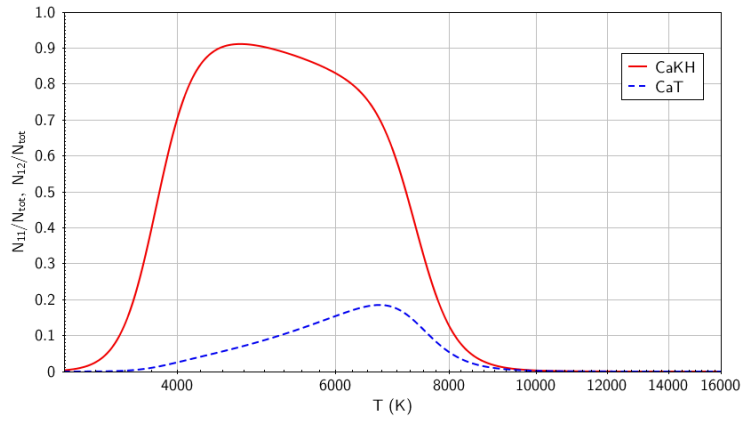


Figure 2.15: The population of the fundamental and the first excited levels of Ca II.

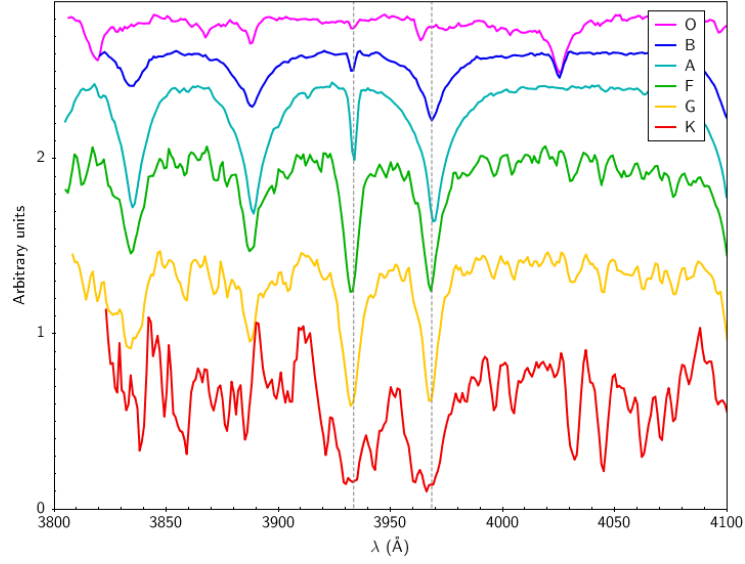


Figure 2.16: Ca II K and H lines in different spectral types. The dashed vertical lines indicate their positions. The Y axis is in arbitrary units to make the comparison easier.

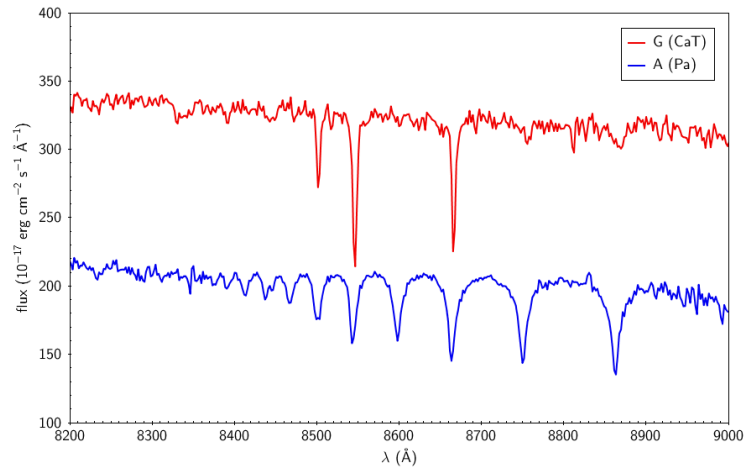


Figure 2.17: A comparison between a G star (red) and an A star (blue) in the region of CaT.

3

Radiation transport

Let us now consider the case of a radiation beam crossing a gas cloud and describe what happens during the interaction between the gas cloud and the radiation field. First we need to introduce some parameters to describe the radiation field (Fig. 3.1). To do this we take into account a surface and an infinitesimal area element dA on it. The energy flowing through the area $dA \cos \theta$ per time unit and frequency unit in a beam propagating in the solid angle $d\Omega$ is given by the **specific intensity**:

$$I_\nu = \frac{dE_\nu}{dA \cos \theta d\nu d\Omega dt} \quad [\text{erg cm}^{-2} \text{ s}^{-1} \text{ Hz}^{-1} \text{ sr}^{-1}]$$

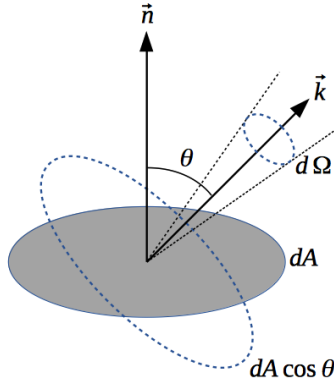


Figure 3.1: A flux of radiation goes through a small portion of a surface. $\vec{n}$ is the normal to the surface, $\vec{k}$ is the direction of the flowing radiation.

As the beam passes through the cloud, energy may either be absorbed or emitted due to the interactions of radiation with the atoms (Fig. 3.2). We define the **emission coefficient**:

$$\varepsilon_\nu = \frac{dE_\nu}{ds dA \cos \theta d\nu d\Omega dt} \quad [\text{erg cm}^{-3} \text{ s}^{-1} \text{ Hz}^{-1} \text{ sr}^{-1}]$$

as the amount of energy added to the beam while crossing the element volume $ds dA \cos \theta$ (per time and frequency unit), and the **absorption coefficient** through the expression:

CHAPTER 3. RADIATION TRANSPORT

$$k_\nu I_\nu = -\frac{dE_\nu}{ds dA \cos \theta d\nu d\Omega dt} \quad [\text{erg cm}^{-3} \text{ s}^{-1} \text{ Hz}^{-1} \text{ sr}^{-1}]$$

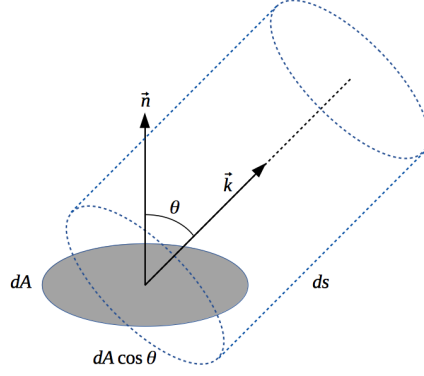


Figure 3.2: A beam of energy flows within a cylindrical volume, where emission and absorption processes take place.

Notice that, while the emission coefficient is an amount of energy, the absorption coefficient k_ν represents the fraction of energy that will be absorbed from a beam of specific intensity I_ν and it is measured in units of cm^{-1} .

If we take into account a radiation beam which propagates into a medium for a path of length ds , in the most general situation its specific intensity will be affected by emission and absorption of energy according to:

$$dI_\nu = -k_\nu I_\nu ds + \varepsilon_\nu ds$$

from which we obtain the **Radiative Trasport equation** in its differential form:

$$\frac{dI_\nu}{ds} = -k_\nu I_\nu + \varepsilon_\nu \quad (3.1)$$

Let us assume that the cloud is located along a line-of-sight to a radiation source. We can fix two symmetrical reference axes, one having the origin in the side of cloud facing the source and increasing toward the observer (coordinate $\mathbf{s}$), the other having the origin in the side of cloud facing the observer and increasing toward the source (coordinate $\mathbf{r}$). In the first case the size of cloud is s^* , in the second is r^* (Fig. 3.3).

We can now introduce the concept of **optical depth** τ_ν of the medium through the definition:

$$d\tau_\nu = k_\nu dr = -k_\nu ds$$

Notice that the optical depth is a dimensionless quantity that, for our geometry, is null at the observer's side of the cloud, while it is maximum on the source's side. The total optical depth of the medium is, thus:

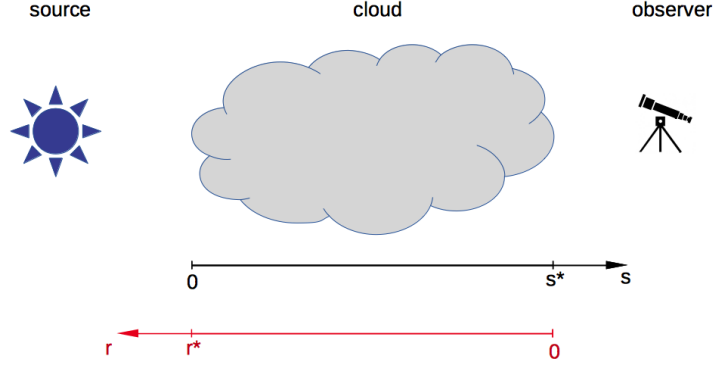


Figure 3.3: Two reference systems applied to describe the emission/absorption processes within the gaseous cloud.

$$\tau_\nu^* = \int_0^{r^*} k_\nu(r) dr = - \int_{s^*}^0 k_\nu(s) ds$$

Dividing Eq. 3.1 by $-k_\nu$, we have:

$$\frac{dI_\nu}{d\tau_\nu} = I_\nu - \frac{\varepsilon_\nu}{k_\nu} \quad (3.2)$$

The quantity $S_\nu = \frac{\varepsilon_\nu}{k_\nu}$ is a characteristic of the cloud, fairly independent from the properties of the radiation source, and it is called **source function**.

Solving Eq. 3.2, we have:

$$\begin{aligned} \frac{dI_\nu}{d\tau_\nu} - I_\nu &= -\frac{\varepsilon_\nu}{k_\nu} \\ e^{-\tau_\nu} \left[\frac{dI_\nu}{d\tau_\nu} - I_\nu \right] &= -\frac{\varepsilon_\nu}{k_\nu} e^{-\tau_\nu} \\ \frac{d}{d\tau_\nu} (I_\nu e^{-\tau_\nu}) &= -S_\nu e^{-\tau_\nu} \\ I_\nu(\tau_\nu = 0) &= \int_0^{\tau_\nu^*} S_\nu(\tau_\nu) e^{-\tau_\nu} d\tau_\nu + I_{\nu_0} e^{-\tau_\nu^*} \end{aligned} \quad (3.3)$$

where $I_{\nu_0} = I_\nu(\tau_\nu^*)$ is the specific intensity of the initial radiation beam, as it strikes the cloud on the source's side. It can be seen that the emergent intensity is the sum of the original specific intensity, emitted by the source and attenuated by the whole path across the cloud, and of any contribution arising within the cloud itself and attenuated by a path corresponding to its own depth. Recalling the expression of source function, indeed, the integral on the right hand side turns out to be:

$$\int_0^{\tau_\nu^*} S_\nu e^{-\tau_\nu} d\tau_\nu = \int_0^{r^*} \varepsilon_\nu e^{-\tau_\nu} dr$$

CHAPTER 3. RADIATION TRANSPORT

that is the overall emitted radiation is a sum of contributions attenuated by the proper $e^{-\tau_\nu}$ factor.

It is clear that to solve the integral we need to know how S_ν depends on τ_ν , that is how ε_ν and k_ν depend on τ_ν . If we make the hypothesis that S_ν is constant inside the cloud, we have:

$$I_\nu(0) = S_\nu(1 - e^{-\tau_\nu^*}) + I_{\nu_0}e^{-\tau_\nu^*}$$

This solution has two fundamental limits:

- $\tau_\nu \gg 1$ (**optically thick case**) $\Rightarrow I_\nu(0) = S_\nu$, that is we can only see the cloud through its source function;
- $\tau_\nu \ll 1$ (**optically thin case**) $\Rightarrow e^{-\tau_\nu} \approx 1 - \tau_\nu \Rightarrow I_\nu(0) = \tau_\nu^*(S_\nu - I_{\nu_0}) + I_{\nu_0}$.

In the optically thin case, if we do not look along the direction to the radiation source, it is $I_{\nu_0} = 0$ and, therefore:

$$I_\nu(0) = \tau_\nu^* S_\nu = \varepsilon_\nu r^*$$

that is all the radiation produced in any element ds emerges from the cloud without being absorbed.

In TE conditions the energy absorbed in a volume element dV must be equal to the energy emitted in the same region:

$$k_\nu I_\nu = \varepsilon_\nu$$

and the source function is given by the Planck's distribution:

$$\frac{\varepsilon_\nu}{k_\nu} = I_\nu = B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{\frac{h\nu}{k_B T}} - 1}$$

3.1 The case of the stellar atmosphere

Let us consider the radiation field of a stationary stellar atmosphere viewed by an outside observer. In this case, the specific intensity I_ν depends on two angles, ϑ and φ (Fig. 3.4), and $d\Omega = \sin \vartheta d\vartheta d\varphi$.

The energy passing through the surface dA per time unit and frequency unit, F_ν , is called **flux of radiation** and it is given by specific intensity integrated over the whole solid angle:

$$\int_{\Omega} I_\nu \cos \vartheta d\Omega = \int_0^{2\pi} \int_0^\pi I_\nu \cos \vartheta \sin \vartheta d\vartheta d\varphi = \pi F_\nu$$

When the radiation field is isotropic $F_\nu = 0$, but in a stellar atmosphere we observe a flow of radiation toward us, therefore in this case $F_\nu \neq 0$.

We now assume that the stellar atmosphere has a plane parallel geometry and we consider a radiation beam passing through a cylindrical volume $dA ds$, as in Fig. 3.2.

3.1. THE CASE OF THE STELLAR ATMOSPHERE

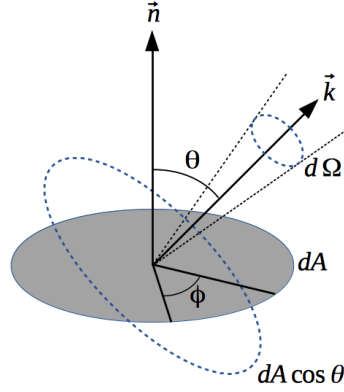


Figure 3.4: The same as in Fig. 3.1, with the azimuthal angle φ in addition.

$$d\tau_\nu = k_\nu dr \quad ds = -\frac{dr}{\cos \vartheta}$$

thus the radiative transport equation becomes:

$$\frac{dI_\nu(\tau_\nu, \vartheta)}{ds} = -k_\nu I_\nu(\tau_\nu, \vartheta) + \varepsilon_\nu$$

$$\cos \vartheta \frac{dI_\nu(\tau_\nu, \vartheta)}{d\tau_\nu} = I_\nu(\tau_\nu, \vartheta) - S_\nu$$

whose solution is the **limb darkening equation**:

$$I_\nu(0, \vartheta) = \int_0^\infty S_\nu(\tau_\nu) e^{-\tau_\nu \sec \vartheta} d(\tau_\nu \sec \vartheta) \quad (3.4)$$

where $I_\nu(0, 0)$ is the specific intensity at the center of the stellar disk.

If we now assume the simple linear case:

$$S_\nu(\tau_\nu) = a_\nu + b_\nu \tau_\nu$$

where a_ν and b_ν are unknown, we obtain:

$$I_\nu(0, \vartheta) = a_\nu \int_0^\infty e^{-\tau_\nu \sec \vartheta} d(\tau_\nu \sec \vartheta) + b_\nu \int_0^\infty \tau_\nu e^{-\tau_\nu \sec \vartheta} d(\tau_\nu \sec \vartheta)$$

$$I_\nu(0, \vartheta) = a_\nu + b_\nu \cos \vartheta$$

Therefore $I_\nu(0, 0) = a_\nu + b_\nu$ at the center of the disk, while it becomes $I_\nu(0, \pi/2) = a_\nu$ at the limb. Obviously, after measuring the intensity at different angular distances from the center it is possible to recover a_ν and b_ν , and so the dependence of the source function from the optical depth.

The observed flux of radiation at the stellar surface ($\tau_\nu = 0$) is:

CHAPTER 3. RADIATION TRANSPORT

$$\pi F_\nu(0) = \int_0^{2\pi} \int_0^{\pi/2} I_\nu(0, \vartheta) \cos \vartheta \sin \vartheta \, d\vartheta \, d\varphi$$

$$\pi F_\nu(0) = 2\pi \int_0^1 I_\nu(0, \vartheta) \cos \vartheta \, d(\cos \vartheta)$$

and in case of the linear source function:

$$\pi F_\nu(0) = 2\pi \int_0^1 (a_\nu + b_\nu \cos \vartheta) \cos \vartheta \, d(\cos \vartheta) = \pi \left(a_\nu + \frac{2}{3} b_\nu \right)$$

From which we obtain the so-called **Eddington-Barbier relation**:

$$F_\nu(0) = S_\nu \left(\tau_\nu = \frac{2}{3} \right)$$

that is the flux emitted at the stellar surface is equal to the source function at optical depth $\tau_\nu = 2/3$.

In case of LTE, $S_\nu(\tau_\nu) = B_\nu(\tau_\nu)$, then:

$$F_\nu(0) = B_\nu(T(\tau_\nu = 2/3))$$

and the total flux of radiation is:

$$\pi F(0) = \pi \int_0^\infty F_\nu(0) \, d\nu = \pi \int_0^\infty B_\nu(T_1) \, d\nu = \sigma_{SB} T_1^4$$

where $T_1 = T(\tau_\nu = 2/3)$.

Since the **effective temperature** of a star (T_{eff}) is defined as the temperature corresponding to the emergent flux:

$$\pi F(0) = \sigma_{SB} T_{eff}^4$$

therefore: $T_{eff} = T_1$, that is the effective temperature is the temperature of the stellar atmosphere at optical depth $\tau_\nu = 2/3$.

4

Absorption lines

An absorption line of an atom is caused by a transition of an electron from a lower energy level m to an upper energy level n . The frequency or the wavelength of the transition are given by the usual formulas:

$$E_n - E_m = h\nu_{mn} = \frac{hc}{\lambda_{mn}}$$

In the previous chapter we have seen that the temperature of the stellar atmosphere increases with increasing depth:

$$T^4(\tau) = \frac{3}{4}T_{eff}^4(\tau + q(\tau))$$

where $q(\tau)$ is a slowly increasing function of τ , such as $q(0) = 0.577$ and $q(\infty) = 0.71$.

Therefore, the continuous radiation emitted by the inner hotter layers is absorbed by the atoms and/or the ions of the outer cooler layers. This mechanism causes the formation of the observed **absorption lines** (Fig. 4.1).

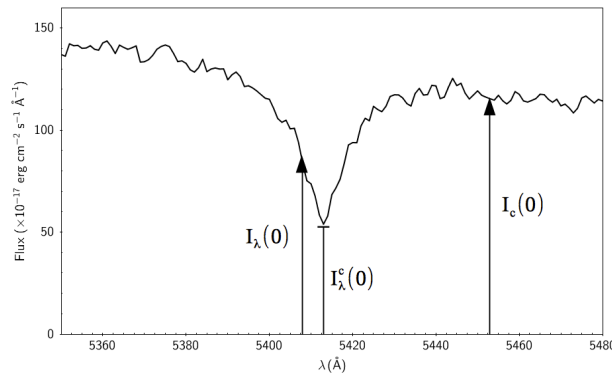


Figure 4.1: Absorption line He II $\lambda 5412$. $I_{\lambda}^c(0)$ is the surface intensity at the center of the line, $I_{\lambda}(0)$ at a wavelength within the line profile, and $I_c(0)$ is the intensity of the adjacent continuum.

If τ_c and τ_{λ} are the optical depths for the continuum and the absorption line:

$$I_c(0) = \int_0^{\infty} S_c(\tau_c) e^{-\tau_c} d\tau_c$$

CHAPTER 4. ABSORPTION LINES

$$I_\lambda(0) = \int_0^\infty S_\lambda(\tau_\lambda) e^{-\tau_\lambda} d\tau_\lambda$$

where we have assumed $\vartheta = 0$. As usual $d\tau_c = k_c dr$ and $d\tau_\lambda = k_\lambda dr$.

The absorption coefficient of the continuum is lower than that of the spectral line ($k_c < k_\lambda$), thus if we consider a value of optical depth (τ_λ) the geometrical depth (r) is larger for the continuum than for the line. As a result, the continuum is formed in a deeper hot layer and the absorption line in a cooler layer near the surface.

The **line depth** R_λ is defined as:

$$R_\lambda = \frac{I_c(0) - I_\lambda(0)}{I_c(0)}$$

If we consider a flux of radiation near the surface

$$I_c(0) = F_c(0) = B_\lambda(\tau_c = 2/3) \approx B_\lambda(T_{eff})$$

while, according to the Eddington-Barbier relation:

$$I_\lambda(0) = \int_0^\infty B_\lambda(T(\tau_\lambda)) e^{-\tau_\lambda} d\tau_\lambda = B_\lambda(T(\tau_\lambda = 2/3)) = B_\lambda(T_1)$$

Therefore,

$$R_\lambda = 1 - \frac{B_\lambda(T_1)}{B_\lambda(T_{eff})}$$

that is the depth of an absorption line depends on the difference between T_1 and T_{eff} . When an absorption line is weak, the absorption coefficients k_λ and k_c are similar, the difference between $B_\lambda(T_{eff})$ and $B_\lambda(T_1)$ is small and thus $R_\lambda \approx 0$. On the contrary, when an absorption line is strong, $k_c \ll k_\lambda$, $B_\lambda(T_1) \ll B_\lambda(T_{eff})$ and thus $R_\lambda \approx 1$.

A more common and used measure of the line depth is the so called **equivalent width**, indicated as **EW** or **W_λ**. It refers to the width of a rectangle having length equal to the continuum intensity and area equal to the area of the spectral line (Fig. 4.2).

It is formally expressed as:

$$EW = \frac{\int_{\lambda_1}^{\lambda_2} I_\lambda d\lambda}{< I_c >} \quad [\text{\AA}]$$

EW is used to quantify the intensity of a spectral line with respect of the continuum and it has the important property to be independent from the flux calibration of the spectrum. Therefore it can be effectively used when the flux calibration is not available or very uncertain. Ratios between EWs of different spectral lines are possible only in case the lines are close in wavelengths.

The EW has nothing to do with the real width of a line. A spectral line is never a monochromatic emission or absorption, it has always a width that depends

4.1. NATURAL BROADENING

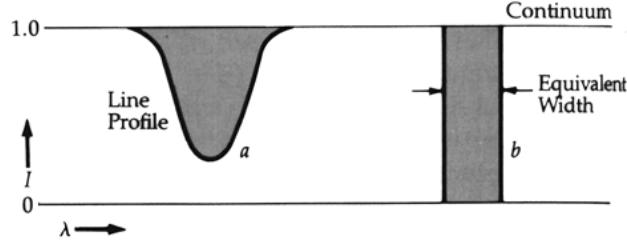


Figure 4.2: This sketch shows how the EW is calculated. The area inside the absorption line is equal to the area of the rectangle. The height of the rectangle is the continuum at the base of the spectral line, and the width is the EW.

on **microscopic** (= atomic) processes, **macroscopic** (= kinematic) processes, and **instrumental** effects. Their combination causes the so called broadening of a spectral line profile. This **broadening** can be evaluated in different ways, but the easiest and most diffuse is the measure of the **Full Width at Half Maximum (FWHM)**, that is the width of the line at half of the peak over (or under) the continuum.

4.1 Natural broadening

Because of the **uncertainty principle**, the energy levels of an atom are **naturally broadened**. Indeed, for a level above the ground one, having energy E and lifetime Δt :

$$\Delta E \Delta t \sim \hbar$$

Therefore, a photon absorbed or emitted in a transition having the ground state as a reference level can range within a frequency interval:

$$\Delta \nu = \frac{\Delta E}{h} \sim \frac{1}{2\pi \Delta t}$$

If the atoms can be considered like dipoles, we can apply the theory of the oscillator. In this frame, the classical cross section is given by:

$$\begin{aligned} a_\omega &= \left(\frac{1}{4\pi\epsilon_0} \right) \frac{4\pi e^2}{m_e c} \frac{\gamma \omega^2}{(\omega^2 - \omega_0^2)^2 + \gamma^2 \omega^2} \\ &= \frac{e^2}{\epsilon_0 m_e c} \frac{\gamma \omega^2}{(\omega^2 - \omega_0^2)^2 + \gamma^2 \omega^2} \end{aligned}$$

where $\omega = 2\pi\nu$ is the frequency, $\omega_0 = 2\pi\nu_0$ is the **resonance frequency**, and

$$\gamma = \left(\frac{1}{4\pi\epsilon_0} \right) \frac{2e^2}{3m_e c^3} \omega_0^2 = \left(\frac{1}{4\pi\epsilon_0} \right) \frac{8\pi^2}{3} \frac{e^2}{m_e c^3} \nu_0^2 \simeq 2.47 \cdot 10^{-22} \nu_0^2 \text{ [s}^{-1}\text{]}$$

CHAPTER 4. ABSORPTION LINES

is the **classical damping constant**.

$$\begin{aligned}
 a_\nu &= \left(\frac{1}{4\pi\epsilon_0} \right) \frac{4\pi e^2}{m_e c} \frac{\gamma 4\pi^2 \nu^2}{16\pi^4 (\nu^2 - \nu_0^2)^2 + \gamma^2 4\pi^2 \nu^2} \\
 &= \left(\frac{1}{4\pi\epsilon_0} \right) \frac{4\pi e^2}{m_e c} \frac{\gamma \nu^2}{4\pi^2 \left[(\nu^2 - \nu_0^2)^2 + \left(\frac{\gamma}{2\pi} \right)^2 \nu^2 \right]} \\
 &= \left(\frac{1}{4\pi\epsilon_0} \right) \frac{2e^2}{m_e c} \frac{\left(\frac{\gamma}{2\pi} \right) \nu^2}{(\nu^2 - \nu_0^2)^2 + \left(\frac{\gamma}{2\pi} \right)^2 \nu^2}
 \end{aligned}$$

This function is significantly different from zero when the frequency is close to the resonance value, therefore:

$$\nu \simeq \nu_0 \Rightarrow (\nu^2 - \nu_0^2)^2 = [(\nu + \nu_0)(\nu - \nu_0)]^2 \simeq 4\nu_0^2 (\nu - \nu_0)^2$$

and

$$\begin{aligned}
 a_\nu &\simeq \left(\frac{1}{4\pi\epsilon_0} \right) \frac{2e^2}{m_e c} \frac{\left(\frac{\gamma}{2\pi} \right) \nu_0^2}{4\nu_0^2 (\nu - \nu_0)^2 + \left(\frac{\gamma}{2\pi} \right)^2 \nu_0^2} \\
 &= \left(\frac{1}{4\pi\epsilon_0} \right) \frac{e^2}{m_e c} \frac{\left(\frac{\gamma}{4\pi} \right)}{(\nu - \nu_0)^2 + \left(\frac{\gamma}{4\pi} \right)^2}
 \end{aligned}$$

Let us assume now that the absorption cross section is the product between a constant and a normalized spectral line profile:

$$a_\nu = a_0 \cdot \varphi(\nu)$$

where in the case

$$\varphi(\nu) = \frac{A}{(\nu - \nu_0)^2 + \left(\frac{\gamma}{4\pi} \right)^2}$$

and we need to determine the value of the constant A that makes $\varphi(\nu)$ a normalized function.

$$\int_{-\infty}^{+\infty} \varphi(\nu) d\nu = \int_{-\infty}^{+\infty} \left(\frac{4\pi}{\gamma} \right)^2 \frac{A}{\left(\frac{4\pi}{\gamma} \right)^2 (\nu - \nu_0)^2 + 1} d\nu = 1$$

If $\frac{4\pi}{\gamma}(\nu - \nu_0) = x$

$$\int_{-\infty}^{+\infty} \varphi(\nu) d\nu = \frac{4\pi}{\gamma} A \int_{-\infty}^{+\infty} \frac{1}{x^2 + 1} dx = \frac{4\pi}{\gamma} A [\arctan x]_{-\infty}^{+\infty} = \frac{4\pi^2}{\gamma} A$$

Therefore

4.1. NATURAL BROADENING

$$A = \frac{\gamma}{4\pi^2}$$

and

$$a_\nu = \left(\frac{1}{4\pi\epsilon_0} \right) \frac{e^2\pi}{m_e c} \cdot \frac{\left(\frac{\gamma}{4\pi^2} \right)}{(\nu - \nu_0)^2 + \left(\frac{\gamma}{4\pi} \right)^2} \simeq 2.65 \cdot 10^{-6} \cdot \frac{\left(\frac{\gamma}{4\pi^2} \right)}{(\nu - \nu_0)^2 + \left(\frac{\gamma}{4\pi} \right)^2} \quad [\text{m}^2]$$

where

$$\varphi(\nu) = \frac{\left(\frac{\gamma}{4\pi^2} \right)}{(\nu - \nu_0)^2 + \left(\frac{\gamma}{4\pi} \right)^2} \quad [\text{s}]$$

In conclusion the profile of the natural broadening is a **Lorentzian function**. We now calculate the FWHM.

$$\varphi(\nu) = \frac{1}{2} \varphi(\nu_0)$$

$$\frac{\left(\frac{\gamma}{4\pi^2} \right)}{(\nu - \nu_0)^2 + \left(\frac{\gamma}{4\pi} \right)^2} = \frac{1}{2} \frac{4}{\gamma}$$

$$(\nu - \nu_0)^2 = \frac{\gamma^2}{16\pi^2} \Rightarrow |\nu - \nu_0| = \frac{\gamma}{4\pi}$$

and finally

$$\text{FWHM}(\nu) = 2|\nu - \nu_0| = \frac{\gamma}{2\pi} \simeq 3.92 \cdot 10^{-23} \nu_0^2 \quad [\text{Hz}]$$

If we need to convert the FWHM from frequency to wavelength, we need to remind that

$$\nu = \frac{c}{\lambda} \Rightarrow \Delta\nu = \frac{c}{\lambda^2} \Delta\lambda$$

therefore

$$\text{FWHM}(\lambda) = \frac{\lambda_0^2}{c} \text{FWHM}(\nu) = \frac{\lambda_0^2}{c} \frac{\gamma}{2\pi} = \frac{c}{\nu_0^2} \cdot 3.92 \cdot 10^{-23} \nu_0^2 \simeq 1.18 \cdot 10^{-4} \quad [\text{\AA}]$$

The natural broadening causes a very narrow Lorentzian profile that is usually not detected in astrophysical spectra.

Moving from classical physics to quantum mechanics the damping constant γ is substituted by Γ that depends on the probability of natural decay of the energy levels. The absorption cross section becomes:

$$a_\nu = \left(\frac{1}{4\pi\epsilon_0} \right)^2 \frac{2\pi e^4}{3m_e^2 c^4} \cdot \frac{\nu_0^2}{(\nu - \nu_0)^2 + \left(\frac{\Gamma}{4\pi} \right)^2} \cdot f$$

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where the constant f is called **oscillator strength**, and the spectral line profile becomes:

$$\varphi(\nu) = \frac{\frac{\Gamma}{4\pi^2}}{(\nu - \nu_0)^2 + \left(\frac{\Gamma}{4\pi}\right)^2}$$

4.2 Thermal Doppler broadening

If a radiation beam at monochromatic frequency ν_0 is absorbed by a gas of particles moving at velocities v , the frequency of the absorption will be affected by the Doppler effect:

$$\nu = \nu_0 \left(1 + \frac{v}{c}\right)$$

that is each particle will cause an absorption at a specific frequency depending on its velocity.

We now assume a Maxwell-Boltzmann distribution along a single direction, that is the direction of the radiation beam:

$$f(v)dv = \sqrt{\frac{m}{2\pi k_B T}} e^{-\frac{mv^2}{2k_B T}} dv$$

From the Doppler formula

$$v = \frac{\nu - \nu_0}{\nu_0} c \Rightarrow dv = \frac{c}{\nu_0} d\nu$$

therefore we convert $f(v)dv$ into $f(\nu)d\nu$

$$\begin{aligned} f(\nu)d\nu &= \sqrt{\frac{m}{2\pi k_B T}} e^{-\frac{m}{2k_B T} \frac{(\nu - \nu_0)^2}{\nu_0^2} c^2} \left(\frac{c}{\nu_0}\right) d\nu \\ &= \frac{1}{\sqrt{\pi}} \frac{1}{\Delta\nu_D} e^{-\left(\frac{\nu - \nu_0}{\Delta\nu_D}\right)^2} d\nu \end{aligned}$$

where

$$\Delta\nu_D = \frac{\nu_0}{c} v_{th}$$

and

$$v_{th} = \sqrt{\frac{2k_B T}{m}}$$

$f(\nu)$ is clearly a **Gaussian function**, $\Delta\nu_D$ is its width, and v_{th} is called **thermal velocity**.

In the previous section we have seen that for a single atom $a_\nu = a_0 \varphi(\nu)$, where $\varphi(\nu)$ is a very narrow Lorentzian function. Therefore we can assume $\varphi(\nu) \sim \delta(\nu - \nu_0)$.

4.2. THERMAL DOPPLER BROADENING

Now the absorption cross section for thermal Doppler is:

$$\begin{aligned} a_\nu^D &= \int_0^\infty a_0 \delta(\nu - \nu_0) \frac{1}{\sqrt{\pi}} \frac{1}{\Delta\nu_D} e^{-\left(\frac{\nu-\nu_0}{\Delta\nu_D}\right)^2} d\nu \\ &= a_0 \frac{1}{\sqrt{\pi}} \frac{1}{\Delta\nu_D} e^{-\left(\frac{\nu-\nu_0}{\Delta\nu_D}\right)^2} \end{aligned}$$

where

$$\varphi(\nu) = \frac{1}{\sqrt{\pi}} \frac{1}{\Delta\nu_D} e^{-\left(\frac{\nu-\nu_0}{\Delta\nu_D}\right)^2}$$

is the Gaussian shape of the spectral line profile, and it is easy to show that it is already a normalized function:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{\pi}} \frac{1}{\Delta\nu_D} e^{-\left(\frac{\nu-\nu_0}{\Delta\nu_D}\right)^2} d\nu = 1$$

We now calculate the FWHM.

$$\frac{1}{2}\varphi(\nu_0) = \frac{1}{2} \frac{1}{\sqrt{\pi}\Delta\nu_D} = \frac{1}{\sqrt{\pi}} \frac{1}{\Delta\nu_D} e^{-\left(\frac{\nu-\nu_0}{\Delta\nu_D}\right)^2}$$

$$(\nu - \nu_0)^2 = \Delta\nu_D^2 \ln 2$$

$$|\nu - \nu_0| = \Delta\nu_D \sqrt{\ln 2}$$

and finally

$$\text{FWHM}(\nu) = 2|\nu - \nu_0| = 2\sqrt{\ln 2}\Delta\nu_D \simeq 1.665 \frac{\nu_0}{c} \sqrt{\frac{2k_B T}{m}}$$

$$\text{FWHM}(\lambda) \simeq 1.665 \frac{\lambda_0}{c} \sqrt{\frac{2k_B T}{m}}$$

and

$$\Delta v_{th} = 1.665 \sqrt{\frac{2k_B T}{m}}$$

is called **thermal Doppler width**.

As an example, let us consider the H α transition and the temperature of the Sun ($T \sim 5780$ K). m will be the hydrogen mass ($\simeq 1.673 \cdot 10^{-27}$ kg), therefore $\text{FWHM}(\lambda) \simeq 0.36$ Å, and the Doppler width is $\Delta v_{th} \simeq 16$ km s $^{-1}$.

It is clear that whenever the atoms have a kinetic energy, the thermal Doppler broadening dominates the natural broadening.

4.3 Microturbulence

This term indicates a random motion of mini-cells of gas whose size is smaller than the mean free path of the photon. Microturbulence affects the absorption coefficient and modify both the profile and the EW of spectral lines. If the mini-cells are moving according to the Maxwell-Boltzmann distribution, the broadening function has a Gaussian shape.

From the previous section:

$$\Delta\nu_D = \frac{\nu_0}{c} \sqrt{\frac{2k_B T}{m}} = \frac{c}{\lambda_0^2} \Delta\lambda_D \Rightarrow \Delta\lambda_D = \frac{\lambda_0}{c} \sqrt{\frac{2k_B T}{m}}$$

The combined effect of thermal Doppler and microturbulence gives:

$$\Delta\lambda = \frac{\lambda_0}{c} \sqrt{\frac{2k_B T}{m} + v_{turb}^2}$$

There is virtually a way to verify if microturbulence is present or not:

$$v_{turb} = 0 \Rightarrow \Delta\lambda \propto \frac{1}{\sqrt{m}}$$

that is lighter atoms should show broader lines. Or:

$$v_{th} = 0 \Rightarrow \Delta\lambda \propto v_{turb}$$

that is the broadening is equal for all atoms.

4.4 Collisional broadening

If atoms are sufficiently close to each other, collisions among them cannot be neglected and produce important broadening effects with a Lorentzian shape. The collisional broadening depends on gas density (or pressure), on the cross-section of the collisional process, and of course on the thermal velocity of the atoms.

Summarizing, natural and collisional broadening (resonance + Van der Waals) functions have a Lorentzian shape, while thermal Doppler and microturbulence broadening have a Gaussian shape. Putting them together the profile of the spectral line has the so-called Voigt shape, that must be convolved with the Stark effect.

The Voigt profile is itself the result of the convolution of the Lorentzian and Gaussian functions. If we assume that natural and microturbulence broadening are negligible:

$$\varphi_V(\nu) = \varphi_C * \varphi_D$$

A convolution between two functions, $f(x)$ and $g(x)$ is:

4.5. ROTATIONAL BROADENING

$$f(x) * g(x) = \int_{-\infty}^{\infty} f(t)g(x-t)dt = \int_{-\infty}^{\infty} f(x-t)g(t)dt$$

In our case:

$$\varphi_C(\nu) = \frac{1}{\pi} \frac{\Delta\nu_C/2}{(\nu - \nu_0)^2 + (\Delta\nu_C/2)^2} \quad \varphi_D(\nu) = \frac{1}{\sqrt{\pi}} \frac{1}{\Delta\nu_D} e^{-\left(\frac{\nu - \nu_0}{\Delta\nu_D}\right)^2}$$

Therefore:

$$\varphi_V(\nu) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{\pi}} \frac{1}{\Delta\nu_D} e^{-\left(\frac{\nu - \nu_0}{\Delta\nu_D}\right)^2} \cdot \frac{1}{\pi} \frac{\Delta\nu_C/2}{(\nu - \nu_0 - u)^2 + (\Delta\nu_C/2)^2} du$$

Now, we make the following substitution:

$$y = \frac{u}{\Delta\nu_D} \Rightarrow u = \Delta\nu_D \cdot y \Rightarrow du = \Delta\nu_D \cdot dy$$

Then,

$$\varphi_V(\nu) = \frac{1}{\sqrt{\pi}} \frac{1}{\Delta\nu_D} \int_{-\infty}^{\infty} \frac{\Delta\nu_D}{\pi} \cdot \frac{\Delta\nu_C/2}{(\nu - \nu_0 - \Delta\nu_D \cdot y)^2 + (\Delta\nu_C/2)^2} e^{-y^2} dy$$

$$\varphi_V(\nu) = \frac{1}{\sqrt{\pi}} \frac{1}{\Delta\nu_D} \int_{-\infty}^{\infty} \frac{1}{\pi} \cdot \frac{\frac{1}{2} \frac{\Delta\nu_C}{\Delta\nu_D}}{\left(\frac{\nu - \nu_0}{\Delta\nu_D} - y\right)^2 + \left(\frac{1}{2} \frac{\Delta\nu_C}{\Delta\nu_D}\right)^2} e^{-y^2} dy$$

$$\varphi_V(\nu) = \varphi_D(\nu_0) \frac{a}{\pi} \int_{-\infty}^{\infty} \frac{e^{-y^2}}{(w - y)^2 + a^2} dy$$

$$\varphi_V(\nu) = \varphi_D(\nu_0) \cdot H(a, w)$$

where $H(a, w)$ is called Hjerting function ($H(a, w) \leq 1$) and

$$a = \frac{1}{2} \frac{\Delta\nu_C}{\Delta\nu_D} \quad w = \frac{\nu - \nu_0}{\Delta\nu_D}$$

It is easy to see that the core of the Voigt profile is dominated by the Gaussian function, in fact $\varphi_D(\nu) \sim e^{-\nu^2}$, while the wings are dominated by the Lorentzian function, in fact $\varphi_C(\nu) \sim \nu^{-2}$.

4.5 Rotational broadening

Let us assume that the star has a spherical shape, that it is rotating as a solid body, and finally that the rotation axis is in the plane of the sky ($i = 90^\circ$). In Fig. 4.3, X is a point belonging to a parallel at latitude ϑ and having circular velocity $\vec{v}_X$. The equatorial circular velocity of the star is $\vec{V}$.

Let us consider the stellar disk projected on the sky (Fig. 4.4, left) and the pole P view of the star (Fig. 4.4, right). $\overline{PQ}$ is the rotation axis, $\overline{EF}$ is the projection of a parallel of the frontal meridian, X has (x, y) coordinates and X' has $(x, 0)$

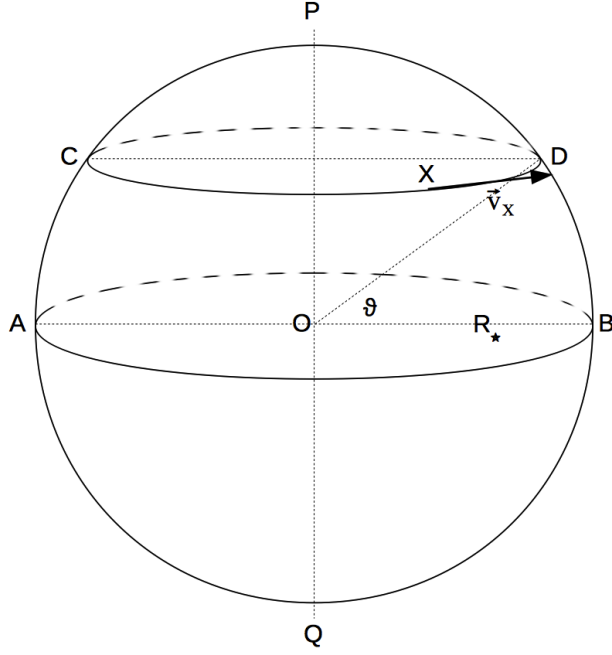


Figure 4.3: P and Q are the poles, $\overline{AB}$ is the stellar diameter, R_* is the stellar radius, $\overline{CD}$ is the diameter a parallel at latitude ϑ and X is a point over this parallel with circular velocity $\vec{v}_X$.

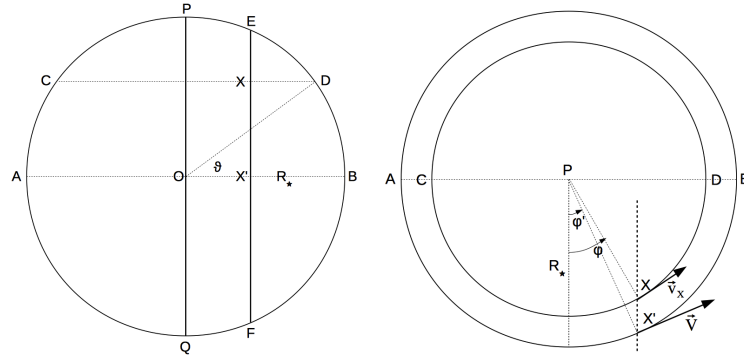


Figure 4.4: On the left: the stellar disk projected on the sky. On the right: the pole P view of the star.

coordinates. φ is the angle between the direction of the observer and $\overline{PX}$, while φ' is the angle between the direction of the observer and $\overline{PX'}$.

The radial velocity measured in X' is $V_r(X') = V \sin \varphi'$. The same in X is $v_r(X) = v_X \sin \varphi = V \cos \vartheta \sin \varphi$.

Now, $x = \overline{PX'} \sin \varphi' = R_* \sin \varphi'$, but also $x = \overline{PX} \sin \varphi = R_* \cos \vartheta \sin \varphi$. Therefore, $\cos \vartheta \sin \varphi = \sin \varphi'$ and so we obtain that $V_r(X') = v_r(X)$, that is all the points of the $\overline{EF}$ chord have the same radial velocity:

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$$V \sin \varphi' = V \frac{x}{R_\star}$$

Because of the rigid rotation all the points of the chord (or better, of the infinitesimal stripe) at distance x from the projected center of the star undergo the same Doppler effect:

$$\frac{\Delta\lambda}{\lambda} = \frac{V}{c} \frac{x}{R_\star}$$

The intensity of the Doppler shifted line with respect to the rest frame is:

$$\frac{I_\lambda(\Delta\lambda)}{I_\lambda(0)} = \frac{2\sqrt{R_\star^2 - x^2}}{2R_\star} = \sqrt{1 - \frac{x^2}{R_\star^2}} = \sqrt{1 - \frac{c^2}{V^2} \left(\frac{\Delta\lambda}{\lambda}\right)^2}$$

In case the rotation axis has $i \neq 0$:

$$\begin{aligned} \frac{I_\lambda(\Delta\lambda)}{I_\lambda(0)} &= \sqrt{1 - \frac{c^2}{(V \sin i)^2} \left(\frac{\Delta\lambda}{\lambda}\right)^2} \\ \frac{\left(\frac{\Delta\lambda}{\lambda}\right)^2}{\left(\frac{V \sin i}{c}\right)^2} + \left(\frac{I_\lambda(\Delta\lambda)}{I_\lambda(0)}\right)^2 &= 1 \end{aligned}$$

This is the equation of an elliptical broadening profile.

$$\frac{I_\lambda(\Delta\lambda)}{I_\lambda(0)} = \frac{1}{2} \Rightarrow \frac{\left(\frac{\Delta\lambda}{\lambda}\right)^2}{\left(\frac{V \sin i}{c}\right)^2} + \frac{1}{4} = 1 \Rightarrow \frac{\Delta\lambda}{\lambda} = \frac{\sqrt{3}}{2} \frac{V \sin i}{c}$$

The width of this profile is:

$$\text{FWHM} = \sqrt{3} \frac{V \sin i}{c}$$

If we take into account the Limb Darkening and the intensity on the stellar disk is $I = \text{constant}(1 + \beta \cos \vartheta)$, where β is the limb darkening coefficient, the broadening profile can be described by the following formula given by Unsöld (1955).

$$Q(y) = \frac{3}{3 + 2\beta} \left[\frac{2}{\pi} \sqrt{1 - y^2} + \frac{\beta}{2} (1 - y^2) \right]$$

This formula corresponds to a pure ellipse only when $\beta = 0$. For increasing values of β the profile becomes more and more parabolic.

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Therefore it is possible to estimate the value of $V \sin i$ comparing the observed spectral absorption lines with a broadened synthetic one (in the assumption that the broadening is dominated by rotation). Typical values for main sequence stars are: 100-200 km s⁻¹ for O, B, and A stars, about 50 km s⁻¹ for F stars, about 10 km s⁻¹ for G stars. Some peculiar stars like Be or Oe (e = emission) show values of 250-400 km s⁻¹. The Sun has a rotation of about 2 km s⁻¹.

5

Emission lines

If a radiation beam flows with the speed of light c through a surface element dA in a direction oriented with an angle ϑ with respect to the surface normal, like it is shown in Fig. 3.1, we have that $dt = ds/c$ and we can write the expression of the specific intensity in the form:

$$I_\nu = \frac{c dE_\nu}{(dA \cos \theta ds) d\nu d\Omega} = \frac{c dE_\nu}{dV d\nu d\Omega}$$

It follows that the presence of a radiation beam implies the existence of an **energy density** per unit frequency:

$$\frac{dE_\nu}{dV d\nu} = \frac{I_\nu d\Omega}{c}$$

Integrating over all the possible propagation directions, we define the **radiation density** as:

$$U_\nu = \frac{1}{c} \int_{4\pi} I_\nu d\Omega$$

or, for an **isotropic radiation field**:

$$U_\nu = \frac{4\pi}{c} I_\nu$$

Now, we consider again the general solution of the transport equation (Eq. 3.3). In order to determine the specific intensity I_ν , we need to know the source function $S_\nu = \varepsilon_\nu/k_\nu$ in each point. In general, the absorption and emission of radiation are due to: 1) **continuous processes**, which arises from **free-free**, **free-bound** and **bound-free** transitions; 2) **discrete processes**, generated only by **bound-bound** transitions.

We can therefore separate the two contributions and write:

$$S_\nu = \frac{\varepsilon_\nu^K + \varepsilon_\nu^L}{k_\nu^K + k_\nu^L}$$

where K and L are superscripts for continuum and spectral line.

If we now focus on the line contribution, the total energy emitted in a certain direction per unit volume and unit time due to a transition $n \rightarrow m$, is:

$$\int_{\nu_1}^{\nu_2} \varepsilon_\nu d\nu = \frac{1}{4\pi} h\nu_{nm} A_{nm} N_n$$

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where the integral extends over the line (considering its intrinsic shape), A_{nm} is the **Einstein spontaneous transition probability coefficient**, numerically equal to the number of transitions $n \rightarrow m$ which can occur per unit time (or, equivalently, to the inverse mean lifetime of the level n), and N_n is the particle density of ions populated at level n .

Since the intrinsic line width is generally much smaller than the typical width observed in spectral line profiles, the emissivity of the line is given by:

$$\varepsilon_\nu^L = \frac{1}{4\pi} h\nu_{nm} A_{nm} N_n \varphi(\nu)$$

where $\varphi(\nu)$ is the normalized line profile:

$$\int_{\nu_1}^{\nu_2} \varphi(\nu) d\nu = 1$$

Similarly, the energy absorbed in a transition $m \rightarrow n$ per unit time, unit volume, and unit solid angle is:

$$\int_{\nu_1}^{\nu_2} k_\nu I_\nu d\nu = \frac{1}{4\pi} h\nu_{nm} N_m B_{mn} U_{\nu_{nm}} - \frac{1}{4\pi} h\nu_{nm} N_n B_{nm} U_{\nu_{nm}}$$

where B_{mn} and B_{nm} are, respectively, the **Einstein absorption probability coefficient** and the **Einstein stimulated emission coefficient**, so that the right hand side of accounts for the energy that is subtracted from the beam by absorption and the energy that is added back to it, because of stimulated emission.

If the specific intensity is constant and isotropic within the intrinsic line profile ($I_\nu = I_{\nu_{nm}}$), we can assume $U_{\nu_{nm}} = 4\pi I_{\nu_{nm}}/c$, therefore we have:

$$\begin{aligned} I_{\nu_{nm}} \int_{\nu_1}^{\nu_2} k_\nu d\nu &= \frac{1}{4\pi} h\nu_{nm} (N_m B_{mn} - N_n B_{nm}) U_{\nu_{nm}} \\ \int_{\nu_1}^{\nu_2} k_\nu d\nu &= \frac{h\nu_{nm}}{c} (N_m B_{mn} - N_n B_{nm}) \end{aligned}$$

To find the line absorption coefficient, we simply have to multiply this integral by the normalized line profile, so that it is:

$$k_\nu^L = \frac{h\nu_{nm}}{c} (N_m B_{mn} - N_n B_{nm}) \varphi(\nu)$$

The Einstein coefficients are related together by general expressions, that can be derived in case of TE. In such conditions, indeed, all the reactions relating levels $n \rightleftharpoons m$ ($n > m$) are perfectly balanced, and we can set up a statistical balance equation between the radiative transitions connecting the levels:

$$U_{\nu_{nm}} (N_m B_{mn} - N_n B_{nm}) = N_n A_{nm}$$

that can be written as:

$$U_{\nu_{nm}} = \frac{A_{nm}}{\frac{N_m}{N_n} B_{mn} - B_{nm}} = \frac{\frac{A_{nm}}{B_{nm}}}{\frac{N_m}{N_n} \frac{B_{mn}}{B_{nm}} - 1}$$

The TE condition implies that:

$$\frac{N_m}{N_n} = \frac{g_m}{g_n} e^{\frac{h\nu_{nm}}{k_B T}}$$

$$U_{\nu_{nm}} = \frac{4\pi}{c} I_{\nu_{nm}} = \frac{8\pi h \nu_{nm}^3}{c^3} \frac{1}{e^{\frac{h\nu_{nm}}{k_B T}} - 1}$$

$$\frac{8\pi h \nu_{nm}^3}{c^3} \frac{1}{e^{\frac{h\nu_{nm}}{k_B T}} - 1} = \frac{\frac{A_{nm}}{B_{nm}}}{\frac{g_m}{g_n} \frac{B_{mn}}{B_{nm}} e^{\frac{h\nu_{nm}}{k_B T}} - 1}$$

This equation has the two following solutions:

$$g_m B_{mn} = g_n B_{nm}$$

$$A_{nm} = \frac{8\pi h \nu_{nm}^3}{c^3} B_{nm}$$

The line absorption coefficient becomes:

$$k_\nu^L = \frac{h\nu_{nm}}{c} N_m B_{mn} \left(1 - \frac{g_m}{g_n} \frac{N_n}{N_m} \right) \varphi(\nu)$$

and the line emissivity coefficient:

$$\varepsilon_\nu^L = \frac{2h^2 \nu_{nm}^4}{c^3} N_n \frac{g_m}{g_n} B_{mn} \varphi(\nu)$$

The source function in the line is then given by the ratio between these two last equations:

$$S_\nu^L = \frac{2h\nu^3}{c^2} \left(\frac{g_n}{g_m} \frac{N_m}{N_n} - 1 \right)^{-1}$$

Since in TE the level population follows the Boltzmann distribution the expression of S_ν^L reduces to:

$$S_\nu^L = \frac{2h\nu^3}{c^2} \frac{1}{e^{\frac{h\nu_{nm}}{k_B T}} - 1} = B_\nu(T)$$

that is the Planck distribution, in agreement with the solution of the transport equation in TE.

5.1 Collisional transitions

In the astrophysical plasma, and in particular in gaseous nebulae and star forming regions, it is frequent to detect emission lines corresponding to forbidden transitions between energy levels that are populated and de-populated by collisions of free electrons with ions.

If v is the relative velocity of an electron with respect to an ion (or a neutral atom), $\sigma(v)$ is the collisional cross-section, and N_e is the electron density, the number of collisions per second, or **collisional rate C** is given by:

$$C = N_e v \sigma(v) = N_e Q(v)$$

where $Q(v)$ is called **collisional rate coefficient** and it is expressed in terms of $\text{cm}^3 \text{s}^{-1}$.

If the elastic collisions dominate the plasma (that is we are in ETE), we can assume the Maxwell-Boltzmann distribution of velocities:

$$f(v)dv = 4\pi \left(\frac{m_e}{2\pi k_B T_e} \right)^{3/2} v^2 e^{-\frac{m_e v^2}{2k_B T_e}} dv$$

and the collisional rate coefficient becomes:

$$Q(v) = \int_0^\infty v \sigma(v) f(v) dv$$

More correctly the reduced mass

$$\mu = \frac{m_A m_e}{m_A + m_e}$$

should be used in place of the electron mass, but since $m_A \gg m_e$ we can affirm that $\mu \simeq m_e$.

The ETE is a correct assumption for H I, He I, and He II up to energies of about 10 eV. Indeed:

$$E_e = k_B T_e \Rightarrow T_e = \frac{E_e}{8.617733 \cdot 10^{-5} \text{ eV K}^{-1}} \simeq 11600 E_e [\text{K}]$$

therefore $10 \text{ eV} \simeq 1.2 \cdot 10^5 \text{ K}$. A typical gaseous nebula has an electron temperature $T_e \sim 10^4 \text{ K}$.

Let us now consider a transition $n \rightarrow m$ with $n > m$. In ETE collisional excitations and de-excitations must be in equilibrium:

$$N_e N_m Q_{mn} = N_e N_n Q_{nm}$$

$$\frac{Q_{mn}}{Q_{nm}} = \frac{N_n}{N_m} = \frac{g_n}{g_m} e^{-\frac{h\nu_{nm}}{k_B T_e}}$$

5.1. COLLISIONAL TRANSITIONS

and we obtain the relation between the collisional rate coefficient of excitation and de-excitation:

$$Q_{mn} = Q_{nm} \frac{g_n}{g_m} e^{-\frac{h\nu_{nm}}{k_B T_e}}$$

From the equilibrium equation we now want to obtain the cross-section.

$$N_e N_m v_m \sigma_{mn}(v_m) f(v_m) dv_m = N_e N_n v_n \sigma_{nm}(v_n) f(v_n) dv_n$$

The collisional excitation means that part of the kinetic energy of the electron is transferred to the ion, therefore:

$$\frac{1}{2} m_e v_m^2 = \chi_{mn} + \frac{1}{2} m_e v_n^2$$

where v_m and v_n are the electron velocities before and after the collision. By differentiating this last equation we obtain:

$$m_e v_m dv_m = m_e v_n dv_n$$

$$v_m dv_m = v_n dv_n$$

Therefore:

$$N_m \sigma_{mn}(v_m) f(v_m) = N_n \sigma_{nm}(v_n) f(v_n)$$

$$\begin{aligned} \sigma_{mn}(v_m) &= \frac{N_n}{N_m} \sigma_{nm}(v_n) \frac{f(v_n)}{f(v_m)} \\ &= \frac{g_n}{g_m} e^{-\frac{h\nu_{nm}}{k_B T_e}} \sigma_{nm}(v_n) \left(\frac{v_n}{v_m} \right)^2 e^{-\frac{m_e(v_n^2 - v_m^2)}{2k_B T_e}} \\ &= \frac{g_n}{g_m} \sigma_{nm}(v_n) \left(\frac{v_n}{v_m} \right)^2 \end{aligned}$$

or, in other words,

$$g_m v_m^2 \sigma_{mn}(v_m) = g_n v_n^2 \sigma_{nm}(v_n)$$

From quantum mechanics the excitation cross-section is given by the following formula:

$$\sigma_{mn}(v_m) = \frac{\pi \hbar^2}{m_e^2 v_m^2} \frac{\Omega(m, n)}{g_m}$$

where $\Omega(m, n)$ is called **collision strength** and it corresponds to the probability that an electron having wavelength λ interacts with an ion, or an atom.

CHAPTER 5. EMISSION LINES

Then, the de-excitation cross-section is:

$$\sigma_{nm}(v_n) = \frac{\pi \hbar^2}{m_e^2 v_n^2} \frac{\Omega(m, n)}{g_n}$$

and now we can calculate the collisional rate coefficient of de-excitation:

$$\begin{aligned} Q_{nm} &= \int_0^\infty v \sigma_{nm}(v) f(v) dv \\ &= \int_0^\infty v \frac{\pi \hbar^2}{m_e^2 v^2} \frac{\Omega(m, n)}{g_n} 4\pi \left(\frac{m_e}{2\pi k_B T_e} \right)^{3/2} v^2 e^{-\frac{m_e v^2}{2k_B T_e}} dv \\ &= \frac{\pi \hbar^2}{m_e^2 g_n} <\Omega(m, n)> 4\pi \left(\frac{m_e}{2\pi k_B T_e} \right)^{3/2} \int_0^\infty v e^{-\frac{m_e v^2}{2k_B T_e}} dv \\ &= \frac{\pi \hbar^2}{m_e^2 g_n} <\Omega(m, n)> 4\pi \left(\frac{m_e}{2\pi k_B T_e} \right)^{3/2} \frac{k_B T_e}{m_e} \\ &= \frac{\hbar^2}{m_e} \sqrt{\frac{2\pi}{m_e k_B T_e}} \frac{<\Omega(m, n)>}{g_n} \\ &\simeq \frac{8.63 \cdot 10^{-6}}{\sqrt{T_e}} \frac{<\Omega(m, n)>}{g_n} \end{aligned}$$

in units of $\text{cm}^3 \text{s}^{-1}$.

Assuming the following values of electron temperature and density, $T_e = 10^4 \text{ K}$ and $N_e = 10^3 \text{ cm}^{-3}$, the collisional rate coefficient of de-excitation is about $10^{-7} \text{ cm}^3 \text{s}^{-1}$ and the collisional rate is about 10^{-4} s^{-1} , that is there is a collisional de-excitation every about 2.8 hours.

In lower temperature regions the numbers of free electrons is much less than the number of neutral atoms (H I regions). In this case the reduced mass must be considered in the Maxwell-Boltzmann distribution.

The average velocity of the collisional partners is

$$\begin{aligned} \bar{v} &= \int_0^\infty v f(v) dv \\ &= \int_0^\infty v 4\pi \left(\frac{\mu}{2\pi k_B T} \right)^{3/2} v^2 e^{-\frac{\mu v^2}{2k_B T}} dv \\ &= 4\pi \left(\frac{\mu}{2\pi k_B T} \right)^{3/2} \int_0^\infty v^3 e^{-\frac{\mu v^2}{2k_B T}} dv \\ &= 2\pi \left(\frac{\mu}{2\pi k_B T} \right)^{3/2} 4 \left(\frac{k_B T}{\mu} \right)^2 \Gamma(2) \\ &= \sqrt{\frac{8k_B T}{\pi \mu}} \end{aligned}$$

In case the collisions are among H atoms the reduced mass is $\mu = \frac{m_H}{2}$, and the average velocity becomes $\bar{v} \simeq 2.05 \cdot 10^4 \sqrt{T} \text{ cm s}^{-1}$. This value is about 30 times lower than that of free electrons ($\bar{v}_e \simeq 6.2 \cdot 10^5 \sqrt{T_e} \text{ cm s}^{-1}$).

5.2. STATISTICAL EQUILIBRIUM EQUATIONS

In addition if the H atoms are populated at the fundamental level, the cross-section is

$$\sigma = \pi a_B^2 = \pi \left(\frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} \right)^2 \simeq 0.88 \cdot 10^{-16} \text{ cm}^2$$

and the collisional rate coefficient of de-excitation is:

$$Q_{nm} = \bar{v} \sigma \simeq 1.8 \cdot 10^{-12} \sqrt{T} \text{ cm}^3 \text{ s}^{-1}$$

If $T = 10^2 \text{ K}$, $Q_{nm} \sim 10^{-11} \text{ cm}^3 \text{ s}^{-1}$, and the collisional rate is $C_{nm} = N_H Q_{nm} \sim 10^{-11} \text{ s}^{-1}$ assuming a density $N_H \sim 1 \text{ cm}^{-3}$, that is a collision every 3170 years.

5.2 Statistical equilibrium equations

We now want to determine the ratio between the level populations, without assuming our system to be in thermodynamic equilibrium. We assume a **steady state**, i.e. a condition where there are no temporal variations of the level populations. The steady state assumption implies the condition:

$$\frac{dN_n}{dt} = 0$$

We define R_{nm} and R_{mn} the **velocities of radiative transitions** $n \rightarrow m$ and $m \rightarrow n$, while C_{nm} and C_{mn} are the **velocity of collisional transitions** $n \rightarrow m$ and $m \rightarrow n$. In case $m > n$ we have the following equations:

$$n \rightarrow m \begin{cases} R_{nm} = U_\nu B_{nm} \\ C_{nm} = N' Q_{nm} \end{cases} \quad m \rightarrow n \begin{cases} R_{mn} = A_{mn} + U_\nu B_{mn} \\ C_{mn} = N' Q_{mn} \end{cases}$$

where N' is the density of collisional partners. In case $m < n$ we have:

$$n \rightarrow m \begin{cases} R_{nm} = A_{nm} + U_\nu B_{nm} \\ C_{nm} = N' Q_{nm} \end{cases} \quad m \rightarrow n \begin{cases} R_{mn} = U_\nu B_{mn} \\ C_{mn} = N' Q_{mn} \end{cases}$$

Now we can write the equation of the steady state:

$$\begin{aligned} \frac{dN_n}{dt} &= N_n \sum_{m>n} (R_{nm} + C_{nm}) - \sum_{m>n} N_m (R_{mn} + C_{mn}) + \\ &\quad + N_n \sum_{m<n} (R_{nm} + C_{nm}) - \sum_{m<n} N_m (R_{mn} + C_{mn}) = 0 \\ N_n \sum_{m>n} (U_\nu B_{nm} + N' Q_{nm}) - \sum_{m>n} N_m (A_{mn} + U_\nu B_{mn} + N' Q_{mn}) + \\ &\quad + N_n \sum_{m<n} (A_{nm} + U_\nu B_{nm} + N' Q_{nm}) - \sum_{m<n} N_m (U_\nu B_{mn} + N' Q_{mn}) = 0 \end{aligned}$$

$$\begin{aligned}
 N_n \sum_m \left(\frac{4\pi}{c} I_\nu B_{nm} + A_{nm} + N' Q_{nm} \right) &= \sum_m N_m \left(\frac{4\pi}{c} I_\nu B_{mn} + A_{mn} + N' Q_{mn} \right) \\
 N_n \left(\frac{4\pi}{c} I_\nu \sum_m B_{nm} + \sum_m A_{nm} + N' \sum_m Q_{nm} \right) &= \\
 = \frac{4\pi}{c} I_\nu \sum_m N_m B_{mn} + \sum_m N_m A_{mn} + N' \sum_m N_m Q_{mn}
 \end{aligned}$$

To obtain the level population, we must solve this system of equations, called **statistical equilibrium** equations, for each level. In addition, the complete solution would require the sums over m to be extended and to include the continuously distributed states, in order to account also for ionizations from the level n . For the moment we consider the statistical equilibrium equations for bound states only, under the effect of **radiative** and **collisional** transitions.

5.3 Dilution of radiation

Let us imagine to have an extended gaseous region ionized by a hot star. The star has radius $R_\star$. An observer in a point P within the region, at distance r from the center of the star receives a flux $\phi_\nu(r)$ (Fig.5.1).

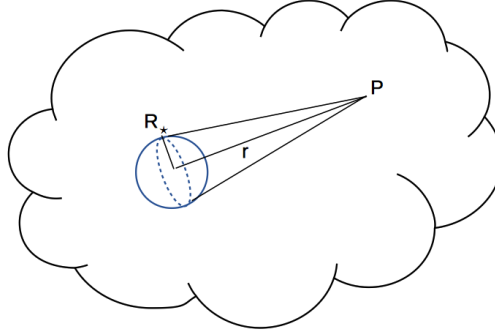


Figure 5.1: An observer in P receives the radiation from the star.

The specific intensity in P is

$$I_\nu^\star(r) = \frac{\phi_\nu(r)}{\Omega_\star}$$

where

$$\Omega_\star = \frac{\pi R_\star^2}{r^2}$$

is the solid angle of the star as seen by the observer in P.

If P was on the surface of the star,

$$I_\nu^*(R_\star) = \frac{\phi_\nu(R_\star)}{\pi}$$

therefore, in absence of absorption of radiation between the star and the observer

$$\frac{I_\nu^*(r)}{I_\nu^*(R_\star)} = \frac{\phi_\nu(r)}{\phi_\nu(R_\star)} \frac{r^2}{R_\star^2} = 1$$

that is the specific intensity does not depend on the distance.

The radiation density is

$$U_\nu = \frac{1}{c} \int_{\Omega_\star} I_\nu^* d\Omega = \frac{I_\nu^*}{c} \Omega_\star$$

Let us define now an isotropic radiation field I_ν having the same energy density:

$$\frac{4\pi}{c} I_\nu = \frac{I_\nu^*}{c} \Omega_\star \Rightarrow I_\nu = \frac{\Omega_\star}{4\pi} I_\nu^* = w \frac{2h\nu^3}{c^2} \frac{1}{e^{\frac{h\nu}{k_B T}} - 1}$$

where w is called **dilution factor**.

For example, if $R_\star = 10R_\odot$ and $r = 0.1$ pc, $w \sim 1.3 \cdot 10^{-14}$, that is a value much lower than 1. The conclusion is that in an extended nebula the radiation field is strongly diluted almost everywhere.

5.4 Two-level atom

We apply now the statistical equilibrium equations to an atom having only two energy levels. We call 1 the lower and 2 the higher.

$$N_1 \left(\frac{4\pi}{c} I_\nu B_{12} + N_e Q_{12} \right) = N_2 \left(\frac{4\pi}{c} I_\nu B_{21} + A_{21} + N_e Q_{21} \right)$$

$$\frac{N_2}{N_1} = \frac{\frac{4\pi}{c} I_\nu B_{12} + N_e Q_{12}}{\frac{4\pi}{c} I_\nu B_{21} + A_{21} + N_e Q_{21}}$$

We assume that there are no ionizations or recombinations, that the $1 \leftrightarrow 2$ transition is in the visible spectral range, and finally that $I_\nu = w B_\nu(T)$. Since $w \ll 1$, $I_\nu \sim 0$ and then

$$\frac{N_2}{N_1} = \frac{N_e Q_{12}}{A_{21} + N_e Q_{21}}$$

We have seen that

$$Q_{12} = Q_{21} \frac{g_2}{g_1} e^{-\frac{h\nu_{12}}{k_B T_e}}$$

therefore

$$\frac{N_2}{N_1} = \frac{N_e Q_{21} \frac{g_2}{g_1} e^{-\frac{h\nu_{12}}{k_B T_e}}}{A_{21} + N_e Q_{21}} = \frac{g_2}{g_1} \frac{e^{-\frac{h\nu_{12}}{k_B T_e}}}{1 + \frac{A_{21}}{N_e Q_{21}}}$$

In TE conditions

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$$\frac{1}{1 + \frac{A_{21}}{N_e Q_{21}}} = 1 \Rightarrow A_{21} \ll N_e Q_{21}$$

that is collisions dominate over spontaneous transitions.

We define now the **departure coefficient from TE**:

$$b_n = \frac{N_n}{N_n^*}$$

where N_n is the number density of atom populated at level n in case of non-TE, while N_n^* in case of TE. Then,

$$\frac{N_2}{N_1} = \frac{N_2^*}{N_1^*} \frac{b_2}{b_1} = \frac{b_2}{b_1} \frac{g_2}{g_1} e^{-\frac{h\nu_{12}}{k_B T_e}}$$

When $b_2/b_1 = 1$ we are in TE, on the contrary we are in deviation from the TE when $b_2/b_1 < 1$.

$$\frac{b_2}{b_1} = \frac{1}{1 + \frac{A_{21}}{N_e Q_{21}}}$$

If $A_{21} \gg N_e Q_{21} \Rightarrow b_2/b_1 \ll 1$, that is the radiative processes dominate over the collisions.

The ratio A_{21}/Q_{21} is called N_c , **critical density**. Therefore, when $N_e > N_c$ we are in TE, while when $N_e < N_c$ we are in non-TE.

Let us make now an example.

The transition is $\text{Ly}\alpha$ ($2^2\text{P}_{3/2} - 1^2\text{S}_{1/2}$) at $\lambda = 1215.7 \text{ \AA}$, and we assume $T_e = 10^4 \text{ K}$ and $N_e = 10^4 \text{ cm}^{-3}$.

In case of TE:

$$\frac{N_2^*}{N_1^*} = \frac{g_2}{g_1} e^{-\frac{h\nu_{12}}{k_B T_e}}$$

where $g_1 = 2 \cdot \frac{1}{2} + 1 = 2$, $g_2 = 2 \cdot \frac{3}{2} + 1 = 4$, and $h\nu_{12} = 10.2 \text{ eV}$. Therefore,

$$\frac{N_2^*}{N_1^*} \simeq 1.44 \cdot 10^{-5}$$

While, in case of non-TE, $A_{21} \simeq 6.3 \cdot 10^8 \text{ s}^{-1}$ and

$$Q_{21} = \frac{8.63 \cdot 10^{-6}}{\sqrt{T_e}} \frac{\Omega(1,2)}{g_2} \simeq 10^{-8} \text{ cm}^3 \text{ s}^{-1}$$

where $\Omega(1,2) \simeq 0.5$ (Giovanardi et al. 1987).

From these terms, the critical density is $N_c \simeq 6.3 \cdot 10^{16} \text{ cm}^{-3}$ and $b_2/b_1 \simeq 1.6 \cdot 10^{-13} \ll 1$.

$$\frac{N_2}{N_1} \simeq 2.3 \cdot 10^{-18}$$

5.5 Optically thin recombination lines

Recombination lines, also called permitted lines, are emission lines obtained when a free electron recombines with an ion. Let us now consider the fundamental level 1, a generic level $m > 1$ and an upper level n , and make the following hypotheses:

- we are in non-TE;
- recombinations occur directly to the level n or through a cascade;
- there is no induced emission (because the radiation field is diluted);
- the level n is not populated by collisions;
- ionizations occur only from the fundamental level;
- there is statistical equilibrium

We remind that the solution of the radiation transport is:

$$I_\nu(\tau_\nu = 0) = \int_0^{\tau_\nu^*} S_\nu(\tau_\nu) e^{-\tau_\nu} d\tau_\nu + I_{\nu 0} e^{-\tau_\nu^*}$$

where $\tau_\nu^* = (k_\nu^L + k_\nu^C)r^*$, that is the maximum optical depth of the cloud takes into account the absorption coefficients of the spectral line and of the continuum. In case of an optically thin line, $k_\nu^L \sim 0$, therefore $\tau_\nu^* \simeq k_\nu^C r^*$. In general, $\varepsilon_\nu^L \gg \varepsilon_\nu^C$, then the source function becomes:

$$S_\nu = \frac{\varepsilon_\nu^L + \varepsilon_\nu^C}{k_\nu^L + k_\nu^C} \simeq \frac{\varepsilon_\nu^L}{k_\nu^C}$$

Since we are in the case $\tau_\nu^* \ll 1$ and assuming $I_{\nu,0} = 0$,

$$I_\nu(0) = \frac{\varepsilon_\nu^L}{k_\nu^C} \tau_\nu^* = \varepsilon_\nu^L r^*$$

Therefore, the intensity of a generic $n \rightarrow m$ transition ($n > m$) is given by:

$$I_{nm} = \frac{h\nu_{nm}}{4\pi} A_{nm} N_n \ell$$

or more correctly:

$$I_{nm} = \frac{h\nu_{nm}}{4\pi} A_{nm} \int_0^\ell N_n dr$$

where the integral term is called **column density**.

The unknown terms are the number density of the atoms populated at level n , the size of the ionized cloud along the line of sight ℓ and the spatial dependence of N_n within the cloud. It is however possible to simplify the problem assuming that N_n is constant through the cloud and that the cloud has a symmetric shape so that width and length are approximately the same.

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From the Boltzmann equation applied to the hydrogen case, we know that:

$$\frac{N_n}{N_1} = \frac{g_n}{g_1} e^{-\frac{\chi_{0n}}{k_B T_e}} = \frac{2n^2}{2} e^{-\frac{\chi_{0n}}{k_B T_e}} = n^2 e^{-\frac{\chi_{0n}}{k_B T_e}}$$

and in the general case of the deviation from LTE:

$$\frac{N_n}{N_1} = \frac{b_n}{b_1} n^2 e^{-\frac{\chi_{0n}}{k_B T_e}}$$

From the Saha equation:

$$\frac{N_p N_e}{N_1} = 2 \frac{g_{11}}{g_{01}} \left(\frac{2\pi m_e k_B T_e}{h^2} \right)^{3/2} e^{-\frac{\chi_0}{k_B T_e}} = \left(\frac{2\pi m_e k_B T_e}{h^2} \right)^{3/2} e^{-\frac{\chi_0}{k_B T_e}}$$

being $g_{11} = 1$ and $g_{01} = 2$.

Again, in case of deviation from LTE:

$$\frac{N_p N_e}{N_1} = \frac{1}{b_1} \left(\frac{2\pi m_e k_B T_e}{h^2} \right)^{3/2} e^{-\frac{\chi_0}{k_B T_e}}$$

Now, combining the two equations, we obtain:

$$\begin{aligned} \frac{N_n}{N_1} \frac{N_1}{N_p N_e} &= \frac{b_n}{b_1} n^2 e^{-\frac{\chi_{0n}}{k_B T_e}} b_1 \left(\frac{2\pi m_e k_B T_e}{h^2} \right)^{-3/2} e^{\frac{\chi_0}{k_B T_e}} \\ N_n &= b_n n^2 \left(\frac{h^2}{2\pi m_e k_B T_e} \right)^{3/2} e^{\frac{\chi_0 - \chi_{0n}}{k_B T_e}} N_p N_e \end{aligned}$$

Using the statistical equilibrium equations applied to recombinations:

$$N_n \sum_{m=1}^{n-1} A_{nm} = \sum_{m=n+1}^{\infty} N_m A_{mn} + N_p N_e \alpha_{0n}$$

where α_{0n} is the **recombination coefficient** directly to the level n , and it depends on T_e . Then,

$$\begin{aligned} b_n n^2 \left(\frac{h^2}{2\pi m_e k_B T_e} \right)^{3/2} e^{\frac{\chi_0 - \chi_{0n}}{k_B T_e}} N_p N_e \sum_{m=1}^{n-1} A_{nm} &= \\ = \sum_{m=n+1}^{\infty} b_m m^2 \left(\frac{h^2}{2\pi m_e k_B T_e} \right)^{3/2} e^{\frac{\chi_0 - \chi_{0m}}{k_B T_e}} N_p N_e A_{mn} &+ N_p N_e \alpha_{0n} \end{aligned}$$

$$b_n \sum_{m=1}^{n-1} A_{nm} = \sum_{m=n+1}^{\infty} \frac{m^2}{n^2} b_m A_{mn} e^{-\frac{\chi_{0m} - \chi_{0n}}{k_B T_e}} + \frac{\alpha_{0n}}{n^2} \left(\frac{2\pi m_e k_B T_e}{h^2} \right)^{3/2} e^{-\frac{\chi_0 - \chi_{0n}}{k_B T_e}}$$

5.5. OPTICALLY THIN RECOMBINATION LINES

This is a system of equations that allows to obtain b_n . As an example, Fig. 5.2 taken from Prozesky & Smits (2018), shows theoretical b_n values as a function of n for different temperatures.

b_n tends to 1 because the average lifetime of the levels increases with n , therefore high energy levels are more stable. Indeed, for example, $A_{32} = A_{H\alpha} \simeq 4.4 \cdot 10^7 \text{ s}^{-1} \Rightarrow \langle t \rangle_{32} \simeq 2 \cdot 10^{-8} \text{ s}$, while $A_{40,2} = A_{H40} \simeq 67.5 \text{ s}^{-1} \Rightarrow \langle t \rangle_{40,2} \simeq 1.5 \cdot 10^{-2} \text{ s}$.

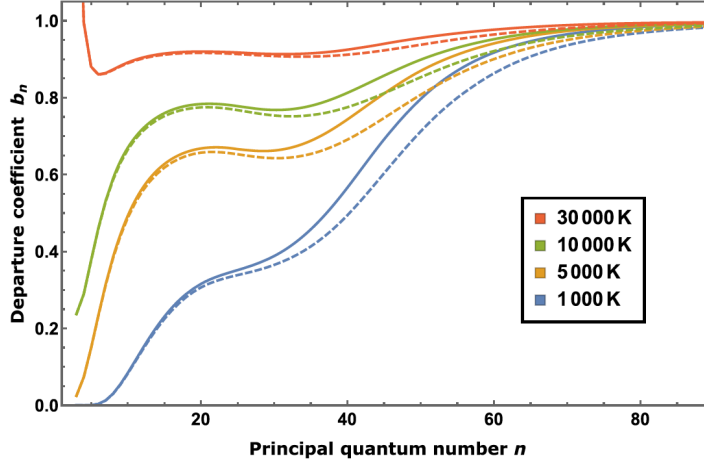


Figure 5.2: Figure 3 from Prozesky & Smits (2018). The departure coefficient b_n is calculated for different temperatures and assuming $N_e = 10^4 \text{ cm}^{-3}$. Solid lines are the results by Prozesky & Smits (2018), while dashed lines from Storey & Hummer (1995).

Substituting the expression of the population level n in the formula of intensity written at the beginning and assuming that $N_e \approx N_p$, we have:

$$I_{nm} = \alpha_{nm}^{eff} \frac{h\nu_{nm}}{4\pi} \int_0^\ell N_p N_e dr$$

where α_{nm}^{eff} is the **effective recombination coefficient** at level n , that takes into account all the recombinations and the cascade processes. It is tabulated as a function of T_e . As an example, $\alpha_{H\beta}^{eff} \sim 10^{-14} \text{ cm}^3 \text{ s}^{-1}$.

In case $N_p = N_e$:

$$I_{nm} = \alpha_{nm}^{eff} \frac{h\nu_{nm}}{4\pi} \int_0^\ell N_e^2 dr$$

and the integral term is called **emission measure**.

If we apply this formula to calculate the intensity ratios between hydrogen Balmer lines (relative to $H\beta$), the so-called **Balmer Decrement**, we obtain:

$$\frac{I_{H\alpha}}{I_{H\beta}} = \frac{\nu_{H\alpha}}{\nu_{H\beta}} \frac{\alpha_{H\alpha}^{eff}}{\alpha_{H\beta}^{eff}} \simeq 2.86$$

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$$\frac{I_{H\gamma}}{I_{H\beta}} \simeq 0.47 \quad , \quad \frac{I_{H\delta}}{I_{H\beta}} \simeq 0.26 \quad , \quad \frac{I_{H\epsilon}}{I_{H\beta}} \simeq 0.16 \quad \dots \quad \frac{I_{H20}}{I_{H\beta}} \simeq 0.006$$

These ratios are valid for $T_e = 10^4$ K and low electron density.

5.6 Dust extinction

One of the relevant applications of the Balmer decrement is the estimate of the dust extinction. Dust is present in the interstellar medium in the form of grains (by silicates, graphite, aluminum, iron, etc.) and has the ability to absorb UV-visible radiation and re-emit at longer wavelengths, typically in infrared.

When we look at a star its intrinsic spectrum appears with a different shape and intensity because the light passes through a layer of dust and undergoes processes of absorption and scattering.

If $I_{0,\lambda}$ is the intrinsic stellar spectrum and I_λ is the detected one:

$$dI_\lambda = -k_\lambda I_\lambda ds = I_\lambda d\tau_\lambda$$

$$\int_{I_{0,\lambda}}^{I_\lambda} \frac{dI_\lambda}{I_\lambda} = \int_{\tau_\lambda^*}^0 d\tau_\lambda$$

$$\ln \left(\frac{I_\lambda}{I_{0,\lambda}} \right) = -\tau_\lambda^*$$

$$I_\lambda = I_{0,\lambda} e^{-\tau_\lambda^*}$$

If we consider a pair of stars having the same spectral type, but located at different distances from us and in different positions in the Galaxy, the ratio of the detected flux is:

$$\frac{F_\lambda(1)}{F_\lambda(2)} = \frac{F_{0,\lambda}(1) e^{-\tau_\lambda(1)}}{F_{0,\lambda}(2) e^{-\tau_\lambda(2)}}$$

where

$$F_{0,\lambda}(1) = \frac{L_\lambda(1)}{4\pi d_1^2} \quad F_{0,\lambda}(2) = \frac{L_\lambda(1)}{4\pi d_2^2}$$

and $L_\lambda(1) = L_\lambda(2)$.

$$\frac{F_\lambda(1)}{F_\lambda(2)} = \left(\frac{d_2}{d_1} \right)^2 e^{-[\tau_\lambda(1) - \tau_\lambda(2)]}$$

If the star number 2 is not (or very low) absorbed: $\tau_\lambda(2) \simeq 0$, and

$$\tau_\lambda(1) = 2 \ln \left(\frac{d_2}{d_1} \right) - \ln \left(\frac{F_\lambda(1)}{F_\lambda(2)} \right)$$

At long wavelengths the absorption is low, that is $\tau_\lambda \rightarrow 0$ when λ increases. Therefore in the case the flux ratio allows to calculate the distance ratio. If we

5.6. DUST EXTINCTION

begin to observe the pair of stars in infrared, then we move progressively to the visible range, and we repeat this method for many pairs of stars we obtain:

$$\tau_\lambda = c \cdot f(\lambda)$$

where the coefficient c depends on distance and direction, while $f(\lambda)$ is the **extinction function**.

$$I_\lambda = I_{0,\lambda} e^{-cf(\lambda)} = I_{0,\lambda} 10^{-Cf(\lambda)}$$

where $C \simeq 0.43429 \cdot c$.

$$\log \left(\frac{I_\lambda}{I_{0,\lambda}} \right) = -Cf(\lambda)$$

$$-2.5 \log \left(\frac{I_\lambda}{I_{0,\lambda}} \right) = 2.5Cf(\lambda)$$

$$A(\lambda) = 2.5Cf(\lambda)$$

where $A(\lambda)$ is the absorption in magnitudes.

Finally,

$$I_\lambda = I_{0,\lambda} 10^{-0.4A(\lambda)}$$

The most popular extinction law is that published by **Cardelli, Clayton & Mathis** in 1989 (CCM89). It is normalized to $\lambda = 5500 \text{ \AA}$ and in the visible range has the following formula:

$$A(\lambda) = A(V) \left[a(x) + \frac{b(x)}{3.1} \right]$$

where $A(V)$ is the absorption in the photometric V band,

$$x = \frac{10000}{\lambda} - 1.82 \quad (\text{with } \lambda \text{ in } \text{\AA})$$

$$a(x) = 1. + 0.17699x - 0.50447x^2 - 0.02427x^3 + 0.72085x^4 + 0.01979x^5 - 0.77530x^6 + 0.32999x^7$$

$$b(x) = 1.41338x + 2.28305x^2 + 1.07233x^3 - 5.38434x^4 - 0.62251x^5 + 5.30260x^6 - 2.09002x^7$$

Fig. 5.3 shows the extinction law by Cardelli et al. If the V-band absorption is $A(V) = 1$ mag, the flux of the observed spectrum is about 40% in V, about 23% in U and about 63% in I. In Fig. 5.4 it is possible to see the effect of the reddening caused by an extinction of that amount for a A0 star. The continuum is all less intense, but the absorption is stronger in the U and B bands, weaker in the R and I bands. This changes the shape of the spectrum, causing the so-called reddening and giving the impression that the star has a lower temperature. Having an emission line spectrum it is possible to use the Balmer decrement coupled to the CCM89. In particular, the most diffuse ratio is between $H\alpha$ and $H\beta$, because these are the brightest lines of the decrement.

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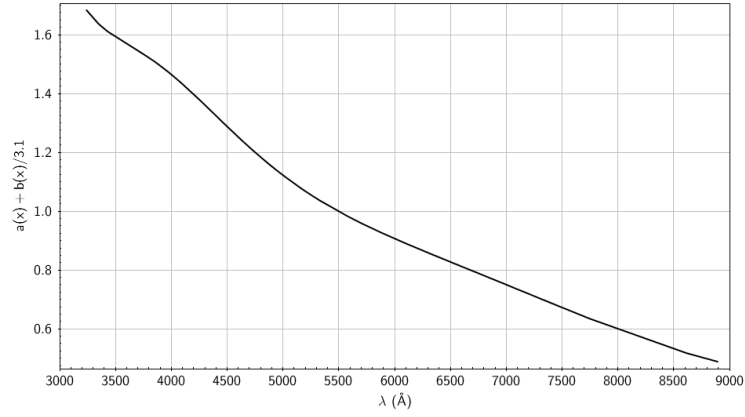


Figure 5.3: The Cardelli et al. (1989) extinction law in the visible wavelength domain.

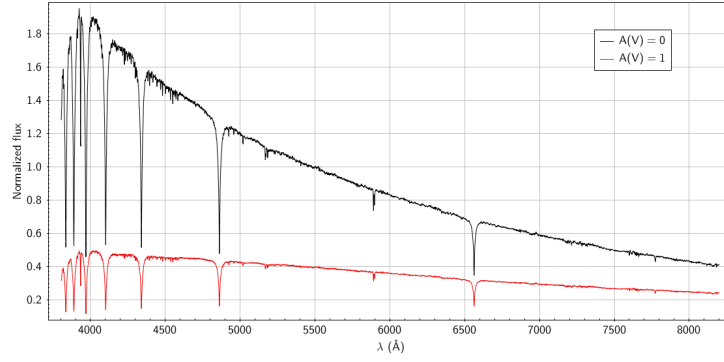


Figure 5.4: In black the spectrum of an A0 star, in red the same spectrum reddened by $A(V) = 1$ mag.

According to the CCM89:

$$\frac{I_{H\alpha}}{I_{H\beta}} = \left(\frac{I_{H\alpha}}{I_{H\beta}} \right)_0 10^{-0.4[A(H\alpha) - A(H\beta)]} \simeq \left(\frac{I_{H\alpha}}{I_{H\beta}} \right)_0 10^{0.1386A(V)}$$

$$A(V) = 7.215 \log \left[\frac{I_{H\alpha}/I_{H\beta}}{(I_{H\alpha}/I_{H\beta})_0} \right]$$

and for other lines:

$$A(V) = 13.713 \log \left[\frac{I_{H\gamma}/I_{H\beta}}{(I_{H\gamma}/I_{H\beta})_0} \right]$$

$$A(V) = 9.362 \log \left[\frac{I_{H\delta}/I_{H\beta}}{(I_{H\delta}/I_{H\beta})_0} \right]$$

5.7 Forbidden lines

The forbidden lines are so-named because they are emitted by forbidden transitions between metastable energy levels. These metastable levels are usually few eV above the fundamental level. Since the transitions are forbidden, the upper levels cannot be populated through absorption of radiation, but through collisional processes between ions and free electrons. And they are de-populated by collisions and spontaneous decays. The forbidden emission lines are indicated by [] around the ion.

5.7.1 The case of [O III] lines

Let us assume to have an atom with 3 levels: $m = 1, n = 2, n' = 3$. We consider the transitions: $3 \rightarrow 1$, $3 \rightarrow 2$, and $2 \rightarrow 1$ (Fig. 5.5).

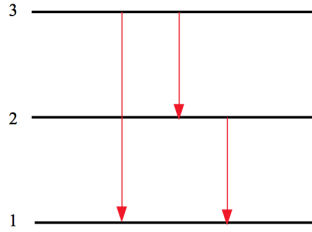


Figure 5.5: Scheme of a 3-level atom: electrons at level 3 decay to level 2 and 1, electrons at level 2 decay to level 1.

Then, we write the statistical equilibrium equation for the level 2 in the assumption that this level is populated by collisional excitation from level 1 and de-excitation from 3, and by radiative decay from level 3. Electron on level 2 move to level 1 through collisional de-excitation and radiative decay, and to level 3 through collisional excitation.

$$N_1 N_e Q_{12} + N_3 N_e Q_{32} + N_3 A_{32} = N_2 A_{21} + N_2 N_e Q_{21} + N_2 N_e Q_{23}$$

Similarly, level 3 is collisionally populated from levels 1 and 2, and de-populated toward levels 1 and 2 by collisional de-excitation and radiative decay.

$$N_1 N_e Q_{13} + N_2 N_e Q_{23} = N_3 A_{31} + N_3 A_{32} + N_3 N_e Q_{31} + N_3 N_e Q_{32}$$

In order to make the calculations easier, we call $a = N_1 N_e Q_{12}$, $b = N_e Q_{32} + A_{32}$, $c = A_{21} + N_e Q_{21} + N_e Q_{23}$, $d = N_1 N_e Q_{13}$, $e = N_e Q_{23}$, $f = A_{31} + A_{32} + N_e Q_{31} + N_e Q_{32}$. We can now write the equation system:

$$\begin{cases} a + bN_3 = cN_2 \\ d + eN_2 = fN_3 \end{cases}$$

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that has the following solution:

$$N_2 = \frac{af + bd}{cf - be} \quad N_3 = \frac{d + eN_2}{f}$$

Let us focus on the full expression for N_2 in case of the [O III] $\lambda 2321$, $\lambda 4363$, $\lambda 4959$, $\lambda 5007$ lines. The last three are the most widely used spectral lines to estimate the electron temperature.

Oxygen has atomic number $Z=8$, so O III has 6 electrons and the configuration is: $1s^2 2s^2 2p^2$, the same as C I. The Grotrian diagram of the [O III] transitions is plotted in Fig. 5.6.

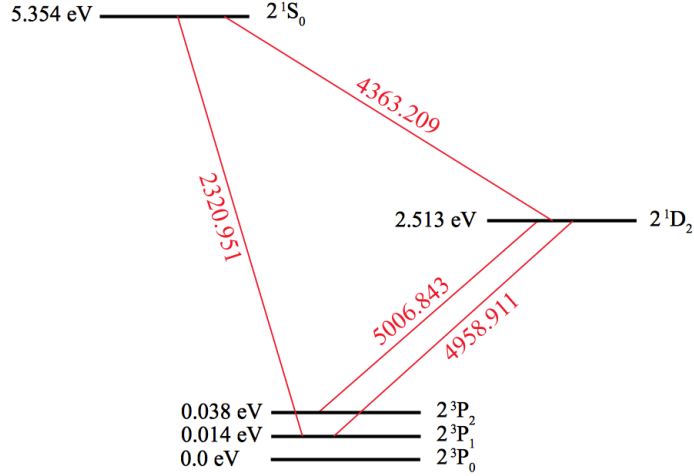


Figure 5.6: The Grotrian diagram of [O III] $\lambda 2321$, $\lambda 4363$, $\lambda 4959$, $\lambda 5007$.

$$N_2 = \frac{N_1 N_e Q_{12} \cdot (A_{31} + A_{32} + N_e Q_{31} + N_e Q_{32}) + (N_e Q_{32} + A_{32}) \cdot N_1 N_e Q_{13}}{(A_{21} + N_e Q_{21} + N_e Q_{23}) \cdot (A_{31} + A_{32} + N_e Q_{31} + N_e Q_{32}) - (N_e Q_{32} + A_{32}) \cdot N_e Q_{23}}$$

Let us compare the collisional excitation coefficients from level 1 to levels 2 and 3.

$$Q_{12} = Q_{21} \frac{g_2}{g_1} e^{-\frac{h\nu_{21}}{k_B T_e}} \quad , \quad Q_{13} = Q_{31} \frac{g_3}{g_1} e^{-\frac{h\nu_{31}}{k_B T_e}}$$

$$\frac{Q_{13}}{Q_{12}} = \frac{Q_{31}}{Q_{21}} \frac{g_3}{g_2} e^{-\frac{(h\nu_{31} - h\nu_{21})}{k_B T_e}} = \frac{\Omega_{31}}{\Omega_{21}} e^{-\frac{(h\nu_{31} - h\nu_{21})}{k_B T_e}}$$

The energy values of the $3 \rightarrow 1$ and $2 \rightarrow 1$ transitions are: $h\nu_{31} \simeq 5.34$ eV and $h\nu_{21} \simeq 2.5$ eV. The collisional strengths are: $\Omega_{31} \simeq 0.29$ and $\Omega_{21} \simeq 2.29$. By assuming $T_e = 10^4$ K, then $k_B T_e \simeq 0.86$ eV, therefore:

5.7. FORBIDDEN LINES

$$\frac{Q_{13}}{Q_{12}} = \frac{0.29}{2.29} e^{-\frac{5.34-2.5}{0.86}} \simeq 4.7 \cdot 10^{-3}$$

$Q_{13} \ll Q_{12}$, that is collisional excitations from level 1 to 2 are more probable than those from level 1 to 3.

Let us compare $N_e(Q_{31} + Q_{32})$ and $A_{31} + A_{32}$.

$$Q_{31} = 8.63 \cdot 10^{-6} \frac{\Omega_{31}}{g_3 \sqrt{T_e}} \quad , \quad Q_{32} = 8.63 \cdot 10^{-6} \frac{\Omega_{32}}{g_3 \sqrt{T_e}}$$

The collisional strengths are: $\Omega_{31} \simeq 0.29$ and $\Omega_{32} \simeq 0.58$. The statistical weight is: $g_3 = 1$. By assuming, again, $T_e = 10^4$ K: $Q_{31} \simeq 2.5 \cdot 10^{-8} \text{ cm}^3 \text{ s}^{-1}$ and $Q_{32} \simeq 5 \cdot 10^{-8} \text{ cm}^3 \text{ s}^{-1}$. The radiative decay probabilities are: $A_{31} = A_{\lambda 2321} = 0.23 \text{ s}^{-1}$ and $A_{32} = A_{\lambda 4363} = 1.6 \text{ s}^{-1}$, therefore we can affirm that $N_e(Q_{31} + Q_{32}) \ll A_{31} + A_{32}$, that is radiative decays from level 3 are more probable than collisional de-excitations from the same level.

Let us compare $N_e Q_{23}$ and $N_e Q_{21}$.

$$Q_{23} = Q_{32} \frac{g_3}{g_2} e^{-\frac{h\nu_{32}}{k_B T_e}} \quad , \quad Q_{21} = 8.63 \cdot 10^{-6} \frac{\Omega_{21}}{g_2 \sqrt{T_e}}$$

Since $h\nu_{32} \simeq 2.84 \text{ eV}$ and $g_2 = 5$, then:

$$Q_{23} = 5 \cdot 10^{-8} \frac{1}{5} e^{-\frac{2.84}{0.86}} \simeq 3.7 \cdot 10^{-10} \text{ cm}^3 \text{ s}^{-1}$$

$$Q_{21} \simeq 4 \cdot 10^{-8} \text{ cm}^3 \text{ s}^{-1}$$

Therefore, $N_e Q_{23} \ll N_e Q_{21}$, that is collisional de-excitations from level 2 to 1 are more probable than collisional excitations from level 2 to 3.

The radiative decay probabilities for $2 \rightarrow 1$ transitions are: $A_{\lambda 4959} = 6.8 \cdot 10^{-3} \text{ s}^{-1}$ and $A_{\lambda 5007} = 2.0 \cdot 10^{-2} \text{ s}^{-1}$. In case of electron densities $10^2 < N_e < 10^4$, the collisional excitation probabilities from level 2 to 3 are: $N_e Q_{23} \sim 10^{-8} - 10^{-6} \text{ s}^{-1} \ll A_{21} \sim 10^{-2} \text{ s}^{-1}$. This means that there are not collisional excitations from level 2 to level 3 during the average lifetime of the level 2 ($Q_{23} \sim 0$).

We can now significantly simplify the expression of N_2 :

$$N_2 \simeq \frac{N_1 N_e Q_{12}}{A_{21} + N_e Q_{21}}$$

And for the level 3:

$$N_3 = \frac{N_1 N_e Q_{13} + N_e Q_{23} \frac{N_1 N_e Q_{12}}{A_{21} + N_e Q_{21}}}{A_{31} + A_{32} + N_e (Q_{31} + Q_{32})} \simeq \frac{N_1 N_e Q_{13}}{A_{31} + A_{32}}$$

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Let us calculate the emission line ratio:

$$\begin{aligned} R_{[\text{O III}]} &= \frac{I_{21}}{I_{32}} = \frac{\nu_{21}}{\nu_{32}} \frac{A_{21}}{A_{32}} \frac{N_2}{N_3} = \frac{\nu_{21}}{\nu_{32}} \frac{A_{21}}{A_{32}} \frac{N_1 N_e Q_{12}}{A_{21} + N_e Q_{21}} \frac{A_{31} + A_{32}}{N_1 N_e Q_{13}} = \\ &= \frac{\nu_{21}}{\nu_{32}} \frac{Q_{12}}{Q_{13}} \frac{A_{31} + A_{32}}{A_{32}} \frac{1}{1 + N_e \frac{Q_{21}}{A_{21}}} \end{aligned}$$

The ratio between the collisional coefficients is:

$$\frac{Q_{12}}{Q_{13}} = \frac{Q_{21}}{Q_{31}} \frac{g_2}{g_3} e^{\frac{(h\nu_{31} - h\nu_{21})}{k_B T_e}} = \frac{\Omega_{21}}{\Omega_{31}} e^{\frac{h\nu_{32}}{k_B T_e}}$$

and finally:

$$R_{[\text{O III}]} = \frac{I_{21}}{I_{32}} = \frac{\nu_{21}}{\nu_{32}} \frac{\Omega_{21}}{\Omega_{31}} \frac{A_{31} + A_{32}}{A_{32}} \frac{e^{\frac{h\nu_{32}}{k_B T_e}}}{1 + 8.63 \cdot 10^{-6} \frac{\Omega_{21}}{g_2} \frac{N_e}{A_{21} \sqrt{T_e}}}$$

By substituting the value relative to the [O III] transitions, we obtain a formula that is sensitive to T_e , as long as $N_e < 10^4 \text{ cm}^{-3}$ (Fig. 5.7):

$$R_{[\text{O III}]} = \frac{I_{4959} + I_{5007}}{I_{4363}} = \frac{7.73 \cdot e^{\frac{3.29 \cdot 10^4}{T_e}}}{1 + 4.5 \cdot 10^{-4} \frac{N_e}{\sqrt{T_e}}}$$

This is a list of emission line ratios used to estimate the electron temperature: [O II] $(\lambda 3726 + \lambda 3729)/(\lambda 7320 + \lambda 7330)$, [Ne III] $(\lambda 3869 + \lambda 3969)/\lambda 3342$, [O III] $(\lambda 4959 + \lambda 5007)/\lambda 4363$, [O I] $(\lambda 6300 + \lambda 6363)/\lambda 5577$, [N II] $(\lambda 6548 + \lambda 6583)/\lambda 5755$, [S II] $(\lambda 6716 + \lambda 6731)/(\lambda 4068 + \lambda 4076)$, [Ar III] $(\lambda 7136 + \lambda 7751)/\lambda 5192$, [S III] $(\lambda 9069 + \lambda 9532)/\lambda 6312$.

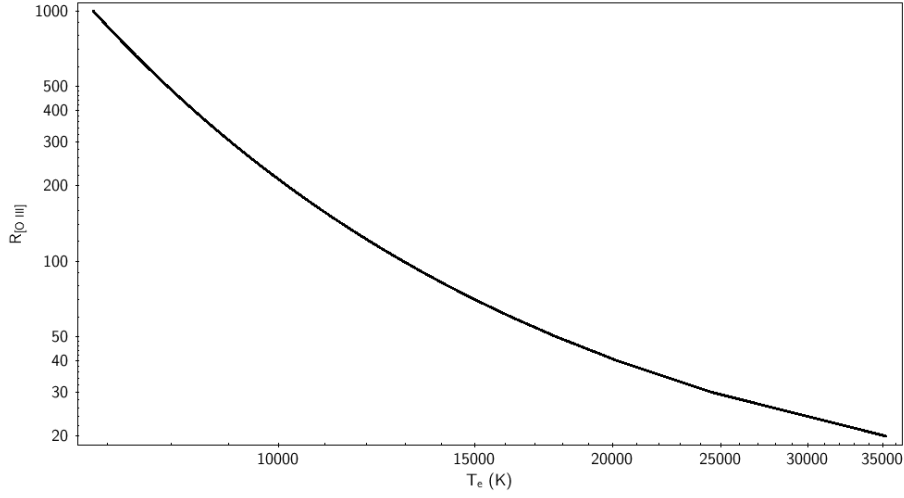


Figure 5.7: The temperature can be estimated from the [O III] lines ratio. This curve was calculated with the IRAF task TEMDEN by assuming $N_e = 10^3 \text{ cm}^{-3}$.

5.7.2 The case of [S II] lines

Let us assume to have an **atom with 3 levels**: $m = 1, n = 2, n' = 3$, where n and n' are very close in energy ($h\nu_{31} \simeq h\nu_{21}$), and let us consider the transitions: $3 \rightarrow 1$ and $2 \rightarrow 1$ (Fig. 5.8).

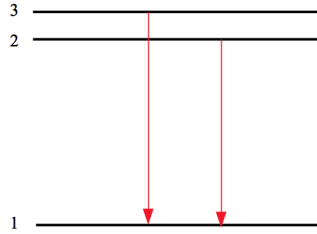


Figure 5.8: Scheme of a 3-level atom: electrons at level 2 and 3 decay to level 1.

The intensities of the two corresponding emission lines are:

$$I_{21} = \frac{h\nu_{21}}{4\pi} A_{21} N_2 \ell \quad I_{31} = \frac{h\nu_{31}}{4\pi} A_{31} N_3 \ell$$

and their ratio:

$$R = \frac{I_{31}}{I_{21}} = \frac{\nu_{31}}{\nu_{21}} \frac{A_{31}}{A_{21}} \frac{N_3}{N_2} \simeq \frac{A_{31}}{A_{21}} \frac{N_3}{N_2}$$

that is, R depends on the population of the levels 2 and 3. In order to calculate the $\frac{N_3}{N_2}$ ratio, we apply the statistical equilibrium equation for the $2 \rightarrow 1$ transition:

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$$N_2(N_e Q_{21} + A_{21} + N_e Q_{23}) = N_1 N_e Q_{12} + N_3(N_e Q_{32} + A_{32})$$

and for the $3 \rightarrow 1$ transition:

$$N_3(N_e Q_{31} + N_e Q_{32} + A_{31} + A_{32}) = N_1 N_e Q_{13} + N_2 N_e Q_{23}$$

As in the previous case, we solve the equation system:

$$\begin{cases} aN_2 = b + cN_3 \\ dN_3 = e + fN_2 \end{cases}$$

obtaining:

$$N_2 = \frac{bd + ce}{ad - cf} \quad N_3 = \frac{e + fN_2}{d}$$

We focus on the case of [S II] $\lambda 6716, 6731$, and in particular on their ratio [S II] $\lambda 6716/\lambda 6731$, which is the most popular emission line ratio in the visible wavelength interval used to estimate electron density.

The atomic number of S is $Z=16$, therefore S II has 15 electrons and the configuration is: $1s^2 2s^2 2p^6 3s^2 3p^3$, that is as N I. The Grotrian diagram for the [S II] transitions is plotted in Fig. 5.9.

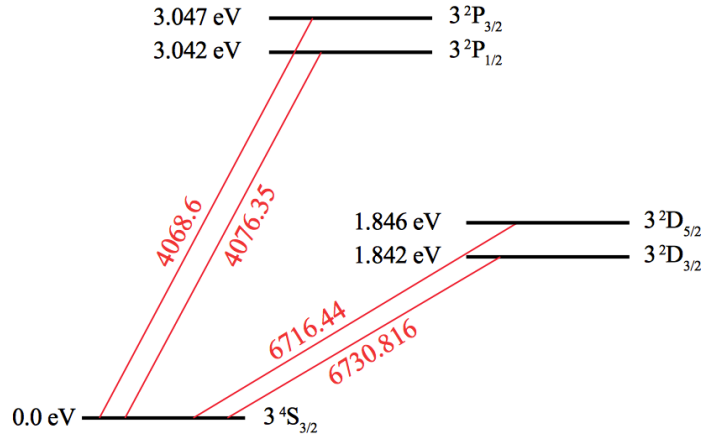


Figure 5.9: The Grotrian diagram of S II. The indicated transitions are forbidden because they take place between terms that are all odd.

The spontaneous transition probabilities of the lines indicated in figure are: $A_{4068} = 0.192 \text{ s}^{-1}$, $A_{4076} = 7.72 \cdot 10^{-2} \text{ s}^{-1}$, $A_{6716} = 2.02 \cdot 10^{-4} \text{ s}^{-1}$, and $A_{6731} = 6.84 \cdot 10^{-4} \text{ s}^{-1}$.

5.7. FORBIDDEN LINES

In order to calculate the collision strengths, we need to apply the following formula:

$$\Omega(SLJ, S'L'J') = \frac{2J' + 1}{(2S' + 1)(2L' + 1)} \Omega(SL, S'L')$$

In our case, for [S II] λ 6731 :

$$\Omega(^4S_{3/2}, ^2D_{3/2}) = \frac{2 \cdot \frac{3}{2} + 1}{2(2 \cdot 2 + 1)} \Omega(^4S, ^2D) = 2.76$$

being $\Omega(^4S, ^2D) = 6.90$ (Osterbrock & Ferland, 2006).

While for [S II] λ 6716 :

$$\Omega(^4S_{3/2}, ^2D_{5/2}) = \frac{2 \cdot \frac{5}{2} + 1}{2(2 \cdot 2 + 1)} \Omega(^4S, ^2D) = 4.14$$

The statistical weights are: $g_{6716} = 2 \cdot \frac{5}{2} + 1 = 6$ and $g_{6731} = 2 \cdot \frac{3}{2} + 1 = 4$. Then, assuming $T_e = 10^4$ K, the collisional coefficients are: $Q_{6716} = Q_{6731} = 5.95 \cdot 10^{-8} \text{ cm}^3 \text{ s}^{-1}$. And the critical densities are: $N_c(6716) \simeq 3.4 \cdot 10^3 \text{ cm}^{-3}$ and $N_c(6731) \simeq 1.15 \cdot 10^4 \text{ cm}^{-3}$.

On the basis of these numbers, we consider the population of the level 2:

$$N_2 = \frac{N_1 N_e Q_{12} (N_e Q_{31} + N_e Q_{32} + A_{31} + A_{32}) + (N_e Q_{32} + A_{32}) N_1 N_e Q_{13}}{(N_e Q_{21} + A_{21} + N_e Q_{23}) (N_e Q_{31} + N_e Q_{32} + A_{31} + A_{32}) - (N_e Q_{32} + A_{32}) N_e Q_{23}}$$

and we evaluate the **low density case** ($N_e \leq 10^2 \text{ cm}^{-3}$).

First, we compare $N_e(Q_{31} + Q_{32})$ with $A_{31} + A_{32}$.

We have seen that $Q_{31} = Q_{6716} = 5.95 \cdot 10^{-8} \text{ cm}^3 \text{ s}^{-1}$, while $Q_{32} = 1.07 \cdot 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ ($\Omega_{23} = 7.47$). In addition, $A_{32} = 3.3 \cdot 10^{-7} \text{ s}^{-1}$. Therefore, we can state that $N_e(Q_{31} + Q_{32}) \ll A_{31} + A_{32}$.

Second, we compare $N_e(Q_{21} + Q_{23})$ with A_{21} .

$Q_{21} = Q_{6731} = 5.95 \cdot 10^{-8} \text{ cm}^3 \text{ s}^{-1}$, while $Q_{23} = 1.6 \cdot 10^{-7} \text{ cm}^3 \text{ s}^{-1}$. We conclude that $N_e(Q_{21} + Q_{23}) \ll A_{21}$.

Then, it is easy to see that $N_e Q_{32} \ll A_{31}$ as well as $N_e Q_{23} \ll A_{21}$.

And finally, we have:

$$N_2 \simeq \frac{N_1 N_e Q_{12}}{A_{21}}$$

$$N_3 \simeq \frac{N_1 N_e Q_{13} + N_e Q_{23} \frac{N_1 N_e Q_{12}}{A_{21}}}{N_e Q_{31} + N_e Q_{32} + A_{31} + A_{32}} \simeq \frac{N_1 N_e Q_{13}}{A_{31}}$$

taking into account that $Q_{12} = 7 \cdot 10^{-9} \text{ cm}^3 \text{ s}^{-1}$.

$$R = \frac{I_{31}}{I_{21}} \simeq \frac{A_{31}}{A_{21}} \frac{N_1 N_e Q_{13}}{A_{31}} \frac{A_{21}}{N_1 N_e Q_{12}} \simeq \frac{\Omega_{13}}{\Omega_{12}}$$

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$$R_{[\text{SII}]} = \frac{I_{6716}}{I_{6731}} \simeq 1.5$$

Now, we explore the **high density case** ($N_e \geq 10^4 \text{ cm}^{-3}$), and we notice that in general the collisional terms are higher or much higher than the spontaneous decay terms.

$$\frac{N_3}{N_2} = \frac{e}{d} \left(\frac{ad - cf}{bd + ce} \right) + \frac{f}{d}$$

$$\frac{N_3}{N_2} \simeq \frac{N_1 N_e Q_{13}}{N_e(Q_{31} + Q_{32})} - \frac{N_e^2 (Q_{21} + Q_{23})(Q_{31} + Q_{32}) - N_e^2 Q_{32} Q_{23}}{N_1 N_e^2 Q_{12}(Q_{31} + Q_{32}) + N_1 N_e^2 Q_{32} Q_{13}} + \frac{N_e Q_{23}}{N_e(Q_{31} + Q_{32})}$$

$$\frac{N_3}{N_2} \simeq \frac{Q_{13}}{Q_{31} + Q_{32}} \frac{Q_{21} Q_{31} + Q_{21} Q_{32} + Q_{23} Q_{31}}{Q_{12} Q_{31} + Q_{12} Q_{32} + Q_{32} Q_{13}} + \frac{Q_{23}}{Q_{31} + Q_{32}}$$

$$\frac{N_3}{N_2} \simeq \frac{Q_{31} \frac{g_3}{g_1} e^{-\frac{h\nu_{31}}{k_B T_e}}}{Q_{31} + Q_{32}} \frac{Q_{21} Q_{31} + Q_{21} Q_{32} + Q_{32} \frac{g_3}{g_2} e^{-\frac{h\nu_{32}}{k_B T_e}} Q_{31}}{Q_{21} \frac{g_2}{g_1} e^{-\frac{h\nu_{21}}{k_B T_e}} Q_{31} + Q_{21} \frac{g_2}{g_1} e^{-\frac{h\nu_{21}}{k_B T_e}} Q_{32} + Q_{32} Q_{31} \frac{g_3}{g_1} e^{-\frac{h\nu_{31}}{k_B T_e}}} + \frac{Q_{32} \frac{g_3}{g_2} e^{-\frac{h\nu_{32}}{k_B T_e}}}{Q_{31} + Q_{32}}$$

Now, reminding that $h\nu_{31} \simeq h\nu_{21}$ and then $h\nu_{32} \simeq 0$, we obtain:

$$\frac{N_3}{N_2} \simeq \frac{Q_{31} \frac{g_3}{g_1}}{Q_{31} + Q_{32}} \frac{Q_{21} Q_{31} + Q_{21} Q_{32} + Q_{32} \frac{g_3}{g_2} Q_{31}}{Q_{21} \frac{g_2}{g_1} Q_{31} + Q_{21} \frac{g_2}{g_1} Q_{32} + Q_{32} Q_{31} \frac{g_3}{g_1}} + \frac{Q_{32} \frac{g_3}{g_2}}{Q_{31} + Q_{32}}$$

Since $g_1 = 2 \cdot \frac{3}{2} + 1 = 4 = g_2$,

$$\frac{N_3}{N_2} \simeq \frac{g_3}{g_2}$$

And finally,

$$R = \frac{I_{31}}{I_{21}} \simeq \frac{A_{31}}{A_{21}} \frac{g_3}{g_2}$$

$$R_{[\text{SII}]} = \frac{I_{6716}}{I_{6731}} \simeq 0.44$$

In conclusion, it is easy to see that when $N_e \leq 10^2$ radiative transitions dominate, while when $N_e \geq 10^4$ collisions dominate (Fig. 5.10). In both cases, the [S II] ratio assumes a constant value preventing to give a precise electron density estimate.

This is a list of emission line ratios used to estimate the electron density: [N I] $\lambda 5198/\lambda 5200$, [O II] $\lambda 3726/\lambda 3729$, [S II] $\lambda 6716/\lambda 6731$, [Cl III] $\lambda 5517/\lambda 5537$, [Ar IV] $\lambda 4711/\lambda 4740$.

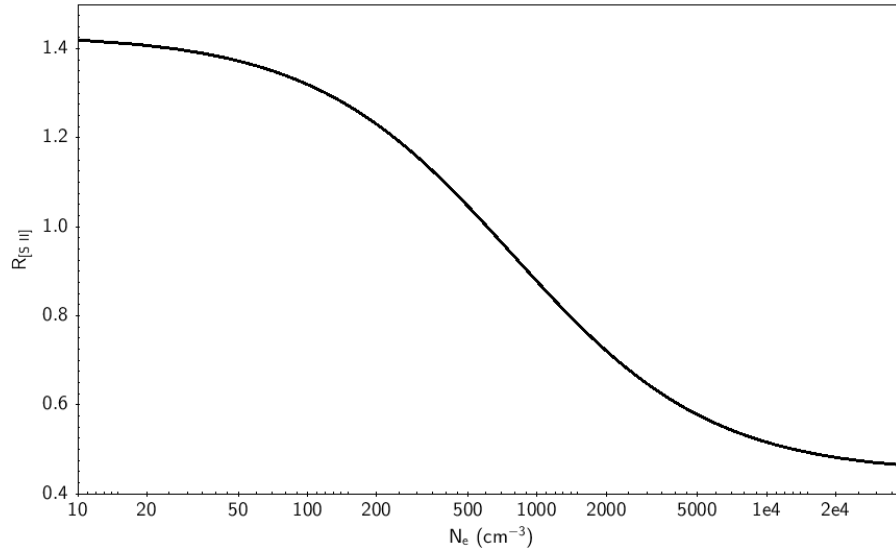


Figure 5.10: More precise calculations made with the IRAF task TEMDEN show slightly different limits of the [S II] emission line ratio. This curve is useful for estimating electron density when $10^2 < N_e < 10^4$.

6

Continuum sources

6.1 Recombination continuum

The **recombination** phenomenon, also called **free-bound** transition (*fb*), is the contrary of the photoionization (**bound-free** transition): a free electron in a plasma is passing at velocity v close to an ion, whose electric field is able to catch it, therefore becoming a neutral or a less ionized atom. The electron changes its energy status and the consequence is the emission of a photon. Since free electrons have all possible kinetic energies, whose distribution depends on temperature, the emitted photons form a continuum spectrum.

In addition, in nebular physical conditions, recombination is the mechanism at the base of the spectral permitted lines, since it allows to populate the high levels of the atoms.

First of all, we consider the photoionization mechanism (*bf*). The probability that a photon of frequency ν is absorbed by an atom or an ion having an electron at level n is given by the **photoionization cross-section**:

$$\sigma_{bf} \simeq \frac{64\pi^4}{3\sqrt{3}} \frac{m_e e^{10} Z^4}{(4\pi\epsilon_0)^5 c h^6} \frac{1}{n^5} \frac{1}{\nu^3} \simeq 2.81 \cdot 10^{25} \frac{Z^4}{n^5 \nu^3} \quad [\text{m}^2]$$

or $2.81 \cdot 10^{29} \dots \text{cm}^2$.

The absorption coefficient for **photoionization** is:

$$k_{\nu}^{bf} = \sigma_{bf} N_{on} g_{nf} \simeq 3 \cdot 10^{25} \frac{Z^4}{n^5 \nu^3} N_{0n} \quad [\text{m}^{-1}]$$

or $3 \cdot 10^{29} \dots \text{cm}^{-1}$. N_{0n} is the number density of atoms populated at level n and g_{nf} is the Gaunt factor for photoionization, a term close to 1. Therefore, the photoionization is favourite in case of atoms populated at low levels and photons with relatively low energies, which must be however larger than the ionization potential χ_n . In any case, since photoionization produces free electrons having a kinetic energy, it is a **heating mechanism**.

The emission coefficient for **recombination** is given by:

$$\varepsilon_{\nu}^{fb} d\nu = \frac{h\nu}{4\pi} N_e \int f(v) dv Q_{fb}(v) N_1$$

CHAPTER 6. CONTINUUM SOURCES

where $h\nu$ is the emitted energy, $f(v)dv$ the velocity distribution of the free electrons, $Q_{fb}(v)$ the collisional rate, N_e and N_1 the electron and ion densities. As function $f(v)$ we assume the Maxwell-Boltzmann distribution, and we have already seen that $Q_{fb}(v) = v\sigma_{fb}(v)$, where σ_{fb} is the **recombination cross-section**. In addition,

$$\frac{1}{2}m_e v^2 = h\nu - \chi_n \Rightarrow m_e v dv = h d\nu$$

Therefore:

$$\begin{aligned}\varepsilon_\nu^{fb} d\nu &= \frac{h\nu}{4\pi} N_e 4\pi \left(\frac{m_e}{2\pi k_B T_e} \right)^{3/2} v^2 e^{-\frac{m_e v^2}{2k_B T_e}} \frac{h d\nu}{m_e v} v \sigma_{fb} N_1 \\ \varepsilon_\nu^{fb} &= \frac{h^2 \nu m_e^{1/2}}{(2\pi k_B T_e)^{3/2}} v^2 \sigma_{fb} e^{-\frac{h\nu - \chi_n}{k_B T_e}} N_e N_1\end{aligned}$$

We know that $\varepsilon_\nu^{fb} = k_\nu^{bf} B_\nu(T)$, and assuming the Wien approximation:

$$\begin{aligned}\frac{h^2 \nu m_e^{1/2}}{(2\pi k_B T_e)^{3/2}} v^2 \sigma_{fb} e^{-\frac{m_e v^2}{2k_B T_e}} N_e N_1 &= k_\nu^{bf} \frac{2h\nu^3}{c^2} e^{-\frac{h\nu}{k_B T_e}} \\ \sigma_{fb} &= \frac{2\nu^2 (2\pi k_B T_e)^{3/2}}{c^2 h m_e^{1/2}} \frac{1}{v^2} \frac{e^{-\frac{\chi_n}{k_B T_e}}}{N_e N_1} k_\nu^{bf}\end{aligned}$$

Now we apply the Saha equation:

$$\frac{N_{11}}{N_{0n}} N_e = 2 \frac{g_{11}}{g_{0n}} \frac{(2\pi m_e k_B T_e)^{3/2}}{h^3} e^{-\frac{\chi_n}{k_B T_e}}$$

where $N_{11} = N_1$, $g_{11} = 1$, and $g_{0n} = 2n^2$ in case of hydrogen or hydrogenoid ions.

$$\begin{aligned}\sigma_{fb} &= \frac{2\nu^2 (2\pi k_B T_e)^{3/2}}{c^2 h m_e^{1/2}} \frac{1}{v^2} e^{-\frac{\chi_n}{k_B T_e}} 2n^2 \frac{h^3}{(2\pi m_e k_B T_e)^{3/2}} e^{\frac{\chi_n}{k_B T_e}} \frac{k_\nu^{bf}}{N_{0n}} \\ \sigma_{fb} &= \frac{2h^2 \nu^2}{c^2 m_e^2} \frac{n^2}{v^2} \frac{k_\nu^{bf}}{N_{0n}}\end{aligned}$$

and substituting the absorption coefficient:

$$\sigma_{fb} = \frac{128\pi^4}{3\sqrt{3}} \frac{e^{10} Z^4}{(4\pi\epsilon_0)^5 c^3 m_e h^4} \frac{1}{v^2} \frac{1}{n^3} \frac{1}{\nu} g_{nf} \simeq 3.3 \cdot 10^2 \frac{Z^4}{v^2 n^3 \nu} g_{nf} \quad [\text{m}^2]$$

or $3.3 \cdot 10^{10} \dots \text{cm}^2$.

In conclusion, recombination is more effective in case of low velocity (and low kinetic energy) electrons, and at lower levels. This result has an important consequence: even in case of ionization equilibrium, that is in case of the same number of photoionizations and recombinations, the lower energy electrons are those more likely to recombine and the final energy balance is a clear increase of gas temperature.

Now we can calculate the emission coefficient for recombination (assuming $Z = 1$):

6.1. RECOMBINATION CONTINUUM

$$\varepsilon_\nu^{fb} = \frac{h^2 \nu m_e^{1/2}}{(2\pi k_B T_e)^{3/2}} v^2 e^{-\frac{h\nu - \chi_n}{k_B T_e}} N_e N_1 \cdot \frac{128\pi^4}{3\sqrt{3}} \frac{e^{10} Z^4}{(4\pi\epsilon_0)^5 c^3 m_e h^4} \frac{1}{v^2} \frac{1}{n^3} \frac{1}{\nu} g_{nf}$$

$$\varepsilon_\nu^{fb} = \frac{128\pi^4 e^{10} m_e}{(6\pi m_e k)^{3/2} (4\pi\epsilon_0)^5 c^3 h^2} \frac{N_e N_1}{T_e^{3/2}} \frac{e^{-\frac{h\nu - \chi_n}{k_B T_e}}}{n^3} g_{nf}$$

$$\varepsilon_\nu^{fb} \simeq 1.7 \cdot 10^{-46} \frac{N_e N_1}{T_e^{3/2}} \frac{e^{-\frac{h\nu - \chi_n}{k_B T_e}}}{n^3} g_{nf} \quad [\text{J m}^{-3} \text{s}^{-1} \text{Hz}^{-1} \text{sr}^{-1}]$$

or $1.7 \cdot 10^{-33} \dots \text{erg cm}^{-3} \text{s}^{-1} \text{Hz}^{-1} \text{sr}^{-1}$.

Let us consider now the hydrogen atom. We know that the ionization potential of the level n is given by $\chi_n = \frac{13.6}{n^2} \text{ eV}$. The following table lists the heads of the Lyman, Balmer and Paschen series:

n	χ_n (eV)	ν_n (Hz)	λ_n (Å)	series
1	13.6	$3.29 \cdot 10^{15}$	912	Lyman
2	3.4	$8.22 \cdot 10^{14}$	3648	Balmer
3	1.5	$3.66 \cdot 10^{14}$	8208	Paschen

For a gas at a given temperature $T_e = 10^4 \text{ K}$ the quantity $k_B T_e \simeq 0.8614 \text{ eV}$. When $n = 1$:

$$\varepsilon_\nu^{fb} \simeq 1.21 \cdot 10^{-32} N_e N_p e^{-1.16h\nu} g_{nf} \quad [\text{erg cm}^{-3} \text{s}^{-1} \text{Hz}^{-1} \text{sr}^{-1}]$$

which is valid for $\nu \geq 3.29 \cdot 10^{15} \text{ Hz}$.

When $n = 2$:

$$\varepsilon_\nu^{fb} \simeq 1.1 \cdot 10^{-38} N_e N_p e^{-1.16h\nu} g_{nf} \quad [\text{erg cm}^{-3} \text{s}^{-1} \text{Hz}^{-1} \text{sr}^{-1}]$$

which is valid for $\nu \geq 8.22 \cdot 10^{14} \text{ Hz}$.

When $n = 3$:

$$\varepsilon_\nu^{fb} \simeq 3.6 \cdot 10^{-40} N_e N_p e^{-1.16h\nu} g_{nf} \quad [\text{erg cm}^{-3} \text{s}^{-1} \text{Hz}^{-1} \text{sr}^{-1}]$$

which is valid for $\nu \geq 3.66 \cdot 10^{14} \text{ Hz}$.

Fig. 6.1 shows the Lyman, Balmer and Paschen continua. It is worth to notice that the Paschen recombination continuum fall in the visible wavelength range.

Now we integrate the emission coefficient from the head of series (ν_n) to infinite:

$$\varepsilon_n^{fb} = 4\pi \int_{\nu_n}^{\infty} \varepsilon_\nu^{fb} d\nu \simeq \frac{2.14 \cdot 10^{-32}}{T_e^{3/2} n^3} \bar{g}_{nf} N_e N_p \int_{\nu_n}^{\infty} e^{-\frac{h\nu - \chi_n}{k_B T_e}} d\nu \quad [\text{erg cm}^{-3} \text{s}^{-1}]$$

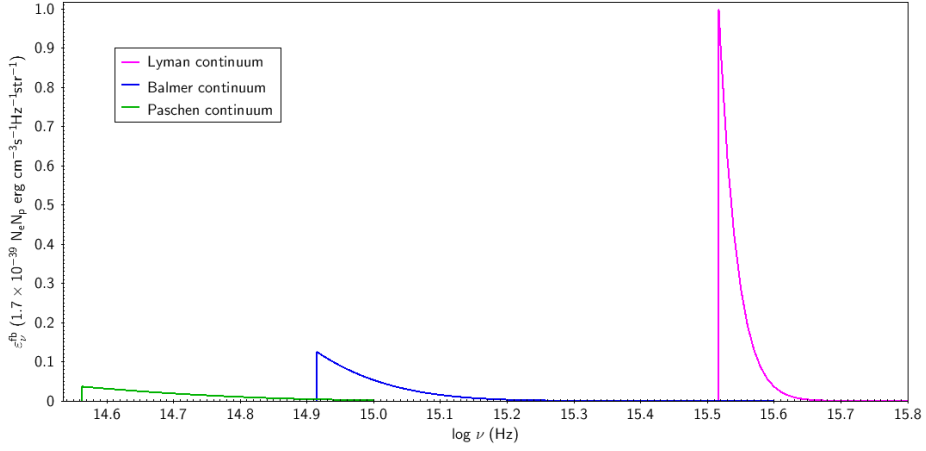


Figure 6.1: The recombination continuum for the three sequences of the hydrogen atom. A temperature $T_e = 10^4$ K is assumed.

where $\bar{g}_{nf}$ is the Gaunt factor averaged over the frequencies interval. Since $\chi_n = h\nu_n$, the solution of the integral is $\frac{k_B T_e}{h}$, therefore:

$$\epsilon_n^{fb} \simeq \frac{4.5 \cdot 10^{-22} \bar{g}_{nf}}{T_e^{1/2} n^3} N_e N_p \quad [\text{erg cm}^{-3} \text{ s}^{-1}]$$

6.2 Free-free continuum

The free-free emission is also called **braking radiation** or **thermal bremsstrahlung**. It refers to the energy emission from free electrons that collide against ions. The changing of their kinetic energy translates into radiation emission and since electron are free, all possible frequencies are permitted. This process produces a continuum spectrum.

The energy emitted by free electrons moving at velocity v in a uniform plasma at temperature T_e is given by:

$$E^{ff} = \frac{8\pi}{3\sqrt{3}} \frac{Z^2 e^6}{(4\pi\epsilon_0)^3 m_e^2 c^3} \frac{N_e N_i}{v} g_{ff}$$

where g_{ff} is the Gaunt factor that depends on velocity of the electron v and the frequency of radiation ν .

$$E^{ff} \approx 2.65 \cdot 10^{-48} Z^2 \frac{N_e N_i}{v} g_{ff} \quad [\text{J m}^{-3} \text{ s}^{-1} \text{ Hz}^{-1}]$$

In cgs units: $E^{ff} \approx 2.65 \cdot 10^{-33} Z^2 \dots \text{erg cm}^{-3} \text{ s}^{-1} \text{ Hz}^{-1}$.

In order to calculate the free-free emission coefficient ϵ_{ν}^{ff} we assume a Maxwell-Boltzmann distribution for velocities and we integrate:

6.2. FREE-FREE CONTINUUM

$$\varepsilon_\nu^{ff} = \int_{v_{min}}^{\infty} E^{ff} \cdot 4\pi \left(\frac{m_e}{2\pi k_B T_e} \right)^{3/2} v^2 e^{-\frac{m_e v^2}{2k_B T_e}} dv$$

where v_{min} is due to the fact that:

$$\frac{m_e v^2}{2} \geq h\nu \Rightarrow v_{min} = \sqrt{\frac{2h\nu}{m_e}}$$

Therefore:

$$\varepsilon_\nu^{ff} = \frac{8\pi}{3\sqrt{3}} \frac{Z^2 e^6}{(4\pi\varepsilon_0)^3 m_e^2 c^3} 4\pi \left(\frac{m_e}{2\pi k_B T_e} \right)^{3/2} N_e N_i g_{ff} \int_{v_{min}}^{\infty} v e^{-\frac{m_e v^2}{2k_B T_e}} dv$$

$$\varepsilon_\nu^{ff} = \frac{8\pi}{3\sqrt{3}} \frac{Z^2 e^6}{(4\pi\varepsilon_0)^3 m_e^2 c^3} 4\pi \left(\frac{m_e}{2\pi k_B T_e} \right)^{3/2} N_e N_i g_{ff} \frac{k_B T_e}{m_e} e^{-\frac{h\nu}{k_B T_e}}$$

$$\varepsilon_\nu^{ff} = \frac{8}{3} \sqrt{\frac{2\pi}{3}} \frac{Z^2 e^6}{(4\pi\varepsilon_0)^3 m_e^{3/2} c^3 (k_B T_e)^{1/2}} N_e N_i g_{ff} e^{-\frac{h\nu}{k_B T_e}}$$

$$\varepsilon_\nu^{ff} \simeq 5.44 \cdot 10^{-52} Z^2 \frac{N_e N_i}{T_e^{1/2}} g_{ff} e^{-\frac{h\nu}{k_B T_e}} \quad [\text{J m}^{-3} \text{ s}^{-1} \text{ Hz}^{-1}]$$

In cgs units, $\varepsilon_\nu^{ff} \simeq 5.44 \cdot 10^{-39} Z^2 \dots [\text{erg cm}^{-3} \text{ s}^{-1} \text{ Hz}^{-1}]$.

According to the formula, the free-free emission coefficient depends on the squared electron density and the inverse square root of temperature. But the exponential term makes ε_ν^{ff} increasing with temperature, at least in a certain range of values (Fig. 6.2). For larger frequencies the ratio between the exponential term and the square root of temperature moves to higher temperature values and to slightly lower ε_ν^{ff} . Viceversa for lower frequencies. Therefore, we can affirm that free-free emission is strong in radio wavelengths at low temperatures, and in optical/UV wavelengths at high temperatures.

When we integrate over all frequencies, we obtain the total free-free emission coefficient ε^{ff} :

$$\varepsilon^{ff} = 4\pi \int_0^\infty \varepsilon_\nu^{ff} d\nu \simeq 4\pi \cdot 5.44 \cdot 10^{-52} Z^2 \frac{N_e N_i}{T_e^{1/2}} \bar{g}_{ff} \int_0^\infty e^{-\frac{h\nu}{k_B T_e}} d\nu$$

$$\varepsilon^{ff} \simeq 4\pi \cdot 5.44 \cdot 10^{-52} Z^2 \frac{N_e N_i}{T_e^{1/2}} \bar{g}_{ff} \frac{k_B T_e}{h}$$

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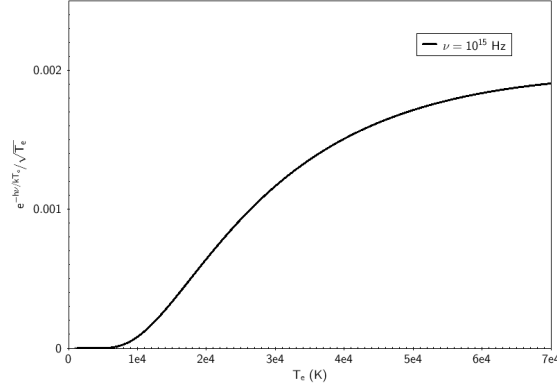


Figure 6.2: The ratio between the exponential term and the square root of temperature for $\nu = 10^{15}$ Hz (visible/UV domain).

$$\varepsilon^{ff} \simeq 1.426 \cdot 10^{-40} Z^2 N_e N_i T_e^{1/2} \bar{g}_{ff} \quad [\text{J m}^{-3} \text{ s}^{-1}]$$

where $\bar{g}_{ff}$ is the Gaunt factor averaged over all frequencies. It is a value around 1.2-1.5 for temperatures around $T = 10^4 \text{ K}$. This formula in cgs units is: $\varepsilon^{ff} \simeq 1.426 \cdot 10^{-27} Z^2 \dots \text{erg cm}^{-3} \text{ s}^{-1}$.

We can compare free-free continuum with recombination continuum in the case of a gas made only by hydrogen (and $n = 1$). The free-free continuum emission is stronger than the recombination continuum emission when:

$$1.426 \cdot 10^{-27} N_e N_p T_e^{1/2} \bar{g}_{ff} > \frac{4.5 \cdot 10^{-22} \bar{g}_{nf}}{T_e^{1/2}} N_e N_p$$

that is:

$$T_e > \frac{4.5 \cdot 10^{-22} \bar{g}_{nf}}{1.426 \cdot 10^{-27} \bar{g}_{ff}} = 3.15 \cdot 10^5 \frac{\bar{g}_{nf}}{\bar{g}_{ff}} \text{ K}$$

Since the electron temperature of nebulae is usually around 10^4 K , we can affirm that in the visible-UV range, the recombination continuum dominates over the free-free continuum.

When an electron passes close to an ion, the two particles system is also able to absorb energy at any frequency. The more free electrons are available, the higher is the probability to cause free-free radiation absorption. In addition, since the emission coefficient depends on the inverse square root of temperature, the higher is the temperature, the shorter is the time interval during which electron and ion are close together. Since free-free is a pure “collisional” phenomenon, we are in LTE, so we can write:

$$\varepsilon_\nu^{ff} = k_\nu^{ff} B_\nu(T_e)$$

6.2. FREE-FREE CONTINUUM

and the free-free absorption coefficient is:

$$k_{\nu}^{ff} = \frac{8}{3} \sqrt{\frac{2\pi}{3}} \frac{Z^2 e^6}{(4\pi\epsilon_0)^3 m_e^{3/2} c^3 (k_B T_e)^{1/2}} N_e N_i g_{ff} e^{-\frac{h\nu}{k_B T_e}} \cdot \frac{c^2}{2h\nu^3} \left(e^{\frac{h\nu}{k_B T_e}} - 1 \right)$$

$$k_{\nu}^{ff} = \frac{4}{3} \sqrt{\frac{2\pi}{3}} \frac{Z^2 e^6}{(4\pi\epsilon_0)^3 m_e^{3/2} c (k_B T_e)^{1/2} h} \frac{N_e N_i}{\nu^3} g_{ff} \left(1 - e^{-\frac{h\nu}{k_B T_e}} \right)$$

In the visible/UV regime $h\nu > k_B T_e$, so that $1 - e^{-\frac{h\nu}{k_B T_e}} \approx 1$ and the absorption coefficient becomes:

$$k_{\nu}^{ff} = \frac{4}{3} \sqrt{\frac{2\pi}{3}} \frac{Z^2 e^6}{(4\pi\epsilon_0)^3 m_e^{3/2} c (k_B T_e)^{1/2} h} \frac{N_e N_i}{\nu^3} g_{ff} \simeq 3.69 \cdot 10^{-2} Z^2 \frac{N_e N_i}{\sqrt{T_e}} \frac{1}{\nu^3} g_{ff} \quad [\text{m}^{-1}]$$

or $k_{\nu}^{ff} \sim 3.69 \cdot 10^8 \dots \text{cm}^{-1}$ in cgs units.

In the radio wavelength domain $h\nu \ll k_B T_e$, so that $1 - e^{-\frac{h\nu}{k_B T_e}} \approx \frac{h\nu}{k_B T_e}$ and the absorption coefficient becomes:

$$k_{\nu}^{ff} = \frac{4}{3} \sqrt{\frac{2\pi}{3}} \frac{Z^2 e^6}{(4\pi\epsilon_0)^3 (m_e k_B T_e)^{3/2} c} \frac{N_e N_i}{\nu^2} g_{ff} \simeq 1.77 \cdot 10^{-12} \frac{N_e N_i}{T_e^{3/2}} \frac{Z^2}{\nu^2} g_{ff} \quad [\text{m}^{-1}]$$

The constant in cgs units is: $1.77 \cdot 10^{-2}$ and k_{ν}^{ff} is expressed in cm^{-1} . In general, we can affirm that the free-free absorption coefficient is proportional to $T_e^{-1.5} \nu^{-2}$, that is it becomes significant at low frequencies and low temperatures.

The Gaunf factor is given by:

$$g_{ff} = 0.5513 \ln \left(\frac{T_e^{3/2}}{Z\nu} \right) + 9.75$$

Assuming $Z = 1$ (hydrogenoid ions) and radio wavelengths, it can be re-written as:

$$g_{ff} = A_0 T_e^a \nu^b \simeq 11.9 T_e^{0.15} \nu^{-0.1}$$

This approximation depends on temperature. Fig. 6.3 shows that when $T_e \approx 10^4$ K the two formulas are similar for $\nu \approx 10^9 - 10^{9.5}$ Hz.

Therefore, the absorption coefficient becomes:

$$k_{\nu}^{ff} \simeq 2.1 \cdot 10^{-11} N_e N_i T_e^{-1.35} \nu^{-2.1} \quad [\text{m}^{-1}]$$

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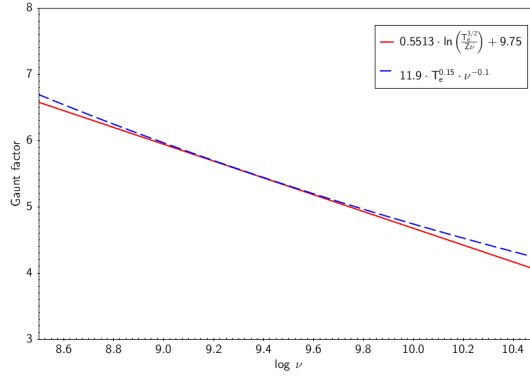


Figure 6.3: The comparison between the Gaunt formula for radio wavelengths and its approximation in case of $T_e \approx 10^4$ K.

Again $k_\nu^{ff} \simeq 0.21 N_e N_i \dots [\text{cm}^{-1}]$.

Now we assume to have a nebula, whose thickness along our line-of-sight is ℓ . We can calculate the optical depth:

$$\tau_\nu^{ff} = \int_0^\ell k_\nu^{ff} dr = 2.1 \cdot 10^{-11} T_e^{-1.35} \nu^{-2.1} \int_0^\ell N_e N_i dr = 2.1 \cdot 10^{-11} T_e^{-1.35} \nu^{-2.1} EM$$

where EM is the emission measure, that is usually expressed in terms of cm^{-6} pc. Therefore we need to use the cgs units:

$$\tau_\nu^{ff} = 0.21 T_e^{-1.35} \nu^{-2.1} \int_0^\ell N_e N_i dr \simeq 0.0824 T_e^{-1.35} \nu_{\text{GHz}}^{-2.1} EM$$

where the frequency is in GHz units.

Assuming a uniform density across the nebula, we can calculate the frequency for which $\tau_\nu^{ff} = 1$:

$$\nu^{2.1} = 0.0824 T_e^{-1.35} N_e^2 \ell$$

If $N_e = 10^2 \text{ cm}^{-3}$, $T_e = 10^4 \text{ K}$, and $\ell = 10 \text{ pc}$, the nebula is optically thick at frequencies $\nu < 0.2 \text{ GHz}$ ($\lambda \simeq 1.5 \text{ m}$), while if $N_e = 10^3 \text{ cm}^{-3}$, $T_e = 10^4 \text{ K}$, and $\ell = 1 \text{ pc}$ this happens at $\nu < 0.6 \text{ GHz}$ ($\lambda \simeq 0.5 \text{ m}$). In any case we can affirm that a gaseous cloud is optically thick to free-free emission at frequencies of some 10^2 MHz .

Using the radiative transport equation, we can evaluate how the **radio-thermal continuum** changes. We have already seen that the equation:

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$$\frac{dI_\nu}{d\tau_\nu} = I_\nu - \frac{\varepsilon_\nu}{k_\nu}$$

has the following solution:

$$I_\nu = S_\nu(1 - e^{-\tau_\nu}) + I_{\nu_0}e^{-\tau_\nu}$$

In our case $I_{\nu_0} = 0$ and the source function S_ν is equal to the Planck function in Rayleigh-Jeans approximation. Therefore, when $\tau_\nu < 1$:

$$I_\nu^{ff} = \tau_\nu B_\nu(T_e) = 0.0824 T_e^{-1.35} \nu_{GHz}^{-2.1} EM \cdot 2 \frac{\nu^2}{c^2} k_B T_e$$

$$I_\nu^{ff} \simeq 2.27 \cdot 10^{-19} T_e^{-0.35} \nu_{GHz}^{-0.1} EM \quad [\text{erg cm}^{-2} \text{ s}^{-1} \text{ Hz}^{-2}]$$

that is, when the cloud is optically thin: $I_\nu^{ff} \propto T_e^{-0.35} \nu^{-0.1}$.

Viceversa, when $\tau_\nu > 1$:

$$I_\nu^{ff} = B_\nu(T_e) = 2 \frac{\nu^2}{c^2} k_B T_e \simeq 3.1 \cdot 10^{-19} T_e \nu_{GHz}^2 \quad [\text{erg cm}^{-2} \text{ s}^{-1} \text{ Hz}^{-2}]$$

that is, when the cloud is optically thick: $I_\nu^{ff} \propto T_e \nu^2$.

Fig. 6.4 shows shape of the radio-thermal continuum and how it depends on the EM or on the electron density, when the size of the cloud is known, or estimated. Radio data at frequencies lower than 1 GHz (optically thick cloud) can be used to determine the electron temperature of the nebula, while radio data at frequencies higher than 1 GHz (optically thin cloud) can be used to derive an estimate of the electron density.

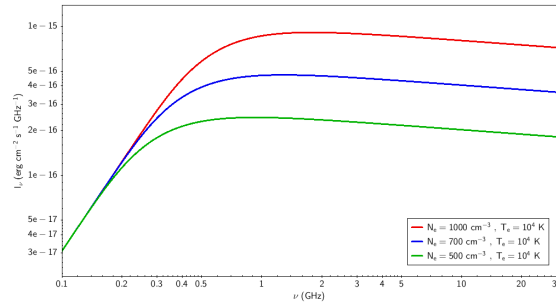


Figure 6.4: The radio-thermal continuum for a nebula with thickness $\ell = 1$ pc. Increasing values of density correspond to increasing values of the radio flux.

6.3 Synchrotron continuum

The synchrotron radiation is the effect of the motion of relativistic electrons through a magnetic field. The intensity of magnetic field is measured in Gauss (G) or in Tesla (T), and $1 \text{ G} = 10^{-4} \text{ T}$. As a unit, 1 T corresponds to $1 \text{ N s C}^{-1} \text{ m}^{-1}$.

The following table shows typical values:

Interstellar medium	$\sim 10^{-6} \text{ G}$
Crab nebula	$\sim 10^{-3} \text{ G}$
Earth Pole	0.5 G
Laboratory	$< 10^6 \text{ G}$
White Dwarf	$< 10^9 \text{ G}$
Neutron star	$< 10^{15} \text{ G}$

If the electron has velocity $\vec{v}$ and energy $\mathcal{E} = \gamma m_e c^2$, where γ is the Lorentz factor, the magnetic field $\vec{B}$ causes the Lorentz force:

$$\vec{F} = e \vec{v} \wedge \vec{B}$$

which makes the electron spiralling along the magnetic field lines (if $\vec{v}$ has a component along $\vec{B}$) and emitting energy. The angular frequency of this circular motion is given by the so called **gyrofrequency**:

$$\nu_G = \frac{eB \sin \theta}{2\pi\gamma m_e}$$

The radiation is emitted within a cone having the $\vec{v}$ vector as axis and aperture $\Delta\theta = \frac{1}{\gamma}$, which means that the higher the electron velocity, the narrower the cone of emission.

Each single electron emits a spectrum:

$$P_\nu d\nu = \frac{1}{4\pi\epsilon_0} \frac{\sqrt{3} e^3 B \sin \theta}{2\pi m_e c} F\left(\frac{\nu}{\nu_c}\right) d\nu$$

where ν_c is called critical frequency and it is equal to:

$$\nu_c = \frac{1}{2\pi} \frac{3\gamma^2 e B \sin \theta}{2m_e} \propto \mathcal{E}^2 B_\perp$$

while the function F is given by:

$$F\left(\frac{\nu}{\nu_c}\right) = \frac{\nu}{\nu_c} \int_{\nu/\nu_c}^{\infty} K_{5/3}(\eta) d\eta$$

The $K_{5/3}$ function is called modified Bessel function of the second kind (Fig. 6.5). At frequencies $\nu \ll \nu_c$, $F \propto \left(\frac{\nu}{\nu_c}\right)^{1/3}$, while at frequencies $\nu \gg \nu_c$, $F \propto \left(\frac{\nu}{\nu_c}\right)^{1/2} e^{-\frac{\nu}{\nu_c}}$. The general behavior of this function is given in Fig. 6.6: there is a narrow peak of emission ($F = 0.92$) at $\nu = 0.29\nu_c$.

6.3. SYNCHROTRON CONTINUUM

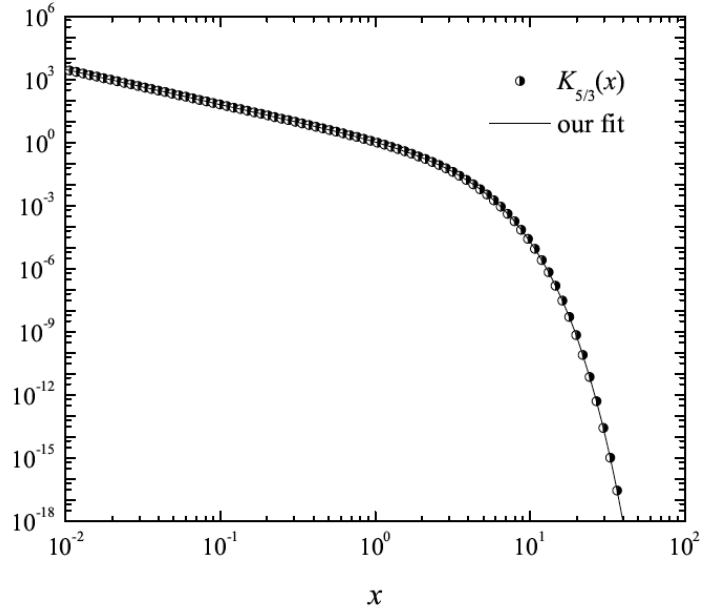


Figure 6.5: The modified Bessel function of the second kind calculated and fitted by Fouka & Ouichaoui (2013, their Fig.1).

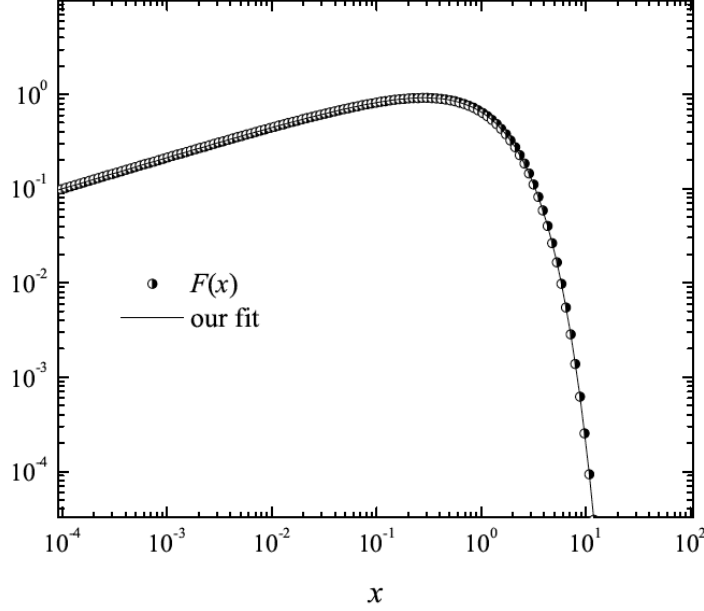


Figure 6.6: The synchrotron function calculated and fitted by Fouka & Ouichaoui (2013, their Fig.5) and plotted in logarithmic scale. $x = \nu/\nu_c$.

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Now if we consider a distribution of electrons per unit volume $N_e(\mathcal{E})d\mathcal{E}$, the emitted synchrotron spectrum is the sum of the spectra emitted by each single electron:

$$\varepsilon_\nu^{syn} = \frac{1}{4\pi} \int_0^\infty P_\nu N_e(\mathcal{E}) d\mathcal{E} \quad [\text{erg cm}^{-3} \text{s}^{-1} \text{Hz}^{-1} \text{sr}^{-1}]$$

In analogy with cosmic radiation, we assume that the distribution follows a power law:

$$N_e(\mathcal{E})d\mathcal{E} = k\mathcal{E}^{-s}d\mathcal{E}$$

where s is called **power index**.

Then we approximate the single electron spectrum to a Dirac function at $\nu = \nu_c$: $P_\nu \approx P\delta(\nu - \nu_c)$, and finally we avoid to have an infinite number of electrons by integrating between a lower limit $\mathcal{E}_1$ and an upper limit $\mathcal{E}_2$.

$$\begin{aligned} \varepsilon_\nu^{syn} &= \frac{1}{4\pi} \int_{\mathcal{E}_1}^{\mathcal{E}_2} \frac{\sqrt{3}}{8\pi^2 \varepsilon_0} \frac{e^3 B_\perp}{m_e c} F\left(\frac{\nu}{\nu_c}\right) \delta(\nu - \nu_c) k \mathcal{E}^{-s} d\mathcal{E} \\ \varepsilon_\nu^{syn} &= \frac{1}{4\pi} \frac{\sqrt{3}}{8\pi^2 \varepsilon_0} \frac{e^3 B_\perp}{m_e c} F\left(\frac{\nu}{\nu_c}\right) \delta(\nu - \nu_c) k \int_{\mathcal{E}_1}^{\mathcal{E}_2} \mathcal{E}^{-s} d\mathcal{E} \\ \varepsilon_\nu^{syn} &= \frac{1}{4\pi} \frac{\sqrt{3}}{8\pi^2 \varepsilon_0} \frac{e^3 B_\perp}{m_e c} F\left(\frac{\nu}{\nu_c}\right) \delta(\nu - \nu_c) k \frac{\mathcal{E}_2^{-s+1} - \mathcal{E}_1^{-s+1}}{1-s} \end{aligned}$$

In general we can affirm that:

$$\varepsilon_\nu^{syn} \propto B \mathcal{E}^{-s+1}$$

The assumption $\nu = \nu_c$ implies that $\mathcal{E} \propto \nu^{1/2} B^{-1/2}$. Therefore:

$$\varepsilon_\nu^{syn} \propto B (\nu^{1/2} B^{-1/2})^{-s+1} \Rightarrow \propto B^{\frac{s+1}{2}} \nu^{-\frac{s-1}{2}}$$

The complete formula of the emission coefficient is:

$$\varepsilon_\nu^{syn} = 0.93 \cdot 10^{-23} a(s) k B^{\frac{s+1}{2}} \left(\frac{6.26 \cdot 10^{18}}{\nu} \right)^{\frac{s-1}{2}} \quad [\text{erg cm}^{-3} \text{s}^{-1} \text{Hz}^{-1} \text{sr}^{-1}]$$

where:

$$a(s) = 2^{\frac{s-3}{2}} \frac{s + \frac{7}{3}}{s + 1} \Gamma\left(\frac{3s-1}{12}\right) \Gamma\left(\frac{3s+17}{12}\right)$$

When s varies from 1 to 5, this function $a(s)$ assumes values from 6 to 2.

If we call **spectral index** $\alpha = \frac{s-1}{2}$ and we integrate the emission coefficient along the size of the emitting gas, we obtain that the synchrotron intensity:

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$$I_\nu^{syn} \propto \nu^{-\alpha}$$

Typical value for the spectral index are: $\alpha = 0.5$ for supernova remnants (SNR), $\alpha = 0.5 - 2$ for extended radio sources, and $\alpha = 0.7$ for cosmic rays.

At radio wavelengths synchrotron radiation is observed also in absorption. This phenomenon is called **synchrotron self-absorption (SSA)**.

The absorption coefficient is given by the following formula:

$$k_\nu^{syn} = 0.019b(s)(3.5 \cdot 10^9)^s k_B^{\frac{s+2}{2}} \nu^{-\frac{s+4}{2}} \quad [\text{cm}^{-1}]$$

where k is again the constant of the electron energy distribution, while the function $b(s)$ is:

$$b(s) = \frac{\sqrt{3}}{16} \left(s + \frac{10}{3} \right) \Gamma \left(\frac{3s+2}{12} \right) \Gamma \left(\frac{3s+10}{12} \right)$$

and it goes from 0.955 to 0.83, when s goes from 1 to 5.

In radio we can assume the Rayleigh-Jeans approximation of the Planck function, $B_\nu \simeq 2 \frac{\nu^2}{c^2} k_B T_e$ (here k is the Boltzmann constant).

Since the energy of the electron is $E = \gamma m_e c^2 \approx k_B T_e$ and $\nu = \nu_c \propto \gamma^2$, we obtain that $T_e \propto \nu^{1/2}$. Therefore, we can write that $B_\nu \propto \nu^{5/2}$.

As we have already see, the solution of the transfer equation in the **optically thin** case is:

$$I_\nu = \tau_\nu B_\nu \propto \nu^{-\frac{s+4}{2}} \nu^{5/2} \propto \nu^{-\frac{s-1}{2}} = \nu^{-\alpha}$$

while in the **optically thick** case is:

$$I_\nu = B_\nu \propto \nu^{2.5}$$

7

Ionization equilibrium

In this chapter we explore what happens in a gaseous nebula when we assume that the number of ionizations is equal to the number of recombinations. The general equation for a single electron involved in both the processes is:

$$N_i R_{i,i+1} + N_i C_{i,i+1} = N_{i+1} R_{i+1,i} + N_{i+1} C_{i+1,i}$$

where $R_{i,i+1}$ is the rate of photoionizations (in units of s^{-1}), $C_{i,i+1}$ is the rate of collisional ionizations, $R_{i+1,i}$ is the rate of recombinations, and $C_{i+1,i}$ is the rate of collisional recombinations. N_i and N_{i+1} are the number density of atoms i times and $i + 1$ times ionized.

Since $C_{i+1,i}$ is three-body collisional process that requires high density, we can neglect it in nebular conditions. In case of a cloud of hydrogen, the equation becomes:

$$N_0(R_{0,1} + C_{0,1}) = N_1 R_{1,0}$$

7.1 Photoionization rate

We now evaluate the $R_{0,1}$ term.

To ionize hydrogen atoms, an energy $\mathcal{E} \geq \mathcal{E}_0 = 13.6 \text{ eV}$ is required (or photons with frequency $\nu \geq \nu_0 = 3.29 \cdot 10^{15} \text{ Hz}$).

$$N_0 R_{0,1} = 4\pi \int_{\nu_0}^{\infty} \frac{k_{\nu}^{bf}}{h\nu} I_{\nu} d\nu$$

We have already seen the bound-free absorption coefficient, that in case of hydrogen ($Z = 1$, $n = 1$) becomes:

$$k_{\nu}^{bf} \simeq \frac{3 \cdot 10^{29} Z^4}{\nu^3 n^5} N_{0n} \Rightarrow k_{\nu}^{bf} \simeq \frac{3 \cdot 10^{29}}{\nu^3} N_0 \quad [\text{cm}^{-1}]$$

In a common nebula photoionized by a hot star, the ionizing radiation is strongly diluted and the dilution factor is very small.

$$N_0 R_{0,1} = 4\pi \int_{\nu_0}^{\infty} \frac{3 \cdot 10^{29}}{\nu^3} N_0 w \frac{B_{\nu}(T)}{h\nu} d\nu$$

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Since we are in UV wavelength range, we can assume the Wien approximation:

$$R_{0,1} = 4\pi \int_{\nu_0}^{\infty} \frac{3 \cdot 10^{29}}{\nu^3} w \frac{1}{h\nu} \frac{2h\nu^3}{c^2} e^{-\frac{h\nu}{k_B T}} d\nu$$

$$R_{0,1} = \frac{24\pi \cdot 10^{29}}{c^2} w \int_{\nu_0}^{\infty} \frac{e^{-\frac{h\nu}{k_B T}}}{\nu} d\nu \simeq 8.4 \cdot 10^9 w \int_{x_0}^{\infty} \frac{e^{-x}}{x} dx$$

where $x_0 = \frac{h\nu_0}{k_B T}$. The integral has no analytic solution and it is called **exponential integral** $E_1(x_0)$.

By assuming a stellar temperature $T = 4 \cdot 10^4$ K, $x_0 \simeq 3.95$ and $E_1(3.95) \simeq 4 \cdot 10^{-3}$. Therefore:

$$R_{0,1} \simeq 8.4 \cdot 10^9 w \int_{x_0}^{\infty} \frac{e^{-x}}{x} dx \simeq 3.36 \cdot 10^7 w \approx 10^{-8} \text{ s}^{-1}$$

in case $w \approx 10^{-15}$.

This result indicates that each atom can experience 1 photoionization every 10^8 s, that is about every 3 yrs. As a comparison, $R_{0,1}$ in case of the Sun is around 10^{-19} s^{-1} , that is the Sun is not able to produce a sufficient number of ionizing photons to photoionize an extended nebula.

In addition to the assumed hot thermal source, we should take into account the presence of cosmic radiation (X-rays and α particles). Since $\sigma_{bf} \propto \frac{1}{\nu^3}$, high energy photons are very unlikely to ionize the nebula. Indeed, when $\mathcal{E} > 1$ keV (hard X-rays), $R_{0,1} \approx 0$, while $0.2 < \mathcal{E} < 1$ keV (soft X-rays), $R_{0,1} \approx 10^{-17} \text{ s}^{-1}$. Therefore, the cosmic radiation contribution is negligible when compared to a hot ionizing star.

7.2 Collisional ionization rate

In order to have an ionization through a collision ($C_{0,1}$), electrons must have enough kinetic energy. Let us now assume a temperature $T_e = 10^4$ K, and study the Maxwell-Boltzmann distribution as a function of the energy $\mathcal{E}$.

$$\varphi(v) = 4\pi \left(\frac{m_e}{2\pi k_B T_e} \right)^{3/2} v^2 e^{-\frac{m_e v^2}{2k_B T_e}} = \frac{8\pi m_e^{1/2}}{(2\pi k_B T_e)^{3/2}} \frac{m_e v^2}{2} e^{-\frac{m_e v^2}{2k_B T_e}}$$

$$\varphi(\mathcal{E}) = \frac{8\pi m_e^{1/2}}{(2\pi k_B T_e)^{3/2}} \mathcal{E} e^{-\frac{\mathcal{E}}{k_B T_e}} = 2\sqrt{\frac{2}{\pi}} m_e^{1/2} \frac{\mathcal{E}}{(k_B T_e)^{3/2}} e^{-\frac{\mathcal{E}}{k_B T_e}}$$

The function $f(\mathcal{E}) = \frac{\mathcal{E}}{(k_B T_e)^{3/2}} e^{-\frac{\mathcal{E}}{k_B T_e}}$ has a maximum at $\mathcal{E} = k_B T_e$, where $f(\mathcal{E}) = \frac{1}{e\sqrt{k_B T_e}}$. Fig. 7.1 shows that increasing the electron temperature, the

7.2. COLLISIONAL IONIZATION RATE

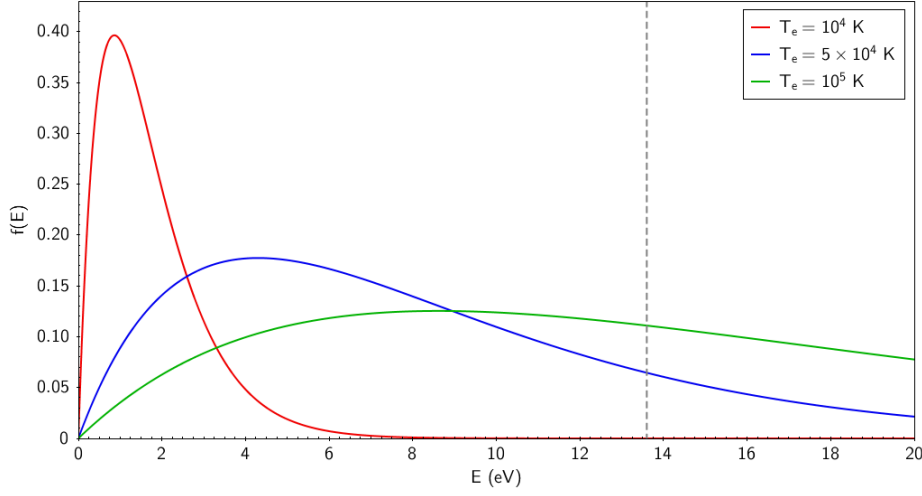


Figure 7.1: The $f(\mathcal{E})$ function for different electron temperatures. The dashed line shows the lower ionization energy limit for hydrogen.

maximum decreases but the amount of ionizing photons increases.

Now, integrating the function we obtain:

$$\int_{\mathcal{E}_0}^{\infty} \varphi(\mathcal{E}) d\mathcal{E} = 2\sqrt{\frac{2}{\pi}} \left(\frac{m_e}{k_B T_e} \right)^{1/2} \int_{\mathcal{E}_0}^{\infty} \frac{\mathcal{E}}{k_B T_e} e^{-\frac{\mathcal{E}}{k_B T_e}} d\mathcal{E}$$

$$\int_{\mathcal{E}_0}^{\infty} \varphi(\mathcal{E}) d\mathcal{E} = 2\sqrt{\frac{2}{\pi}} (m_e k_B T_e)^{1/2} \int_{x_0}^{\infty} x e^{-x} dx$$

where $x_0 = \frac{\mathcal{E}_0}{k_B T_e}$. The integral has no analytic solution and it is called **incomplete gamma function** $\Gamma(2, x_0)$. Therefore:

$$\int_{\mathcal{E}_0}^{\infty} \varphi(\mathcal{E}) d\mathcal{E} = 2\sqrt{\frac{2}{\pi}} (m_e k_B T_e)^{1/2} \Gamma(2, x_0)$$

Let us now consider the function $f(x_0, T_e) = (k_B T_e)^{1/2} \Gamma(2, x_0)$ and evaluate it for different electron temperatures:

T_e (K)	$k_B T_e$ (eV)	$f(x_0, T_e)$
10^4	0.86	$2.14 \cdot 10^{-6}$
$2 \cdot 10^4$	1.72	$4.3 \cdot 10^{-3}$
$4 \cdot 10^4$	3.44	0.17
10^5	8.61	1.56

The conclusion is that an electron temperature $T_e > 10^5$ K is mandatory to cause ionization through collision. This temperature is much larger of that typically observed in nebulae, therefore we can affirm that in nebular conditions,

CHAPTER 7. IONIZATION EQUILIBRIUM

$C_{0,1}$ can be neglected.

As in the previous section, the cosmic radiation should be taken into account. High energy α particles can collide with neutral atoms causing ionization. Calculations based on the Bethe cross-section (for $0.3 < \mathcal{E} < 100$ MeV) suggest that $C_{0,1} \approx 10^{-17} \text{ s}^{-1}$.

7.3 Recombination rate

The rate of recombinations ($R_{1,0}$) depends on the electron density and the recombination coefficient at any level n :

$$R_{i+1,i} = N_e \alpha_i = N_e \sum_{n=1}^{\infty} \alpha_{in}$$

where α_i and α_{in} are collisional terms (in units of $\text{cm}^3 \text{ s}^{-1}$) that depend on the electron velocity, the recombination cross section, and the velocity distribution (we assume a Maxwell-Boltzmann distribution as usual).

$$\alpha_{in} = \overline{v \sigma_{fb}} = \int_0^{\infty} v \sigma_{fb} \varphi(v) dv$$

In our case $i = 0$, therefore:

$$\alpha_{0n} = \int_0^{\infty} v \frac{128\pi^4 e^{10} Z^4}{3\sqrt{3} m_e c^3 h^4 (4\pi\epsilon_0)^5 \nu} \frac{1}{n^3} \frac{1}{v^2} g_{nf} 4\pi \left(\frac{m_e}{2\pi k_B T_e} \right)^{3/2} v^2 e^{-\frac{m_e v^2}{2k_B T_e}} dv$$

Since any recombination produces a photon:

$$h\nu = \frac{1}{2} m_e v^2 + \chi_n \Rightarrow h d\nu = m_e v dv$$

where χ_n is the ionization potential of level n . For $v = 0$ we have $h\nu_0 = \chi_n$.

$$\begin{aligned} \alpha_{0n} &= \frac{512\pi^5 e^{10} Z^4}{m_e^2 c^3 h^3 (4\pi\epsilon_0)^5} \left(\frac{m_e}{6\pi k_B T_e} \right)^{3/2} \frac{g_{nf}}{n^3} e^{\frac{\chi_n}{k_B T_e}} \int_{\nu_0}^{\infty} \frac{e^{-\frac{h\nu}{k_B T_e}}}{\nu} d\nu \\ \alpha_{0n} &= \frac{512\pi^5 e^{10} Z^4}{m_e^2 c^3 h^3 (4\pi\epsilon_0)^5} \left(\frac{m_e}{6\pi k_B T_e} \right)^{3/2} \frac{g_{nf}}{n^3} e^{\frac{\chi_n}{k_B T_e}} E_1(x_0) \end{aligned}$$

where $x_0 = \frac{h\nu_0}{k_B T_e} = \frac{\chi_n}{k_B T_e}$.

$$\alpha_{0n} \simeq 3.26 \cdot 10^{-6} \frac{Z^4}{T_e^{3/2} n^3} g_{nf} e^{\frac{\chi_n}{k_B T_e}} E_1 \left(\frac{\chi_n}{k_B T_e} \right)$$

7.3. RECOMBINATION RATE

When $x_0 \gg 1$ the exponential integral can be approximated by the following expression: $E_1(x_0) \approx \frac{e^{-x_0}}{x_0}$. The following table shows a comparison between the real value of the integral and of its approximation for different n values ($T_e = 10^4$ K is assumed).

n	x_0	$E_1(x_0)$	e^{-x_0}/x_0
1	15.8	$8.208 \cdot 10^{-9}$	$8.7 \cdot 10^{-9}$
2	3.95	$4.016 \cdot 10^{-3}$	$4.87 \cdot 10^{-3}$
3	1.76	$6.85 \cdot 10^{-2}$	$9.8 \cdot 10^{-2}$
4	0.99	0.223	0.37

It is clear that only for $n = 1$ and $n = 2$ the approximation is reasonable. In these two specific cases:

$$\alpha_{0n} \simeq 3.26 \cdot 10^{-6} \frac{Z^4}{T_e^{3/2} n^3} g_{nf} e^{\frac{\chi_n}{k_B T_e}} \frac{e^{-\frac{\chi_n}{k_B T_e}}}{\frac{\chi_n}{k_B T_e}} \simeq 2.06 \cdot 10^{-11} \frac{Z^2}{T_e^{1/2} n} g_{nf}$$

Now we can define the **partial recombination coefficient**: $\alpha_0^{(j)}$, that is the recombination coefficient at all levels $n \geq j$.

$$\begin{aligned} \alpha_0^{(1)} &= \sum_{n=1}^{\infty} \alpha_{0n} = \alpha_0 \\ \alpha_0^{(2)} &= \sum_{n=2}^{\infty} \alpha_{0n} = \alpha_0 - \alpha_{01} \\ \alpha_0^{(3)} &= \sum_{n=3}^{\infty} \alpha_{0n} = \alpha_0 - (\alpha_{01} + \alpha_{02}) \\ &\dots = \dots \\ \alpha_0^{(j)} &= \sum_{n=j}^{\infty} \alpha_{0n} = \alpha_0 - \sum_{n=1}^{j-1} \alpha_{0n} \end{aligned}$$

In general, for hydrogen and hydrogenoid atoms (and $n = 1$):

$$\alpha_0^{(j)} \simeq 2.06 \cdot 10^{-11} \frac{Z^2}{\sqrt{T_e}} \phi_j \left(\frac{\chi_1}{k_B T_e} \right)$$

The function ϕ_j is plotted in Fig. 7.2: at $T_e = 10^4$ K, $\phi_1 \simeq 2$ and $\phi_2 \simeq 1.27$. For the hydrogen case (or hydrogenoid atoms), calculations give $\alpha_0 = \alpha_0^{(1)} \simeq 4 \cdot 10^{-13} \text{ cm}^3 \text{ s}^{-1}$, and $\alpha_0^{(2)} \simeq 2.6 \cdot 10^{-13} \text{ cm}^3 \text{ s}^{-1}$. Therefore, we can conclude that assuming $N_e \approx 10^2 - 10^3 \text{ cm}^{-3}$, the recombination rate is:

$$R_{1,0} = N_e \alpha_0 \approx 10^{-11} - 10^{-10} \text{ s}^{-1}$$

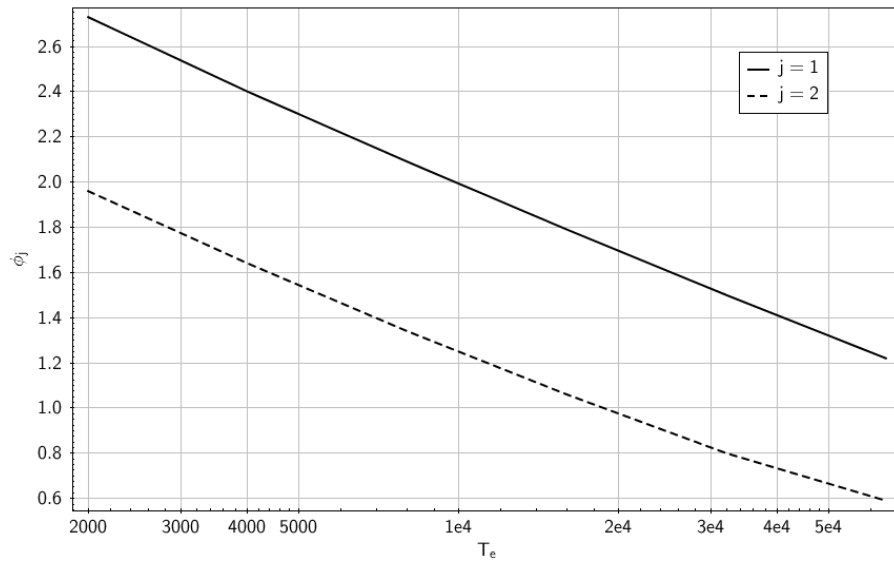


Figure 7.2: The ϕ_j function. Data have been taken by Spitzer (2004, Table 5.2).

8

Ionization structure

In the last sections we have seen that $C_{0,1} \ll R_{0,1}$, therefore the ionization equilibrium equation is very simple:

$$N_0 R_{0,1} = N_1 R_{1,0}$$

where $R_{0,1} \gg R_{1,0}$. This means that ionizations are much more probable than recombinations. Therefore, starting from a neutral gaseous cloud, and then switching on a hot thermal source, even in case of an equal number of ionizations and recombinations, a large number of free electrons still move through the nebula with some kinetic energy. In conclusion, photoionization is a heating mechanism for a nebula.

$$\frac{N_1}{N_0} = \frac{R_{0,1}}{N_e \alpha_0} \simeq \frac{2.5 \cdot 10^4}{N_e} \Rightarrow N_0 \simeq \frac{N_e^2}{2.5 \cdot 10^4}$$

where we assumed $N_1 = N_e$ since the cloud until now considered is made only by hydrogen.

If the electron density varies from 10^2 to 10^3 cm^{-3} , the ratio between ions and neutral atoms varies from 250 to 25, so in general we can affirm that $N_1 > N_0$ or $N_1 \gg N_0$. Obviously, when N_e increases the recombination rate increases and so the density of neutral atoms.

Since the average lifetime of an energy level is $t \approx 10^{-8} \text{ s}$, we can assume that almost all the neutral hydrogen atoms are populated at level $n = 1$, therefore $N_0 \simeq N_{01}$, and we can calculate the optical depth τ_ν at the head of the Lyman series ($\nu_0 = 3.29 \cdot 10^{15} \text{ Hz}$).

$$k_\nu^{bf} \simeq \frac{3 \cdot 10^{29}}{\nu^3} N_{01} \simeq 8.4 \cdot 10^{-18} N_{01} \quad [\text{cm}^{-1}]$$

where the Gaunt factor $g_{1f} \approx 1$. Therefore the absorption coefficient is around $3.37 \cdot 10^{-18} \text{ cm}^{-1}$ when $N_e = 10^2 \text{ cm}^{-3}$, and $3.37 \cdot 10^{-16} \text{ cm}^{-1}$ when $N_e = 10^3 \text{ cm}^{-3}$.

$$\tau_\nu(1 \rightarrow f) = k_\nu^{1f} \ell \simeq 3.37 \cdot 10^{-22} N_e^2 \ell \simeq 1.04 \cdot 10^{-3} N_e^2 \ell_{\text{pc}}$$

$$\tau_\nu = 1 \Rightarrow \ell = \frac{1}{k_\nu^{bf}}$$

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The size ℓ , within which the head of the Lyman series is absorbed, is around 0.1 pc, when $N_e = 10^2 \text{ cm}^{-3}$, and 0.001 pc, when $N_e = 10^3 \text{ cm}^{-3}$. In other words, the larger is the electron density and the smaller is the volume in which the head of the series is absorbed. When ℓ is much smaller than the size of the nebula, we can make the so called **on-the-spot** assumption. This assumption is very important, because every time a free electron having $v \geq 0$ recombines with an ion at level $n = 1$, an ionizing photon is emitted. Therefore, recombinations to level 1 produce a field of **diffuse ionizing radiation** in addition to that of the ionizing source, which is rather complicate to take into account.

We know that

$$N_1 R_{1,0} = N_1 N_e \alpha_0 = N_1 N_e (\alpha_0^{(2)} + \alpha_{01})$$

Given the values of α_0 and $\alpha_0^{(2)}$, we obtain that $\alpha_{01} \simeq 1.4 \cdot 10^{-13} \text{ cm}^3 \text{ s}^{-1}$. Therefore $\frac{\alpha_{01}}{\alpha_0} \simeq 0.35$ and $\frac{\alpha_0^{(2)}}{\alpha_0} \simeq 0.65$, that is more than 1/3 of the recombinations occur at level 1, while the others occur at level 2 and upper levels. This is expected since we know that $\sigma_{fb} \propto \frac{1}{n^3}$, and so the diffuse ionizing radiation field is not negligible, unless we are in the favourable condition to assume that this radiation is re-absorbed on-the-spot.

Interestingly, when a nebula is optically thick to the head of the Lyman series, it is optically thick also to the whole series, with the exception of $\text{Ly}\alpha$.

When a $\text{Ly}\beta$ photon is absorbed, in 10^{-8} s the electron moves from level 3 to level 1 and the $\text{Ly}\beta$ photon is re-emitted and finds another neutral hydrogen atom with the electron at level 1. This sort of scattering continues up to 8 times when the electron jumps from level 3 to level 2 producing a $\text{H}\alpha$ photon and from level 2 to level 1 producing a $\text{Ly}\alpha$ photon. The $\text{H}\alpha$ photon escape from the nebula since the number of hydrogen atoms populated at level 2 is very scarce.

The same happens to $\text{Ly}\gamma$, that after some scattering decays into $\text{Pa}\alpha + \text{H}\alpha + \text{Ly}\alpha$. Also $\text{Pa}\alpha$ escapes from the nebula, while $\text{Ly}\alpha$ continues its scattering causing a overpopulation of the level 2, until it escapes too.

Let us consider now the recombinations at levels $n \geq 2$:

$$N_{02} A_{21} = N_1 N_e \alpha_0^{(2)}$$

where $A_{21} = 4.7 \cdot 10^8 \text{ s}^{-1}$ and $\alpha_0^{(2)} = 2.6 \cdot 10^{-13} \text{ cm}^3 \text{ s}^{-1}$.

The optical depth at the head of the Balmer series ($\nu_{0,2} = 8.23 \cdot 10^{14} \text{ Hz}$) is:

$$\tau_\nu(2 \rightarrow f) \simeq 9.3 \cdot 10^{-39} N_e^2 \ell \simeq 2.87 \cdot 10^{-20} N_e^2 \ell_{\text{pc}}$$

that is $\tau_\nu(2 \rightarrow f) \ll 1$ in the physical conditions of a H II region. When the electron density increases to $10^{10} - 10^{12} \text{ cm}^{-3}$ the number of hydrogen atoms populated at level 2 is sufficient for the nebula to become optically thick to the Balmer series, except $\text{H}\alpha$.

8.1 The Strömgren sphere

According to the ionization equilibrium:

$$N_0 R_{0,1} = 4\pi \int_{\nu_0}^{\infty} \frac{k_{\nu}^{bf} (I_{\nu}^s + I_{\nu}^d)}{h\nu} d\nu = N_1 N_e \alpha_0^{(2)} + N_1 N_e \alpha_{01}$$

where $I_{\nu} = I_{\nu}^s + I_{\nu}^d$, that is the specific intensity is the sum of the radiation coming from the star (assumed to be the ionization source) and the diffused radiation caused by the recombinations to the fundamental level.

We define now the **ionization degree** (X) of a hydrogen cloud:

$$X = \frac{N_1}{N_0 + N_1} = \frac{N_1}{N_H}$$

If the diffuse radiation is reabsorbed on-the-spot ($I_{\nu}^d = 0$):

$$N_0 R_{0,1} = N_1 N_e \alpha_0^{(2)}$$

$$(1 - X) N_H \cdot 4\pi \int_{\nu_0}^{\infty} \frac{k_{\nu}' I_{\nu}^s}{h\nu} d\nu = X^2 N_H^2 \alpha_0^{(2)}$$

where $k_{\nu}^{bf} = k_{\nu}' N_{01} = k_{\nu}' N_0 = (1 - X) N_H k_{\nu}'$.

Let us consider a point P within the nebula at distance r from the ionizing star. What is the power of radiation measured in P when we vary the distance between r and $r + dr$?

$$P_{\nu} = \phi_{\nu} \cdot 4\pi r^2$$

$$dP_{\nu} = -k_{\nu} 4\pi I_{\nu} dV + 4\pi \varepsilon_{\nu} dV$$

where ϕ_{ν} is the ionizing flux in P and V is a spherical volume centered on the star.

$$d(\phi_{\nu} \cdot 4\pi r^2) = 4\pi r^2 (-k_{\nu} 4\pi I_{\nu} + 4\pi \varepsilon_{\nu}) dr$$

We know that $\phi_{\nu} = \phi_{\nu}^s + \phi_{\nu}^d$ and $I_{\nu} = I_{\nu}^s + I_{\nu}^d$. In addition $\varepsilon_{\nu} = \varepsilon_{\nu}^d = \varepsilon_{\nu}^{d,loc} + \varepsilon_{\nu}^{d,ext}$, that is in every infinitesimal volume we must consider the ionizing radiation diffused locally and that one coming from the adjacent regions. Therefore,

$$\frac{d}{dr} [4\pi r^2 (\phi_{\nu}^s + \phi_{\nu}^d)] = 4\pi r^2 [-4\pi k_{\nu} (I_{\nu}^s + I_{\nu}^d) + 4\pi (\varepsilon_{\nu}^{d,loc} + \varepsilon_{\nu}^{d,ext})]$$

$$\frac{d}{dr} \left[4\pi r^2 \int_{\nu_0}^{\infty} \frac{\phi_{\nu}^s + \phi_{\nu}^d}{h\nu} d\nu \right] = 4\pi r^2 \left[-4\pi \int_{\nu_0}^{\infty} \frac{k_{\nu} (I_{\nu}^s + I_{\nu}^d)}{h\nu} d\nu + 4\pi \int_{\nu_0}^{\infty} \frac{\varepsilon_{\nu}^{d,loc} + \varepsilon_{\nu}^{d,ext}}{h\nu} d\nu \right]$$

We know that

$$4\pi \int_{\nu_0}^{\infty} \frac{k_{\nu} (I_{\nu}^s + I_{\nu}^d)}{h\nu} d\nu = N_0 R_{0,1} = N_1 N_e \alpha_0 = X^2 N_H^2 \alpha_0$$

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is the number of absorbed ionizing photons per unit volume and time, while

$$4\pi \int_{\nu_0}^{\infty} \frac{\varepsilon_{\nu}^{d,loc}}{h\nu} d\nu = N_1 N_e \alpha_{01} = X^2 N_H^2 \alpha_{01}$$

is the number of ionizing photons diffused locally per unit volume and time. Much more complicated is to solve the integral involving the ionizing photons diffused around the considered volume. In this case it is mandatory to solve the transport equation.

Whenever it is possible to assume the on-the-spot approximation, $\varepsilon_{\nu}^{d,ext} = 0$ and $\phi_{\nu}^d = 0$. Then,

$$\frac{d}{dr} \left[4\pi r^2 \int_{\nu_0}^{\infty} \frac{\phi_{\nu}^s}{h\nu} d\nu \right] = -4\pi r^2 X^2 N_H^2 (\alpha_0 - \alpha_{01}) = -4\pi r^2 X^2 N_H^2 \alpha_0^{(2)}$$

We define the number of ionizing photons per unit time as:

$$Q_{ion} = 4\pi r^2 \int_{\nu_0}^{\infty} \frac{\phi_{\nu}^s}{h\nu} d\nu$$

Therefore:

$$\frac{dQ_{ion}}{dr} = -4\pi r^2 X^2 N_H^2 \alpha_0^{(2)}$$

that is the decrement of ionizing photons is equal to the number of recombinations at levels $n > 1$.

Now we calculate the distance from the star at which the number of ionizing photons drops to zero: $Q_{ion}(r_s) = 0$, where r_s is called **radius of the Strömgren sphere**.

$$\int_{R_{\star}}^r \left[\frac{dQ_{ion}}{dr} \right] dr = - \int_{R_{\star}}^r 4\pi r^2 X^2 N_H^2 \alpha_0^{(2)} dr$$

$$Q_{ion}(r) - Q_{ion}(R_{\star}) = -\frac{4}{3} \pi X^2 N_H^2 \alpha_0^{(2)} (r^3 - R_{\star}^3)$$

We can reasonably assume that $r \gg R_{\star}$, then

$$-Q_{ion}(R_{\star}) = -\frac{4}{3} \pi X^2 N_H^2 \alpha_0^{(2)} r_s^3$$

so the Strömgren radius is:

$$r_s = \left[\frac{3}{4\pi} \frac{Q_{ion}(R_{\star})}{X^2 N_H^2 \alpha_0^{(2)}} \right]^{1/3}$$

with $X = 1$.

Now we calculate the number of ionizing photons emitted by stars with different spectral types.

$$Q_{ion} = \int_{\nu_0}^{\infty} \frac{L_{\nu}}{h\nu} d\nu = 4\pi R_{\star}^2 \int_{\nu_0}^{\infty} \frac{\pi B_{\nu}(T_{\star})}{h\nu} d\nu$$

8.1. THE STRÖMGREN SPHERE

where π before $B_\nu(T_\star)$ is the solid angle at the surface of the star. As usual we use the Wien approximation, then

$$Q_{ion} = 4\pi R_\star^2 \int_{\nu_0}^{\infty} \pi \frac{2h\nu^3}{c^2} \frac{e^{-\frac{h\nu}{k_B T_\star}}}{h\nu} d\nu$$

$$Q_{ion} = \frac{8\pi^2 R_\star^2}{c^2} \int_{\nu_0}^{\infty} \nu^2 e^{-\frac{h\nu}{k_B T_\star}} d\nu = \frac{8\pi^2 k_B^3}{c^2 h^3} R_\star^2 T_\star^3 \Gamma(3, x_0)$$

where $x_0 = h\nu_0/k_B T_\star$.

$$Q_{ion} \simeq 7.92 \cdot 10^{11} R_\star^2 T_\star^3 \Gamma(3, x_0) \simeq 3.88 \cdot 10^{33} \left(\frac{R_\star}{R_\odot} \right)^2 T_\star^3 \Gamma(3, x_0)$$

In case of a O5 star with $R_\star = 10R_\odot$ and $T_\star = 47000$ K, we obtain $Q_{ion} \simeq 2.8 \cdot 10^{49} \text{ s}^{-1}$. If we have a white dwarf with $R_\star = 0.01R_\odot$ and $T_\star = 120000$ K, the number of ionizing photons is $Q_{ion} \simeq 1.14 \cdot 10^{45} \text{ s}^{-1}$. In this case we should be careful because it must be verified that the assumption of the Wien approximation is correct.

Finally, if the star is the Sun, $Q_{ion} \simeq 8.35 \cdot 10^{35} \text{ s}^{-1}$, a value much lower than the previous ones, as expected.

Correct calculations based on models of stellar atmospheres give the following results:

Sp.T.	$T_\star$ (K)	$Q_{ion} (\text{s}^{-1})$
O5	47000	$5 \cdot 10^{49}$
O7	38500	$7 \cdot 10^{48}$
O9	34500	$2 \cdot 10^{48}$
B1	22600	$3 \cdot 10^{45}$

that can be used to calculate the Strömgren radius for each star. We assume $X = 1$ and a temperature $T_e = 10^4$ K, so that $\alpha_0^{(2)} = 2.6 \cdot 10^{-13} \text{ cm}^3 \text{ s}^{-1}$.

$$r_s \simeq 0.972 \cdot 10^4 \frac{Q_{ion}^{1/3}(R_\star)}{N_H^{2/3}} \simeq 1.527 \cdot 10^{15} \left(\frac{R_\star}{R_\odot} \right)^{2/3} \frac{T_\star}{N_H^{2/3}} \sqrt[3]{\Gamma\left(3, \frac{h\nu_0}{k_B T_\star}\right)}$$

	N_H	O5	O9	B1	Sun
r_s	10	25 pc	8.4 pc	0.93 pc	$6.5 \cdot 10^{-4} \text{ pc}$
r_s	10^2	5.4 pc	1.8 pc	0.2 pc	$1.4 \cdot 10^{-4} \text{ pc}$
r_s	10^3	1.15 pc	0.4 pc	0.04 pc	$3 \cdot 10^{-5} \text{ pc}$

What does it happen in the **recombination zone**, where X drops quickly from 1 to 0 (Fig. 8.1) ? The answer is easy, the optical depth increases rapidly. So now we evaluate the thickness of this region calculating the size of layer within which the optical depth is equal to 1.

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$$\tau_\nu = \frac{3 \cdot 10^{29}}{\nu^3} (1 - X) N_H \Delta r$$

Until $X = 1$ the optical depth is 0, but when $X = 0.5$ and for example $N_H = 10^2 \text{ cm}^{-3}$,

$$\tau_\nu = 1 \quad \Rightarrow \quad 1 = 0.5 \cdot 10^2 \frac{3 \cdot 10^{29}}{\nu_0^3} \Delta r$$

where $\nu_0 = 3.29 \cdot 10^{15} \text{ Hz}$.

The result is $\Delta r \simeq 2.4 \cdot 10^{15} \text{ cm}$ or $\approx 10^{-3} \text{ pc}$. If the density is higher, as $N_H = 10^3 \text{ cm}^{-3}$, $\Delta r \simeq 2.4 \cdot 10^{14} \text{ cm}$ or $\approx 10^{-4} \text{ pc}$. This last size is comparable to the dimension of the Solar System. In conclusion, the recombination zone is very thin with respect to the whole H II region ($\Delta r \ll r_s$). Pay attention, this depends strongly on the shape of the ionizing continuum!

If the physical border of the cloud corresponds to the Strömgren radius plus the recombination zone the cloud is called **matter bounded**. Otherwise the cloud is **ionization bounded** and the gas extends beyond the recombination zone forming the so-called H I region whose temperature is low and whose emission is detected at longer wavelengths (infrared and radio).

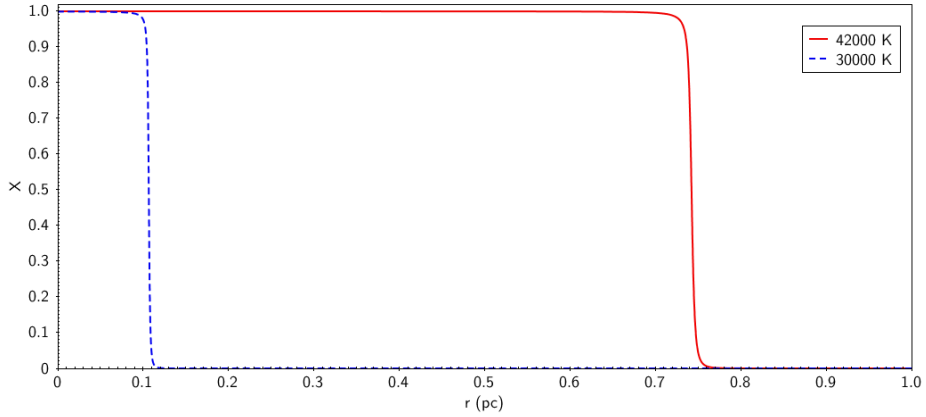


Figure 8.1: The ionization degree as a function of the distance from the ionizing source in pc units. The red solid line corresponds to a black-body of 42000 K and a radius of $12 R_\odot$ (O5 V type star). The blue dashed line corresponds to a black-body of 30000 K and a radius of $7 R_\odot$ (B0 V type star). A hydrogen density $N_H = 10^3 \text{ cm}^{-3}$ has been used. These models have been calculated with CLOUDY (version C17.01).

8.2 Ionization stratification

A nebula is not made by hydrogen only. The second more abundant element is helium: $N(\text{He})/N(\text{H}) \sim 0.1$. If we remind that the first ionization potential of He is 24.6 eV ($\text{He}^0 \rightarrow \text{He}^+$) and the second is 54.4 eV ($\text{He}^+ \rightarrow \text{He}^{++}$), it is

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easy to understand that in order to ionize He hot or very hot stars are necessary (Fig. 8.2).

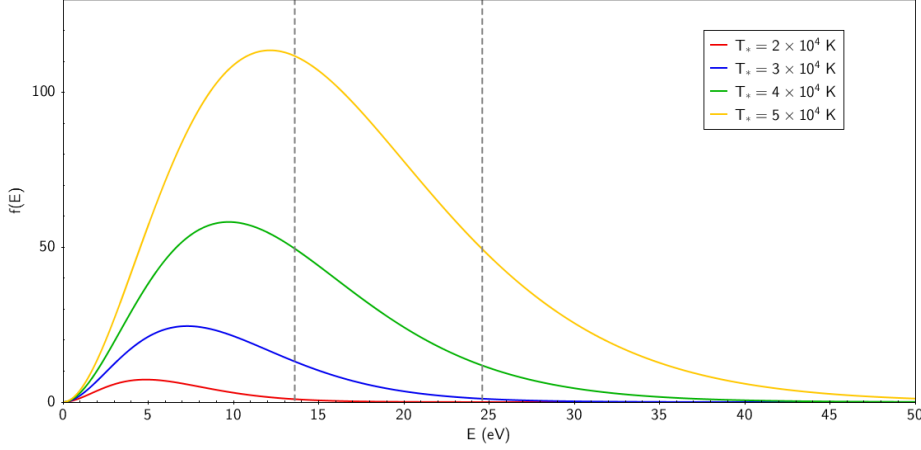


Figure 8.2: Black-body spectrum of increasing temperatures. The dashed lines indicate the hydrogen ionization potential (13.6 eV) and the helium first ionization potential (24.6 eV).

Of course the number of ionizing photons is not sufficient to justify He absorption and emission, but it is mandatory to check the photoionization cross-section. From Fig. 8.3 we can infer that photons with energies $\mathcal{E} > 24.6$ eV ionize mostly atoms of He^0 . In addition the recombinations of He^+ and He^{++} are able to photoionize H^0 , therefore the presence of He affects a lot the radiation field within the nebula.

Varying the temperature of the ionizing source, that is varying the stellar spectral type, the ionization structure of the nebula changes. Let us consider to have a star with temperature $T_* \leq 40000$ K. Many photons with $13.6 < \mathcal{E} < 24.6$ eV are available, while those with $\mathcal{E} \geq 24.6$ eV are few.

The fraction of photons with $\mathcal{E} \geq 24.6$ eV absorbed by hydrogen atoms is:

$$y = \frac{N_H \sigma_H(24.6)}{N_H \sigma_H(24.6) + N_{He} \sigma_{He}(24.6)}$$

From Fig. 8.3, $\sigma_H(24.6) \sim 10^{-18} \text{ cm}^2$ and $\sigma_{He}(24.6) \sim 8 \cdot 10^{-18} \text{ cm}^2$. In addition $N(He) \sim 0.1N(H)$, therefore $y \sim 0.56$. The fraction of photons with $\mathcal{E} \geq 24.6$ eV absorbed by helium neutral atoms is $1 - y \sim 0.44$.

Fig. 8.4 shows the ionization degree in a nebula of hydrogen and helium, where the ionizing star has a temperature $T_* = 30000$ K. We observe that the Strömgren radius of He II is smaller than that of H II and also the abundance of He II drops to zero more smoothly than H II.

If we consider a higher temperature star, like in Fig. 8.5, where $T_* = 47000$ K, the Strömgren radius of He II reaches that of H II, and He III appears very close to the ionizing source.

This can be explained in the following way. The Strömgren radius for hydrogen is ($X = 1$):

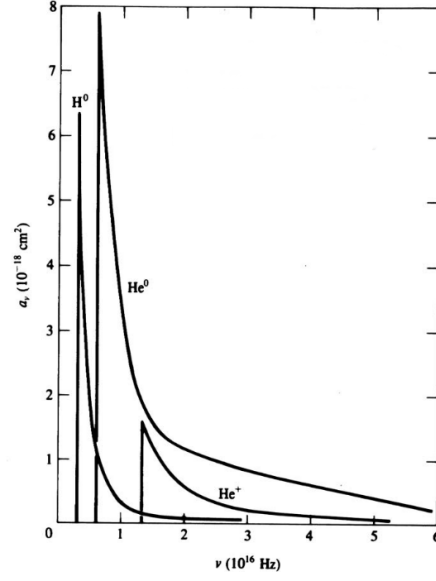


Figure 8.3: This plot, taken from Osterbrock & Ferland (2006, Fig. 2.2), shows the photoionization cross-sections of H^0 , He^0 , and He^+ .

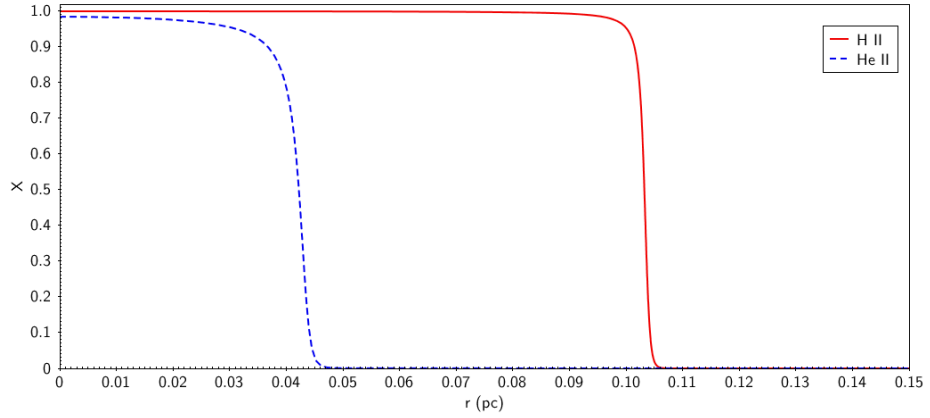


Figure 8.4: The ionization degree as a function of the distance in pc units. The ionizing source is a black-body of 30000 K and a radius of $7 R_{\odot}$ (B0 V type star). The red solid line corresponds to H II, while the blue dashed line corresponds He II. A hydrogen density $N_H = 10^3 \text{ cm}^{-3}$ has been used. These models have been calculated with CLOUDY (version C17.01).

$$r_s(H) = \left[\frac{3}{4\pi} \frac{Q_{ion}(\mathcal{E} \geq 13.6)}{N_H^2 \alpha_0^{(2)}(H)} \right]^{1/3}$$

while the same radius for helium is:

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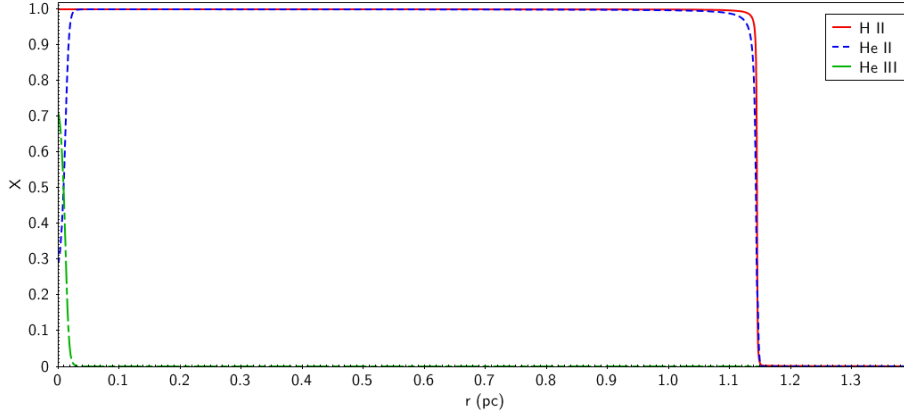


Figure 8.5: The ionization degree as a function of the distance in pc units. The ionizing source is a black-body of 47000 K and a radius of 15 $R_{\odot}$ (O3 V type star). The red solid line corresponds to H II, the blue dashed line corresponds to He II, and the green dot-dashed line corresponds to He III. A hydrogen density $N_H = 10^3 \text{ cm}^{-3}$ has been used. These models have been calculated with CLOUDY (version C17.01).

$$r_s(He) = \left[\frac{3}{4\pi} \frac{Q_{ion}(\mathcal{E} \geq 24.6)}{N_1(He^+) N_e \alpha_0^{(2)}(He)} \right]^{1/3}$$

Now, $N_1(He^+) = N_{He}$ and $N_e = N_{He} + N_H$, therefore the ratio between the two radii is:

$$\left(\frac{r_s(H)}{r_s(He)} \right)^3 = \frac{Q_{ion}(\mathcal{E} \geq 13.6)}{Q_{ion}(\mathcal{E} \geq 24.6)} \frac{N_{He}(N_{He} + N_H)}{N_H^2} \frac{\alpha_0^{(2)}(He)}{\alpha_0^{(2)}(H)}$$

We know that the abundance of He is about a tenth of the H, therefore:

$$\frac{N_{He}(N_{He} + N_H)}{N_H^2} \sim 0.1$$

In addition, $\alpha_0^{(2)}(H) \sim \alpha_0^{(2)}(He)$, then

$$\frac{r_s(H)}{r_s(He)} \simeq 0.46^3 \sqrt[3]{\frac{\Gamma\left(3, \frac{13.6}{k_B T_{\star}}\right)}{\Gamma\left(3, \frac{24.6}{k_B T_{\star}}\right)}}$$

For example, when $T_{\star} = 20000 \text{ K}$, $r_s(H) \simeq 2.7 \cdot r_s(He)$, when $T_{\star} = 30000 \text{ K}$, $r_s(H) \simeq 1.36 \cdot r_s(He)$, and finally when $T_{\star} = 40000 \text{ K}$, $r_s(H) \simeq r_s(He)$.

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If we consider a high temperature star, like a white dwarf in a planetary nebula, the ionization structure changes again and the region where He is two times ionized becomes larger (Fig. 8.6).

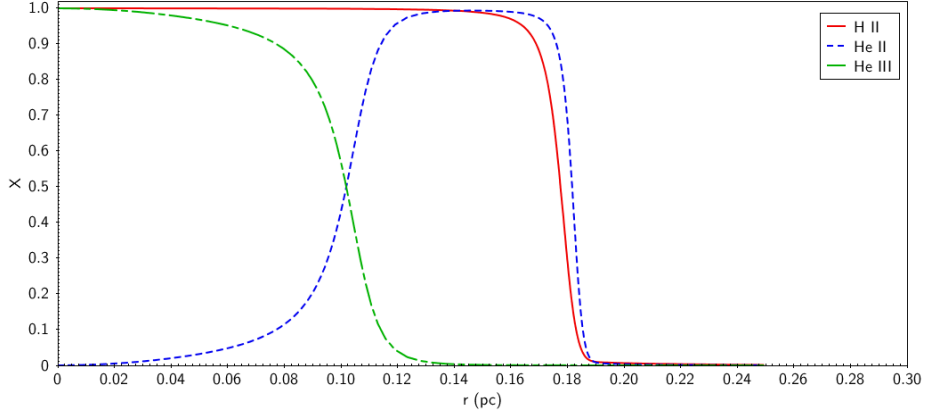


Figure 8.6: The ionization degree as a function of the distance in pc units. The ionizing source is a black-body of 120000 K and a radius of $0.1 R_{\odot}$ (WD star). The red solid line corresponds to H II, the blue dashed line corresponds He II, and the green dot-dashed line corresponds to He III. A hydrogen density $N_H = 10^3 \text{ cm}^{-3}$ has been used. These models have been calculated with CLOUDY (version C17.01).

Clearly nebulae are more complicated, since we should consider all the other elements. Their abundances are low with respect to hydrogen and helium ($N(X)/N(H) \sim 10^{-7} - 10^{-4}$), therefore the absorption of ionizing photons from heavier atoms and ions modify only slightly the radiation field. Who produces free electrons is mostly hydrogen, then helium, carbon and oxygen.

Once understood that each atomic species has its own Strömgren radius, the obvious consequence is that ions are spatially stratified within the ionized cloud and so the detected emission lines are produced in different parts of the nebula, that can have different physical conditions.

Let us consider, for example, the following elements and their first and second ionization potentials (in eV):

	I	II
N	14.5	29.6
O	13.6	35.1
Ne	21.6	41
S	10.4	23.3
Ar	15.8	27.6

In case of a star with $T_{\star} < 40000 \text{ K}$, beyond the Strömgren sphere of H II, that is where the photons have $\mathcal{E} < 13.6 \text{ eV}$, we find H^0 , He^0 , N^0 , O^0 , Ne^0 , S^0 and S^+ , and Ar^0 .

Between the Strömgren spheres of H II and He II ($13.6 \leq \mathcal{E} < 24.6 \text{ eV}$) we find H^+ , He^0 , N^0 and N^+ , O^+ , Ne^0 and Ne^+ , S^+ , and Ar^+ .

Finally, within the Strömgren sphere of He II ($\mathcal{E} \geq 24.6$ eV) we find H^+ , He^+ , N^+ and N^{++} , O^+ and O^{++} , Ne^+ and Ne^{++} , S^+ and S^{++} , Ar^+ and Ar^{++} .

In case of a star with $40000 < T_\star < 100000$ K, the Strömgren spheres of H II and He II are coincident, therefore outside the H and He recombination zone we find mostly neutral species, while ions are inside. However, they are not randomly mixed, since there is always a Strömgren sphere for each ion and the most ionized species will be located close to the ionizing source.

It is worth to note that in the spectrum of a H II region we detect forbidden lines from neutral atoms like N I and O I. These atoms, together with S II can survive in the recombination zone and therefore they can collisionally interact with free electrons that can populate their metastable levels. Since the recombination region is very thin in nebulae ionized by stars, these emission lines are usually weak.

8.3 H I regions

As already mentioned, beyond the recombination zone, that is in the so-called **H I region**, only photons with energies $\mathcal{E} < 13.6$ eV can flow. But it does not mean that all atomic species are neutral, since other elements can have at least the first ionization potential lower than 13.6 eV. The following table shows an example:

	I	II
C	11.3	24.4
Na	5.1	47.3
Si	8.2	16.3
Ca	6.1	11.9
Fe	7.9	16.2

In addition we should not forget the presence of the X-ray background photons and of the cosmic rays, that are able to cause ionization, even if we have already seen that $R_{0,1}$ and $C_{0,1}$ are about 10^{-17} s^{-1} .

Let us call $X_0 = N_e/N_H$ the fraction of free electron produced by metals. We can now write the ionization equilibrium equation in the following way:

$$N_H(R_{0,1} + C_{0,1}) = N_1(N_1 + X_0 N_H) \alpha_0^{(2)}$$

$$R_{0,1} + C_{0,1} = \frac{N_1^2}{N_H} \alpha_0^{(2)} + \frac{X_0 N_1 N_H}{N_H} \alpha_0^{(2)}$$

$$(N_H \alpha_0^{(2)}) X^2 + (X_0 N_H \alpha_0^{(2)}) X - (R_{0,1} + C_{0,1}) = 0$$

The solution of the equation is:

$$X = \frac{X_0}{2} \left(\sqrt{1 + \frac{4(R_{0,1} + C_{0,1})}{X_0^2 N_H \alpha_0^{(2)}}} - 1 \right)$$

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If we consider solar abundances: $N(C)/N(H) \sim 2.4 \cdot 10^{-4}$, $N(Na)/N(H) \sim 2.1 \cdot 10^{-6}$, $N(Si)/N(H) \sim 3.5 \cdot 10^{-5}$, $N(Ca)/N(H) \sim 2.3 \cdot 10^{-6}$, $N(Fe)/N(H) \sim 2.8 \cdot 10^{-5}$, we can give the following approximation: $X_0 \sim 3 \cdot 10^{-4}$, and we observe that carbon is the most abundant element in a H I region.

We can now evaluate the ionization degree and the electron density in a cloud with the following physical parameters: $T_e = 100$ K and $N_H = 20 \text{ cm}^{-3}$.

In this case, $\alpha_0^{(2)} \simeq 6.8 \cdot 10^{-12} \text{ cm}^3 \text{ s}^{-1}$, therefore we obtain: $X = N_1/N_H \simeq 1.2 \cdot 10^{-4}$, and $N_e = N_1 + X_0 N_H = N_H(X + X_0) \simeq 8.4 \cdot 10^{-3} \text{ cm}^{-3}$.

A similar result is obtained increasing the temperature to $T_e = 4000$ K and reducing the density to $N_H = 1 \text{ cm}^{-3}$. Indeed, $\alpha_0^{(2)} \simeq 5.21 \cdot 10^{-13} \text{ cm}^3 \text{ s}^{-1}$, then: $X = 4 \cdot 10^{-3}$, and $N_e \simeq 4.3 \cdot 10^{-3} \text{ cm}^{-3}$.

Let us try to investigate what happens to sodium and calcium.
We remind that

$$R_{0,1} = 4\pi \int_{\nu_0}^{\infty} \frac{k'_\nu I_\nu}{h\nu} d\nu$$

where $\nu_0(Na) = 1.23 \cdot 10^{15} \text{ Hz}$ and $\nu_0(Ca) = 1.48 \cdot 10^{15} \text{ Hz}$. The result of ionization rate is $R_{0,1} \simeq 4 \cdot 10^{-7} \text{ s}^{-1}$ for Na and $R_{0,1} \simeq 2.4 \cdot 10^{-7} \text{ s}^{-1}$ for Ca. The ionization equilibrium equation is:

$$N_0 R_{0,1} = N_1 N_e \alpha_0 \Rightarrow N_e \frac{N_1}{N_0} = \frac{R_{0,1}}{\alpha_0}$$

where $\alpha_0(Na) \simeq 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ and $\alpha_0(Ca) \simeq 10^{-9} \text{ cm}^3 \text{ s}^{-1}$. Therefore, if our H I region has electron temperature $T_e = 100$ K and electron density $N_e = 10^{-2} \text{ cm}^{-3}$, the ionization degree N_1/N_0 is $\sim 4 \cdot 10^2$ for Na and $\sim 2.4 \cdot 10^4$ for Ca.

For comparison in a H II region with $T_e = 10^4$ K and $N_e = 10^2 - 10^3 \text{ cm}^{-3}$, $\alpha_0(Na) \simeq 10^{-8} \text{ cm}^3 \text{ s}^{-1}$ and $\alpha_0(Ca) \simeq 10^{-11} \text{ cm}^3 \text{ s}^{-1}$, therefore $N_1/N_0 \simeq 0.4 \div 0.04$ for Na and $N_1/N_0 \simeq 240 \div 24$ for Ca.

Because of the higher electron density there is also higher probability that recombination occur. And so sodium and calcium are less ionized in a H II region.

9

Thermal equilibrium

In a H II region the thermal equilibrium is reached when the **rate of heating** Γ is balanced by the **rate of cooling** Λ (both in units of s^{-1}):

$$\Gamma = \Lambda \quad (9.1)$$

The result of this equilibrium is the stability of the temperature.

Heating processes are:

- photoionization
- ionization of H I by cosmic rays
- photoelectric effect on the surface of dust grains
- evaporation of H_2 molecules on the surface of dust grains

while cooling processes are:

- collisional excitation of metastable levels in metals
- free-free
- collisions of free electrons with dust grains

Among them, the main heating mechanism is for sure photoionization, while free-free is the dominant cooling process in a pure H nebula, and forbidden lines plus free-free are the main mechanisms when also heavier elements are present.

We explore now these processes calculating their heating and cooling rates. The first one is **photoionization** or better the energy budget of the photoionization-recombination equilibrium. The number of photoionizations corresponds to the number of released electrons:

$$\int_{\nu_0}^{\infty} \frac{k_{\nu} I_{\nu}}{h\nu} d\nu$$

The total kinetic energy of these free electrons is:

$$\int_{\nu_0}^{\infty} h(\nu - \nu_0) \frac{k_{\nu} I_{\nu}}{h\nu} d\nu$$

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and the average kinetic energy of a single free electron is:

$$\bar{\mathcal{E}}_e = \frac{\int_{\nu_0}^{\infty} h(\nu - \nu_0) \frac{k_{\nu} I_{\nu}}{h\nu} d\nu}{\int_{\nu_0}^{\infty} \frac{k_{\nu} I_{\nu}}{h\nu} d\nu}$$

It is clear that this energy depends on the spectrum of the ionizing source and not on its intensity.

If we are close to the source, we can assume a black-body emission and since $h\nu_0 \gg k_B T_{\star}$ we can apply the Wien approximation. In addition we know that the absorption coefficient $k_{\nu} \propto \nu^{-3}$, that is the absorption probability decreases when the frequency increases. The resulting effect is that high energy photons flow through the cloud and the average energy of electrons grows moving away from the ionizing source. Finally the electron temperature increases with distance, but we should remind that this is true only for a pure H nebula.

$$\bar{\mathcal{E}}_e = \frac{\int_{\nu_0}^{\infty} h(\nu - \nu_0) \frac{1}{\nu^3} \frac{\nu^3 e^{-\frac{h\nu}{k_B T_{\star}}}}{h\nu} d\nu}{\int_{\nu_0}^{\infty} \frac{1}{\nu^3} \frac{\nu^3 e^{-\frac{h\nu}{k_B T_{\star}}}}{h\nu} d\nu}$$

$$\bar{\mathcal{E}}_e = k_B T_{\star} \left(\frac{e^{-x_0}}{E_1(x_0)} - x_0 \right)$$

where $x_0 = h\nu_0/k_B T_{\star}$ and $E_1(x_0)$ is the exponential integral. We can finally write:

$$\bar{\mathcal{E}}_e = \psi_0 k_B T_{\star}$$

where ψ_0 is a function ranging from about 0.98 at $T_{\star} = 4000$ K to 0.86 at $T_{\star} = 32000$ K (from Spitzer 2004, Tab. 6.1).

The energy gained through photoionizations is: $N_1 N_e \alpha_0 \bar{\mathcal{E}}_e$, while the energy lost because of recombinations is: $\frac{1}{2} m_e N_1 N_e \sum_n \sigma_{fb} v^3$. Thus, the rate of photoionization heating is:

$$\Gamma_{ph} = N_1 N_e \left(\alpha_0 \bar{\mathcal{E}}_e - \frac{1}{2} m_e \sum_n \sigma_{fb} v^3 \right)$$

where $N_1 = N_p$, $\bar{\mathcal{E}}_e = \psi_0 k_B T_{\star}$ (close to the source),

$$\alpha_0 = \alpha_0^{(1)} = \frac{2.06 \cdot 10^{-11}}{\sqrt{T_e}} \phi_1 \left(\frac{\chi_1}{k_B T_e} \right)$$

and

$$\sum_{n=1}^{\infty} v^3 \sigma_{fb} = \frac{2A_r}{\sqrt{\pi}} \left(\frac{2k_B T_e}{m_e} \right)^{3/2} \frac{h\nu_0}{k_B T_e} \zeta_1 \left(\frac{h\nu_0}{k_B T_e} \right) \simeq 6.26 \zeta_1 \sqrt{T_e}$$

where

$$A_r = \frac{16he^2}{3\sqrt{3}(4\pi\epsilon_0)m_e^2 c^3} = 2.105 \cdot 10^{-22} \text{ cm}^2$$

$$\Gamma_{ph} = N_e N_p \left(\frac{2.06 \cdot 10^{-11}}{\sqrt{T_e}} \phi_1 \psi_0 k_B T_\star - \frac{1}{2} m_e 6.26 \zeta_1 \sqrt{T_e} \right)$$

$$\simeq 2.84 \cdot 10^{-27} \frac{N_e N_p}{\sqrt{T_e}} (\psi_0 \phi_1 T_\star - \zeta_1 T_e)$$

Both ϕ_1 and ζ_1 depends on T_e according to the following table (from Spitzer 2004, Tab. 5.2 and 6.2):

T_e	4000	8000	16000	32000	64000
ϕ_1	2.40	2.09	1.79	1.50	1.22
ζ_1	1.95	1.65	1.37	1.10	0.84

Now we want to calculate Γ_{ph} for the entire H II region.

$$\Gamma_{ph} = N_e N_p \left(\alpha_0^{(2)} \bar{\mathcal{E}}_e - \frac{1}{2} m_e \sum_{n=2}^{\infty} v^3 \sigma_{fb} \right)$$

where $\alpha_0^{(2)}$ is used in place of $\alpha_0^{(1)}$ because we are using the on-the-spot approximation, and $\mathcal{E}_e = \bar{\psi} k_B T_\star$. The function $\bar{\psi}$ assumes the values reported in the following table (from Spitzer 2004, Tab. 6.1):

$T_\star$	4000	8000	16000	32000
$\bar{\psi}$	1.05	1.1	1.2	1.38

$$\Gamma_{ph} = N_e N_p \left(\frac{2.06 \cdot 10^{-11}}{\sqrt{T_e}} \phi_2 \bar{\psi} k_B T_\star - \frac{1}{2} m_e 6.26 \zeta_2 \sqrt{T_e} \right)$$

$$\simeq 2.84 \cdot 10^{-27} \frac{N_e N_p}{\sqrt{T_e}} (\bar{\psi} \phi_2 T_\star - \zeta_2 T_e)$$

where ϕ_2 and ζ_2 depends on T_e according to the following table (from Spitzer 2004, Tab. 5.2 and 6.2):

T_e	4000	8000	16000	32000	64000
ϕ_2	1.64	1.34	1.06	0.80	0.59
ζ_2	1.20	0.92	0.67	0.46	0.30

Now we explore the cooling processes. The first one is caused by collisions between free electrons and ions that cause the formation of **forbidden lines**. If 1 is the fundamental level and m a metastable level, the rate of collisional cooling is given by:

$$\Lambda_{coll} = N_e \sum_m (\mathcal{E}_m - \mathcal{E}_1) (N_{i1} Q_{1m} - N_{im} Q_{m1})$$

CHAPTER 9. THERMAL EQUILIBRIUM

Let us make the example of [O III] $\lambda 5007$ and consider the rate of collisional excitation. The energy lost by electrons is:

$$\begin{aligned}\Lambda_{coll}^{ex}([\text{O III}]) &= N_e \Delta \mathcal{E}_{5007} N_1 Q_{12} \\ &= N_e \Delta \mathcal{E}_{5007} N_1 Q_{21} \frac{g_2}{g_1} e^{-\frac{\Delta \mathcal{E}_{5007}}{k_B T_e}} \\ &= N_e \Delta \mathcal{E}_{5007} N_1 \frac{8.63 \cdot 10^{-6} \Omega_{12}}{g_2 \sqrt{T_e}} \frac{g_2}{g_1} e^{-\frac{\Delta \mathcal{E}_{5007}}{k_B T_e}}\end{aligned}$$

N_1 is the density of O III ions populated at the fundamental level. Since the abundance of O is $N(\text{O})/N(\text{H}) \sim 10^{-4}$ in H II regions, we can reasonably assume $N_1 \sim 10^{-4} N_p$. Therefore:

$$\Lambda_{coll}^{ex}([\text{O III}]) \simeq 1.57 \cdot 10^{-21} N_e N_p \frac{e^{-\frac{2.877 \cdot 10^4}{T_e}}}{\sqrt{T_e}}$$

The second cooling process is free-free, as already mentioned.

In this case the rate of free-free cooling corresponds to the emissivity that we have already calculated:

$$\Lambda_{ff} = 4\pi \varepsilon_{ff} \simeq 1.426 \cdot 10^{-27} N_e N_p \sqrt{T_e} \bar{g}_{ff}$$

Plotting the logarithm of these rates normalized by $N_e N_p$ (Fig. 9.1) we observe that the heating rate for photoionization decreases as electron temperature increases, while both cooling rates for collisions and free do the contrary. In case of a pure hydrogen nebula, photoionization is balanced only by free-free, which is clearly less efficient than collisions. Indeed the equilibrium temperature is significantly higher in pure H II regions.

It is interesting to investigate what happens to a H II region if the ionizing source suddenly switches off. Obviously the nebula cools down, the hydrogen and the other elements recombine, the emission lines disappear and so the continuum with the result that the nebula is no more detected in visible light. But it is still there and longer wavelengths are mandatory to observe it.

If the average energy of gas particles is $\frac{3}{2} k_B T$ the **cooling time** is:

$$t_c = \frac{\frac{3}{2} k_B T N_p}{\Lambda}$$

Let us assume $T = 10^4$ K and $\Lambda = \Lambda_{coll} \sim 10^{-24} N_p^2$. We obtain:

$$t_c \sim \frac{10^{12}}{N_p}$$

Thus, if $N_p = N_e = 10^2 \text{ cm}^{-3}$ the cooling time is $t_c \sim 300$ yrs, that is the nebula disappear in few tens years. In case we have a pure H nebula, being free-free less efficient, the cooling time is approximately 10 times longer.

As a comparison we can estimate the recombination time.

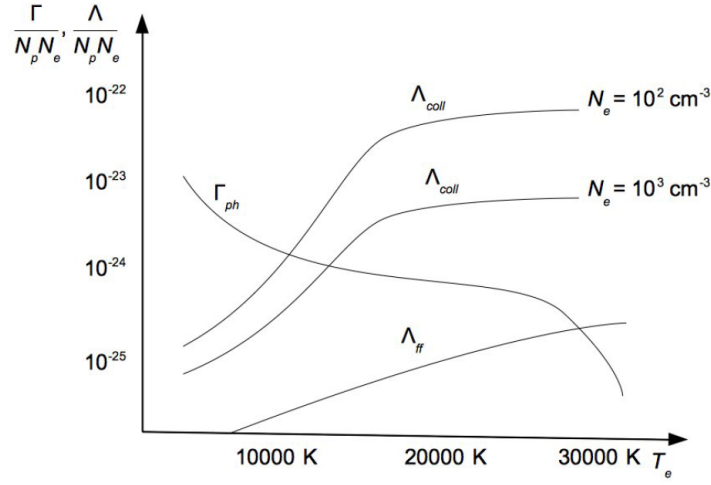


Figure 9.1: Thermal balance between the main heating and cooling mechanisms working in a H II region. The equilibrium temperatures correspond to the values where the efficiencies of heating and cooling processes become equal.

$$t_r = \frac{1}{R_{1,0}} = \frac{1}{N_e \alpha_0} = \frac{\sqrt{T_e}}{N_e \cdot 2.06 \cdot 10^{-11} \phi_1}$$

If $T_e = 10^4$ K, $\phi_1 \sim 2$, and the recombination time is around 800 yrs, having assumed $N_e = 10^2 \text{ cm}^{-3}$. Also in this case, a higher density nebula recombines more quickly.

Appendix A

Here a list of important physical constants:

$$h = 6.626 \cdot 10^{-34} \text{ J s}$$

$$k_B = 1.38 \cdot 10^{-23} \text{ J K}^{-1}$$

$$e = 1.602 \cdot 10^{-19} \text{ C}$$

$$m_e = 9.109 \cdot 10^{-31} \text{ kg}$$

$$\sigma_{SB} = 5.67 \cdot 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

$$\frac{1}{4\pi\epsilon_0} = 8.9876 \cdot 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$\epsilon_0 = 8.8542 \cdot 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

Appendix B

Tabella periodica degli elementi

1 1 A	2 2 A	3-10 3 A 4 A 5 A 6 A 7 A 8 VIII 9 10 11 IB 12 IIB										13 III B	14 IV B	15 V B	16 VI B	17 VII B	18 0																																																																																					
1 H 1,0079 +1	2 He 4,002 0																	19 K 39,098 +1	20 Ca 40,078 +2	21 Sc 44,956 +3	22 Ti 47,88 +3, +4	23 V 50,94 +3, +4	24 Cr 51,996 +3, +4	25 Mn 54,94 +3, +4	26 Fe 55,847 +2, +3	27 Co 58,933 +2, +3	28 Ni 58,69 +2, +3	29 Cu 63,546 +1, +2	30 Zn 65,38 +2	31 Ga 69,723 +3	32 Ge 72,61 +4	33 As 74,922 +3, +5	34 Se 78,96 -2, +4	35 Br 79,9 -1, +3	36 Kr 83,8 0	37 Rb 85,468 +1	38 Sr 87,62 +2	39 Y 88,906 +3	40 Zr 91,224 +3, +4	41 Nb 92,906 +3, +5	42 Mo 95,94 +3, +4, +5, +6	43 Tc [98] +3, +4, +5, +6, +7	44 Ru 101,07 +3, +4	45 Rh 102,905 +3	46 Pd 106,42 +2, +4	47 Ag 107,868 +1	48 Cd 112,411 +2	49 In 114,82 +3	50 Sn 118,69 +2, +4	51 Sb 121,75 -3, +3, +5	52 Te 127,6 -2, +4	53 I 126,9 -1, +3, +5	54 Xe 131,29 0	55 Cs 132,905 +1	56 Ba 137,327 +2	57 La ⁽¹⁾ 138,905 +3	58 Ce ⁽²⁾ 140,12 +3	59 Pr ⁽²⁾ 140,908 +3	60 Nd ⁽²⁾ 144,24 +3	61 Pm ⁽²⁾ [145] +3	62 Sm ⁽²⁾ 150,36 +3	63 Eu ⁽²⁾ 151,965 +3	64 Gd ⁽²⁾ 157,25 +3	65 Tb ⁽²⁾ 158,925 +3	66 Dy ⁽²⁾ 162,5 +3	67 Ho ⁽²⁾ 164,93 +3	68 Er ⁽²⁾ 167,26 +3	69 Tm ⁽²⁾ 168,934 +3	70 Yb ⁽²⁾ 173,04 +3	71 Lu ⁽²⁾ 174,967 +3	72 Hf ⁽²⁾ 178,49 +3, +4	73 Ta ⁽²⁾ 180,948 +3, +5	74 W ⁽²⁾ 183,85 +3, +4, +5, +6	75 Re ⁽²⁾ 186,207 +3, +4, +5, +6, +7	76 Os ⁽²⁾ 190,2 +3, +4, +5, +6	77 Ir ⁽²⁾ 192,22 +3, +4	78 Pt ⁽²⁾ 195,08 +2, +4	79 Au ⁽²⁾ 196,967 +1, +3	80 Hg ⁽²⁾ 200,59 +2, +4	81 Tl ⁽²⁾ 204,383 +1, +3	82 Pb ⁽²⁾ 207,2 +2, +4	83 Bi ⁽²⁾ 208,98 +3, +5	84 Po ⁽²⁾ [209] +2, +4	85 At ⁽²⁾ [210] +3, +5	86 Rn ⁽²⁾ [222] 0	87 Fr ⁽²⁾ [223] +1	88 Ra ⁽²⁾ 226,025 +2	89 Ac ⁽²⁾ 227,028 +3													101 VII B	102 0
																		ELEMENTI SUPERPESANTI (numero atomico > 103)																																				118 0																																																
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(3) anche Ku e Fl
(4) anche Ha

Figure 9.2: Periodic table of Mendeleev.